Chapter 3
Motion

Abstract The study of human motion begins with classifying the many types of locomotion and continues with descriptions of muscles in the body. The analysis of standing starts with a discussion of the overall and local stability of the body and of friction, and continues with models of walking. This includes a review of harmonic motion and pendulums. These concepts are applied to running, which is also compared to walking. The energetics and dynamics of jumping and the pole vault are followed by modeling how to throw balls, such as baseballs, and swing objects, such as golf clubs. This is then related to a more formal treatment of multisegment modeling. Collisions of the body due to falls, in contact sports such as boxing and karate, and in hitting balls are investigated. The flight of balls that are thrown or hit is analyzed, including how it is affected by drag and ball spinning. The effect of friction is probed in skiing and in the bouncing and rolling of balls.

We now discuss human motion. We start with the absence of motion, standing, and then progress to walking and running. Another type of motion is jumping, and we will examine the vertical jump, high jump, long jump, and pole vault. We will develop a model of throwing a ball and then examine collisions of the human body with other objects and collisions caused by the body. In each case we will ask: *What does the body do? What can the body do? How does this translate into performance?* In many instances we will focus on the body itself, but in some we will need to consider the system of the body plus other objects such as running shoes, the floor surface, a ball, or a baseball bat. We will delay our discussion of swimming until the discussion of fluids in Chap. 7.

In this chapter we will follow (1) the motion itself—the kinematics of motion, (2) the forces causing this motion and the ensuing motion—the dynamics or kinetics, (3) the muscles that provide these forces, and (4) the energetics of the motion. Both linear and rotational motion can be important. The discussion of pendulums in the section on walking includes a brief review of the harmonic oscillator and rotational kinematics and kinetics. In Chap. 5 we will address how muscles work. We will be guided by the need to include external propulsion forces on the body itself, internal forces due to muscles, the effect of gravity acting at its center of mass, reaction forces due to the ground and other objects and within the body, and friction. Friction can due

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to contact with surfaces and motion in fluids (gases and liquids), which is called drag. In fluids, motion is also affected by buoyancy and lift (Chap. 7) [64]. Our analysis of statics in Chap. 2 did not include those forces that occur only with motion, such as kinetic friction, drag and lift. In our discussion of motion we willl focus on the many physical aspects of kinesiology, the scientific study of human movement, which encompasses the interaction of anatomy, physiology, biomechanics and psychology [45, 64]. Many of the motions we will describe are analyzed in the wonderful books by Alexander [5] and Hay [66].

There are many types of motions, both in everyday activities and sporting events. One way to classify them is as: (1) moving objects by pushing or pulling, (2) moving objects by throwing, striking, and kicking, (3) locomotion on the ground, (4) locomotion when suspended and/or free of support, and (5) locomotion in water. The stopping of motion is also important, which can involve impact (of people, of objects (as in catching balls)) [64]. We will not be able to examine every type of motion in these five categories, but will cover the core issues and examine important examples of the first four categories, along with the stopping of motion, in the text and problems in this chapter, and then the last type of motion in Chap. 7.

The motions in each category are quite different:

1. In pushing and pulling, there is extension or flexion of an articular joint, usually in an extremity (elbow, shoulder, hip, knee, ankle) that leads to linear (rectilinear) motion of the segment at the end. Examples include pushing a table, lifting a suitcase, shooting an arrow with a bow, opening a window, a boxer punching, heading a soccer ball, and pushing and pulling in football and wrestling. Some involve momentary contact (shot putting, or making a chest pass or free throw in basketball) and some continuous action (pushing furniture, pulling rollable luggage, lifting a garage door, weightlifting). Some involve contact with an object (heading a soccer ball) and some contact on an instrument (pulling a bow) that is in contact with an object (an arrow). Lifting is a type of pulling; holding is the static limit of lifting, and during locomotion holding becomes carrying.
2. In moving objects by throwing, striking, and kicking, there is a sequential series of movements of different body segments whose velocities add to induce rapid curvilinear motion of a distal segment such as your hand or foot [64]. This distal segment may contain an object (throwing) or it may not (striking and kicking). If it contains an object, the overall motion could end with the object being released at high speed (throwing a ball or knife) or with it remaining in the segment (stabbing with a knife). These motions include throwing a ball, batting a ball, kicking a soccer ball, driving a golf ball, swinging a tennis racket, and so on. Throws can be overhand (pitching a baseball, passing a football, throwing a javelin, serving in volleyball or tennis), underhand (pitching a softball, pitching a baseball in a submarine manner, serving in badminton, driving a golf ball), or sidearm (pitching a baseball, swinging a bat, hitting with a ball in racketball or tennis). There are commonalities in most of these motions. They start with a backswing or windup that gets the participating segments in favorable positions for the full range of

motion (muscles that will be shortened are initially lengthened) and that increases the potential duration of applying forces during the event, and therefore the final speed of the hand or foot. They proceed with establishing a stable base (taking a forward step in pitching a baseball) and then applying a force (through leg, torso, shoulder, elbow, wrist motion), and end with the follow-through. In baseball pitching, about half of the final ball speed comes from the forward step and body rotation, and about half from the rotation of the shoulder, elbow, wrist and fingers. The longer the follow-through in slowing down the arm, the less the force required from external rotator muscles in the shoulder (mostly the rotator cuff muscles).

3. Locomotion on the ground includes that (a) with direct foot contact, such as walking, running, climbing and descending (stairs, ladders, hills), jumping, leaping, skipping, hopping, sliding, and dancing; (b) with other parts of the body and sometimes the hands, such as walking on hands and creeping, crawling, and walking with crutches; (c) involving rotary motion, such as rolling and doing cartwheels and handsprings; and (d) through an intermediary instrument, such as bicycling, moving with a wheelchair, skating or skateboarding on the ground, skiing on snow, and skating on ice.
4. Locomotion when suspended includes hanging, climbing, and swinging, which includes trapeze activities and gymnastic events, such as the high bar, parallel bars, uneven bars, and rings. All rely on hand, arm and shoulder muscles. The swinging motion is in part determined by angular momentum and centripetal force considerations, as in pendulums. Locomotion free of support includes the long jump, high jump, springboard diving, somersault, and the free fall. Tracking the motion of the center of mass and rotary motion (how body orientation and shape are related to angular momentum) is important. Any locomotion on the ground that leads to takeoff and the free fall after takeoff are also important.
5. Locomotion in water includes human locomotion, namely swimming, and human-powered locomotion, such as rowing, canoeing and kayaking. Swimming differs from locomotion on the ground because buoyancy needs to be considered rather than only gravity acting on the body. Also, pushing off water gives less resistance than pushing off from the ground, water provides more resistance to motion than does air, and the body is usually horizontal rather than vertical due to the last two factors. Similar principles apply for human-powered aquatic locomotion. The paddles in canoeing and kayaking are used much as arms are in swimming. In rowing, the motion is different because the oars are kept in oarlocks, and much of the propulsive force is provided by the legs.

Several of these motions are parts of athletic events. The physics of these motions will be described in this chapter and Chap. 7. Appendix G will give an overview of the physics of several sports, and will show how to locate specific references to different sports in this text. Also, for more on general lifting see Problems 2.34–2.36.

3.1 Kinematics and Musculature

We can analyze the body itself, a part of the body, or an object held or propelled by the body as it undergoes a range of motions [64]. In *linear or rectilinear motion*, the entire object moves in a straight line. In *angular or rotary motion* there is motion about a fixed point; this motion could be an arc, such as motion about articular joints in the body during motion or throwing, or in a complete circle. More specifically, in *curvilinear motion* the object is translated along a curved path, which the wrist and the bowling ball undergo during the swing phase in bowling, and in *circular motion* the object moves in a circular curvilinear path, as in swinging the hammer in a hammer throw before release. The path of the center of mass is tracked in each. Each of these motions can be repetitive, such as bouncing a ball, which is called *reciprocating motion*. Many actions are combinations of these types of movements. After release, the initial trajectories of a bowling ball and hammer are rectilinear. The rectilinear motion of a bicycle is due to the circular motion of its wheels, which is caused by the reciprocating motion of the legs that are powered by angular motion about the articular joints.

Following the motion in walking, running, cycling, and jumping means tracking the center of mass of the body, as well as the angles the thigh makes with the hip, the lower leg with the thigh, and the foot with the lower leg, as shown in Fig. 3.1 [21]. (Some conventions for defining these angles are somewhat different than those shown in this figure.) The thigh angle should be that with respect to the trunk, because the trunk usually tilts forward during running; this differs from the angle from the vertical (which is sometimes given instead). These three angles correspond to rotation in the sagittal plane of Fig. 3.1 about the hip (coxofemoral), knee, and ankle (talocrural) joints. Increasing θ_{thigh} is abduction, decreasing it is adduction (as in Figs. 1.9 and 1.10). Increasing θ_{knee} is flexion, decreasing it is extension. Increasing θ_{ankle} is dorsiflexion (flexion), decreasing it is plantarflexion (extension).

The hip has three degrees of freedom of rotation, as does a ball-and-socket joint (Fig. 1.4). This sagittal plane flexion/extension motion about the mediolateral axis of the hip in Fig. 3.1 can be larger in amplitude than the other two rotations (adduction–abduction about an anteroposterior axis and internal–external rotation about the lon-

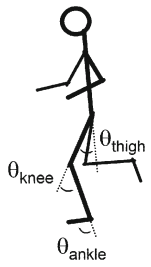


Fig. 3.1 One set of conventions for hip (thigh), knee (lower leg), and ankle (foot) angles. Other conventions are also used, such as with the thigh angle being relative to the vertical and the ankle angle defined as 90° less than here (based on [98])

gitudinal axis of the femur) (Table 1.10). While these two rotations are not negligible, we can ignore them here. The knee sagittal plane flexion/extension motion about the mediolateral axis is the one angular degree of freedom of the knee described in Fig. 1.4 as a "hinge." The knee joint actually has three degrees of freedom, but the other two (internal–external rotation and adduction–abduction (varus-valgus)) are much less important because of soft tissue and bony constraints. Ankle motion of interest here is restricted to extension (plantarflexion) and flexion (dorsiflexion). These three major sagittal plane rotations of hip, knee, and ankle constitute a three-segment model of the leg, which is in contrast to the one segment models of levers examined in Chap. 2. The head, arms, and trunk also move relative to the rest of the body during the walking and running cycles.

Kinematics entails tracking the motion of the center of mass of the body, whether or not it is rigid, and tracking these angles (described here only in the sagittal plane) versus time and versus each other during a walking or running cycle. These three angles are changed by forces controlled by the different sets of muscles outlined in Tables 3.1, 3.2 and 3.3. Several of these muscles are depicted in Figs. 1.8, 2.24, 3.2, 3.3 and 3.4 (see [133] for more details).

Table 3.1 Muscles that move the thigh (upper leg) (θ_{thigh}, femur relative to hip)

Location, muscle	Action	Approximate origin/insertion
Pelvic girdle		
Psoas major[a]	Flexes thigh	Ilium/upper femur
Iliacus[a]	Flexes thigh	Ilium/upper femur
Anterior thigh		
Pectineus	Flexes, adducts thigh	Pubis/upper femur
Adductor longus	Adducts, rotates, flexes thigh	Pubis/upper femur
Adductor brevis	Adducts thigh	Pubis/upper femur
Adductor magnus	Adducts, rotates thigh	Pubis, ischium/upper femur
Gracilis	Adducts thigh, flexes lower leg	Pubis/upper tibia
Lateral thigh		
Tensor fasciae latae	Aids thigh flex, abduction, rotation	Ilium/upper tibia
Posterior thigh		
Gluteus maximus	Extends, rotates thigh; stabilizes knee	Hip/top femur
Gluteus medius	Abducts, rotates thigh	Ilium/GT femur
Gluteus minimus	Abducts, rotates thigh	Ilium/GT femur
Piriformis	rotates, abducts flexed thigh	Sacrum, ilium/GT femur
Obturator internus	Rotates, abducts flexed thigh	Obdurator foramen/GT femur
Obturator externus	Rotates thigh	Obdurator foramen/upper femur
Quadratus femoris	Rotates thigh	Ischium/GT femur

GT is the greater trochanter of the femur
[a]The iliopsoas consists of the psoas major, ilicacus, and psoas minor
See [102] and [133] for more details

Table 3.2 Muscles that move the lower leg (θ_{knee}, lower leg relative to thigh)

Location, muscle	Action	Approximate origin/insertion
Anterior		
Rectus femoris[a]	Extends leg, flexes thigh	Hip/patellar tendon
Vastus lateralis[a]	Extends leg	Upper femur/patellar tendon
Vastus medialis[a]	Extends leg	Upper femur/patellar tendon
Vastus intermedius[a]	Extends leg	Upper femur/patellar tendon
Sartorius	flexes leg; flexes, rotates thigh	Ilium/upper tibia
Gracilis (Table 3.1)		
Posterior		
Biceps femoris[b]	Flexes leg, extends thigh	Ischium, upper femur/upper fibula, tibia
Semitendinosus[b]	Flexes leg, extends thigh	Ischium/upper tibia
Semimembranosus[b]	Flexes leg, extends thigh	Ischium/upper tibia

[a]The quadriceps femoris consists of the rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius
[b]The hamstrings consist of the biceps femoris, semitendinosus, and semimembranosus (see [102] and [133] for more details)

Table 3.3 Muscles that move the foot (θ_{ankle}, foot relative to lower leg) and toe

Location, muscle	Action	Approximate origin/insertion
Anterior		
Tibialis anterior	Dorsiflexes, inverts foot	Upper tibia/first metatarsal
Extensor hallucis longus	Extends great toe, dorsiflexes foot	Mid fibula/great toe
Extensor digitorum longus	Extends toes, dorsiflexes foot	Tibia, fibula/small toe phalanges
Peroneus tertius	Dorsiflexes, everts foot	Lower fibula/metatarsal V
Peroneus longus	Plantar flexes, everts foot	Fibula/metatarsal I
Peroneus brevus	Plantar flexes, everts foot	Fibula/metatarsal V
Posterior		
Gastrocnemius[a]	Plantar flexes foot, flexes leg	Lower femur/calcaneal tendon
Soleus[a]	Plantar flexes foot	Fibula, tibia/calcaneal tendon
Plantaris	Plantar flexion of foot/flexion of leg	Femur/calcaneal tendon
Popliteus	Rotates leg	Lower femur/upper tibia
Tibialis posterior	Inverts foot, plantar flexion of foot	Tibia, fibula/foot
Flexor hallucis longus	Flexes great toe, plantar flexes foot[b]	Fibula/phalanx of great toe
Flexor digitorum longus	Flexes toes, plantar flexes foot	Tibia/small toe distal phalanges

The calcaneal (Achilles) tendon attaches to the calcaneus (heel)
[a]The triceps surae consists of the lateral and medial heads of the gastrocnemius and the soleus
[b]Active in takeoff and tip-toeing
See [102] and [133] for more details

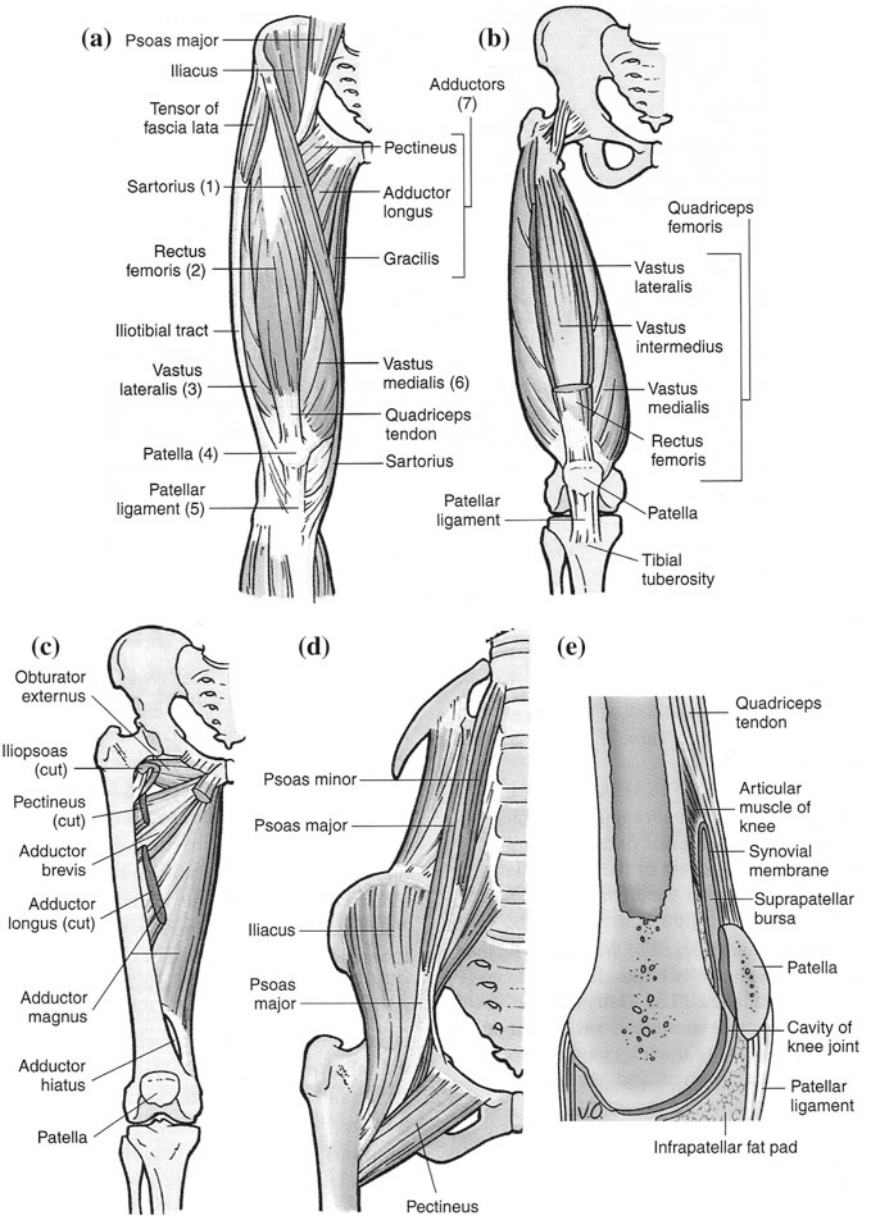


Fig. 3.2 **a** Anterior and medial muscles of the thigh, with **b** quadriceps femoris, with much of the rectus femoris removed, **c** deeper muscles of the medial thigh muscles, **d** the iliopsoas (psoas major and iliacus) and pectineus muscles, and **e** articular muscles of the knee (from [101]. Used with permission)

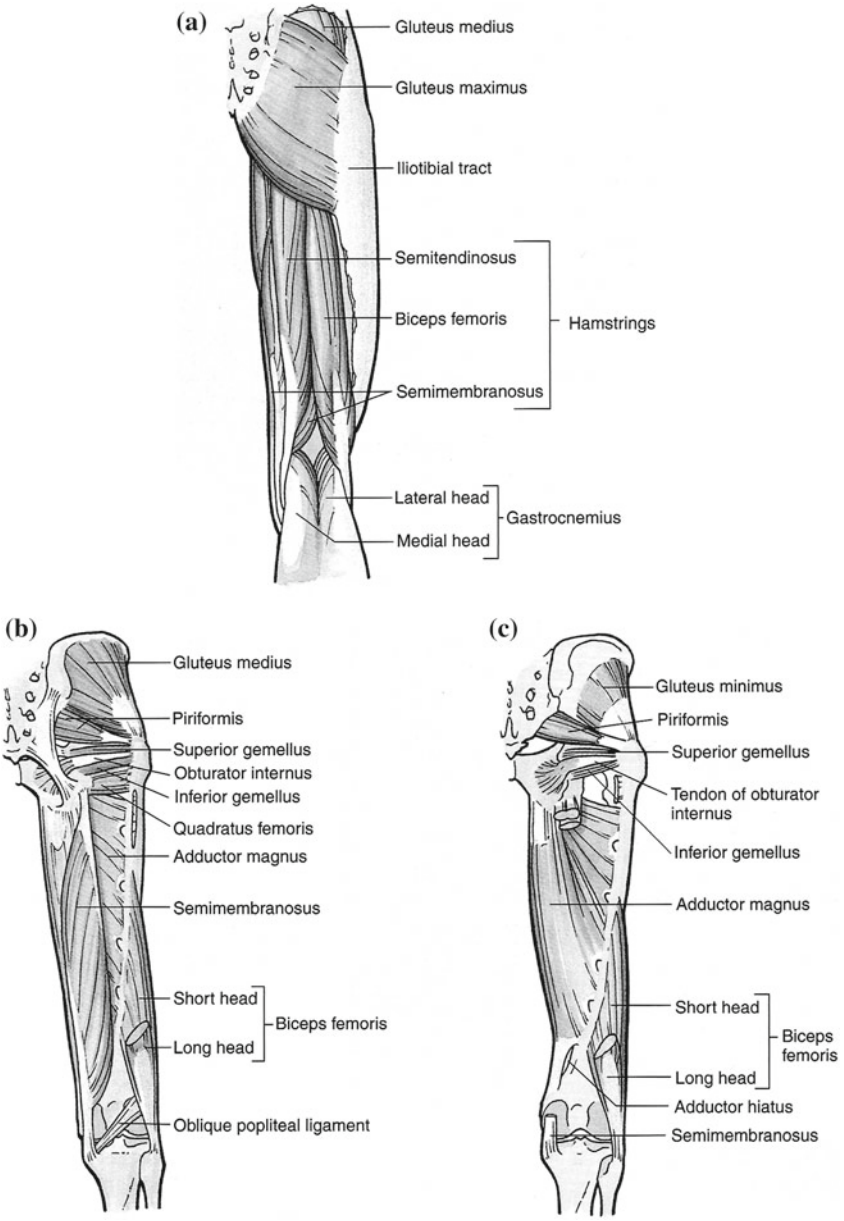


Fig. 3.3 Posterior thigh and gluteal region muscles, showing **a** superficial muscles, **b** deeper muscles, and **c** even deeper muscles (from [101]. Used with permission)

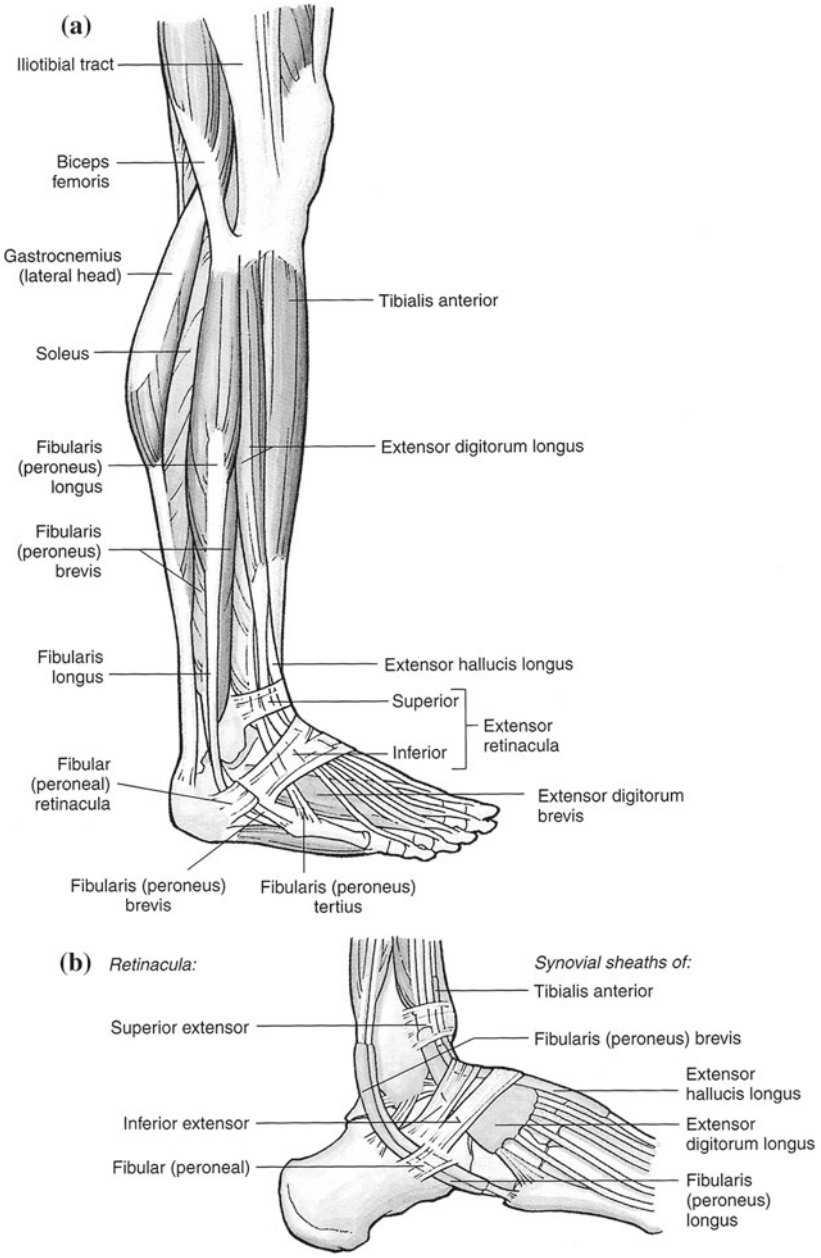


Fig. 3.4 Lateral views of the right leg, showing **a** muscles of the lower leg and foot and **b** the tendons extending into the foot (from [101]. Used with permission)

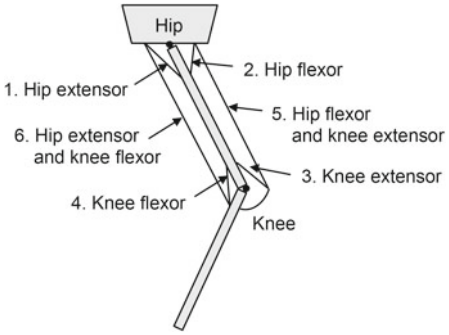


Fig. 3.5 Model of human leg with six effective muscles that connect the hip and knee to the upper and lower legs. Muscles 1–4 cross one joint and muscles 5 and 6 cross two joints (based on [43])

Many of these muscles cross only one joint. However, muscles that cross more than one joint can be very important in motion and in multisegment modeling of motion. For example, in Chap. 2, we modeled the forces on systems with the leg by focusing only on the hip (with the leg straight), knee, or ankle joint and ignored the multisegment nature of the leg and multi-joint muscles. A model of the leg involving the knee and hip is shown in Fig. 3.5 [43]. Muscles 1 and 3 are hip and knee extensors and muscles 2 and 4 are hip and knee flexors. Each of these muscles crosses one joint. Muscles 5 and 6 cross two joints, and so they are biarticulate muscles. Muscle 5 is both a hip flexor and knee extensor, while muscle 6 is both a hip extensor and knee flexor. Each model muscle corresponds to one or more real muscles, and can be activated to different degrees at different times to achieve the desired torques and motions of the leg. Muscles 1 and 3 can be activated to extend the hip and the knee, either with or without muscle 5 activation. With muscle 5 activated, the torque at the hip is reduced and that at the knee is increased, so this two-joint muscle can be thought to be redistributing some muscle torque and joint power from the hip to the knee. If instead muscle 6 is activated along with muscles 1 and 3, torque could be thought to be redistributed from the knee to the hip. The torque about one joint can be increased without affecting the other joint by activating one of these two-joint muscles in conjunction with one-joint muscles, as in Problem 3.21. (The importance of two-joint muscles in bicycling is addressed in Chap. 5 and Fig. 5.39.)

Multiple muscles are activated in a coordinated manner during motion. For example, the muscle and joints involved in the up phase of a pushup are shown in Table 3.4. The concentric action of these muscles provide the force for the upward motion [64]. (As seen in Chap. 5, concentric and eccentric muscle action entail muscle shortening and lengthening, respectively.) In the dip phase the eccentric action of these same muscles provide the braking action acting to control the downward force of gravity. In addition, several muscles are kept static to keep the body straight from head to heels during the dip: the cervical extensors (to keep the head and neck fixed), rectus abdominus and obliques (lumbar spine), and hip flexors (hips).

Table 3.4 Joints and muscles involved in the up phase of a push-up

Joint	Joint action	Segment moved	Major muscles
Shoulder joint	Horizontal adduction	Trunk	Pectoralis major
			Anterior deltoid
			Coracobrachialis
Shoulder girdle	Abduction	Scapula	Pectoralis minor
			Serratus anterior
Elbows	Extension	Upper arm	Triceps
			Anconeus
Wrists	Hyperextension	Forearm	Extensor carpi radialis
			Longus and brevis
			Extensor carpi ulnaris

Using data from [64]

3.2 Standing

3.2.1 Stability

Stability is essential during standing, as well as during any type of motion. We will examine the overall stability of the body and then local stability.

Overall Stability

First consider the body as a rigid mass. The criterion for *overall stability* during standing is for the center of mass to be over the area spanned by the feet, which means that a vertical line passing through the center of mass (*the line of gravity*) passes in this area of the support base (Figs. 3.6 and 3.7). Otherwise there would be torques that would not be balanced, and a "rigid" human would be unstable and topple over. Let us examine the torques about a rotation axis emanating from the center of mass. We see in Fig. 3.6 that when the center of mass is above this area spanned by the feet, the right foot causes a negative torque and the left foot causes a positive torque, and they cancel. (The torque due to the weight of the center of mass

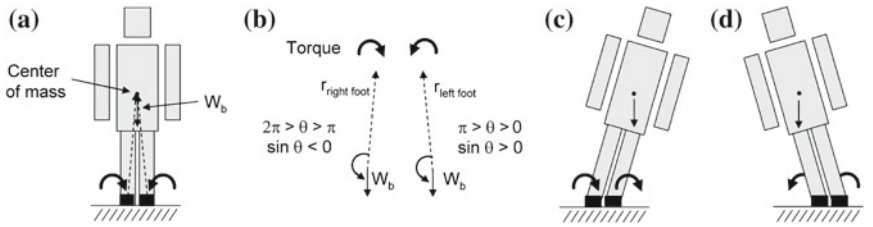


Fig. 3.6 Stability during standing for a model person **a** standing upright (stable) and with **b** torque diagrams, **c** leaning to her *left* (unstable), and **d** leaning to her *right* (unstable)



Fig. 3.7 A standing person is stable when her center of mass is over the cross-hatched region spanned by her feet, as shown here with her feet **a** together, **b** apart, **c** and apart, with stability also provided by a cane or crutch (based on [20])

about this axis is zero.) When the center of mass is to the left of the foot area both torques are negative, and when it is to the right both torques are positive. The torques cannot balance in either case, and there is instability. Try it! (Stand and try to lean over.) We are most stable when the line of gravity is near the center of the support base. When we lean (to one side, forward, and so on) the line of gravity passes can pass through the periphery of this base (a less stable position) or outside this base (an unstable position). (The term center of gravity is also often used in stability analysis; technically it refers to the center of mass in the direction of gravity, which is better expressed as the line of gravity.)

We routinely alter our positions to try to maintain stability. When we carry an object in one hand, we lean or bend our bodies to the other side to keep the line of gravity of our body and the load within this support base. When we walk into the wind, we improve stability by leaning forward so our line of gravity shifts to the edge of the support base nearest the oncoming wind (or in general, nearest the force we are encountering). Then, if the force pushes us back a bit, we can still maintain stability [64].

We generally try to improve stability by adopting these and related strategies by (1) lowering our center of mass, such as a jumper preparing to land (which also increases the collision time, as is discussed later this chapter), a wrestler trying to remain stable, and a gymnast on a balance beam who feels balance being lost; (2) widening the base of support in the direction of force, such as a boxer adopting a forward-backward stance when delivering or preparing to receive a punch and someone pushing and pulling heavy objects or catching a ball; (3) modifying our posture so the line of gravity is near the edge of the support base, such as a football player bending toward the person about to hit him, a person in a tug-of-war leaning backward when expecting to be pulled forward, and those running and changing directions frequently, as in basketball; (4) purposely increasing our mass, such as football players who frequently suffer collisions deciding to gain weight; (5) aligning all segmented body parts or other components along the line of gravity and the support base, such as for people standing on each other; and (6) increasing friction between all components in contact with the ground or other contacting surface, such as runners choosing to wear cleats and rubber-soled shoes and baseball hitters deciding to wear batting gloves [64].

Mobility is often achieved at the expense of some stability. For example, as runners and swimmers prepare to initiate forward motion, they lean forward to a barely stable position. In fact, wrestlers and judo practitioners (judokas) try to throw their opponents so their centers of mass move outside the stability regions [16]. In contrast,

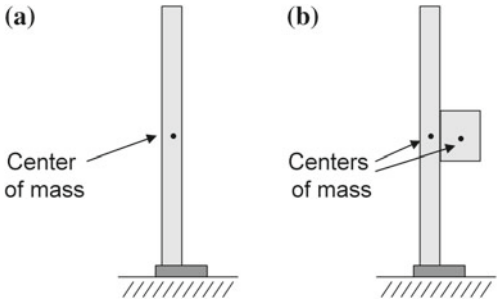


Fig. 3.8 Simple models of people facing to the right: **a** an upright person with mass fairly well balanced in front and back, with the person's center of mass shown—which is normally 58% of the person's height over the soles of his feet, and **b** one with unbalanced mass, such as a person with a beer belly or a woman who is pregnant, with separate centers of mass shown for the main body and the additional mass (which are approximately, but not exactly, at the same height). Clearly, the overall center of mass of the person in (**b**) is displaced to the right relative to that in (**a**). To prevent this being in front of the balls of the feet, the person often contorts his position, which can lead to muscle strain and a bad back (see [20] for more details)

when we try to stop we adopt positions to make us more stable, such as when we adopt the stability position with one leg forward and one backward to help stop our forward motion when we throw a ball forward.

Figure 3.8 suggests different stability conditions for an upright person with mass fairly well balanced in front and back and one with unbalanced mass, such as one with a "beer belly" or a woman who is pregnant. With its five disks, the lumbar curve in our spines (Fig. 2.34) is long and curved enough to keep our centers of mass over our hips. This curvature is further enabled by the dorsal wedge shape of the two lowest lumbar disks (L4 and L5) in men and the three lowest in women (L3, L4 and L5). (These wedges make them thinner in the dorsal, back region.) This extra wedge-shaped disk enables standing pregnant women to change their spine curvatures to keep their centers of mass over their hips. Otherwise these centers of mass would move forward by ∼3.2 cm by the end of pregnancy. Women do this by extending their lower backs to increase the lumbar lordosis (which is the angle between the tops of the highest and lowest lumbar disks) from ∼32° as pregnancy starts to ∼50° at full-term pregnancy [87, 139].

Local Stability

Such overall stability presumes a rigid body and consequently rigid joints. Such stable joints indicate *local stability*. The design of the human knee is an example of good human design. Four ligaments provide much of the needed lateral and cross support.

The knee is the junction of the femur in the upper leg, and the tibia and fibula bones in the lower leg (Fig. 1.3, right knee). The *tibial (or medial) collateral ligament* connects the femur and tibia and the *fibular (or lateral) collateral ligament* connects the femur and the fibula. These collateral ligaments prevent left/right sliding and

provide overall leg tautness. They are slightly posterior, so they are taut during extension, and can produce the straight leg needed during standing.

These two ligaments are not sufficient for stability. There are two cruciate ligaments that cross each other, and prevent twisting and forward/backward sliding of the knee. They cross in between the *lateral and medial menisci*, which are two curved half sockets of cartilage on the tibia. The *posterior cruciate ligament* connects the medial side of the femur to a more lateral part of the tibia. The *anterior cruciate ligament* connects a lateral part of the femur to a more medial part of the tibia. The anterior cruciate ligament is so-named because it attaches to the tibia at a position anterior to that of the posterior cruciate ligament (which is attached more posteriorly).

Local stability of the knee is also provided by the oblique ligaments, as well as others. The kneecap (patella), attached below to the tibia by the patellar ligament and above to the tendon of the quadriceps femoris muscle, also provides stability. Other functions of the kneecap are to lessen the compressivestress on the femur by increasing the area of contact between the patellar tendon and the femur and to aid knee extension by lengthening the lever arm of the quadriceps muscle via an anterior displacement of the quadriceps tendon (Fig. 2.26).

The directions of muscles (Chap. 5) can also affect stability and joint motion. For example, the biceps brachii provide a force that has a component normal to the lower arm that causes rotary motion and a component parallel to the lower arm that tends to stabilize the lower arm for joint angles $>90°$, as in Fig. 3.46c below, or destabilize it (with destabilization tending to cause dislocation) for joint angles $<90°$, as in Fig. 3.46a [48]. This is explored in Problem 3.9.

How do our bone and muscle structures help promote local stability during locomotion? One reason humans are stable when upright on one or two feet is because we have relatively narrow hips and a hip ilium that is short and faces sideways, so that the small gluteal muscles emanate from it at the side of the hip [87]. This stabilizes the upper body over each leg during walking when only leg is on the ground (the stance phase). (In chimpanzees, the ilium is longer and points backwards so these muscles are in the back and do not provide stability when they try to stand and to walk. To promote some stability they need to tilt the hip above the leg off the ground and this swaying motion costs energy during standing and walking.) Stability during running is aided by the gluteus maximus, the largest muscle in the body in terms of mass and strength, which contracts during running to prevent the trunk from falling forward (flexion) and to decelerate the swinging leg; it is fairly inactive during level walking and even uphill walking [84, 87]. Our heads and gaze are stabilized during running by eye and neck muscles that counter head rotational accelerations sensed by the semicircular canals and, each time your foot hits the ground, by the nuchal (neck) ligaments to the shoulder and arm on that same side that pull back your head as they lower [15, 87].

3.2.2 *Forces on the Feet*

Consider a person weighing 700 N (160 lb). During standing, each foot of the person must support 350 N. The total cross-sectional area of the two feet is about 350 cm^2, so the average force/area or pressure on the feet is about 2 N/cm^2. However, not all of the foot touches the ground. Most of the contact is at the ball and heel of each foot. Because of this much smaller area, the pressure on these more limited points of contact is much higher, about 10 N/cm^2 during standing on both feet.

The peak-forces on the feet are much higher during walking. During walking there is usually only one foot on the ground, so the force on that foot is twice that with both feet on the ground and consequently the pressure is twice that during standing. Also, the normal forces involved in braking and forward propulsion during walking can be twice those during standing, which are only those needed to balance gravity. Overall, this leads to a peak force of about 1,400 N per foot, which translates to a pressure of about 40 N/cm^2. Because the whole foot is not flat on the ground during most phases of walking, the contact area is less than during standing and the peak-pressure during walking can be about 60 N/cm^2. Clearly, these forces are larger during running. Force plots are routinely measured during motion by using pressure sensors. (One such force plot is shown later in Fig. 3.19.)

3.3 Walking

We now examine the physics of walking and then running. But before we start this, let us let us look at our physiological features that help us move on land. As bipeds with long legs, we humans are very well designed to walk stably and to do so over long distances with low energy consumption (see Chap. 6). The multiple bones in the human foot are suspended by a series of muscles and ligaments, like cables as in a suspension bridge, to form the foot arch that assists stability during standing. It also helps walking; when the foot hits the ground the arch flattens, and as the arch is restored the foot is propelled forward by the toes. Standing and walking on two feet are also enabled by our vertical necks, our knees being angled under the hips, large hip and knee joints, and by our large heel bones [87].

We are not optimized to run very fast or stably, but we can run long distances at moderate speeds very well, as in marathons. Our long legs (with diameters large enough to minimize stress fracture, Chap. 4), large hip, knee and ankle joints (to minimize damage to cartilage, Chap. 4) and protruding noses (to control water vapor levels, Chap. 9) benefit both walking and running. In addition, our short toes, foot arch, long Achilles tendon (for energy storage, Chap. 4), large gluteus maximus muscles (for torso and leg control), narrow waists, low shoulders, stabilized heads during running, slow twitch muscles in the leg (that favor endurance, Chap. 5), and ability to lose heat quickly for long times (by sweating, Chap. 6) benefit our running.

The downsides of spreading our weight on two feet instead of four include proclivity to lower back pain, sprained ankles, and knee problems [87].

In contrast, chimpanzees use $\sim 4\times$ as much energy as we do to move a given distance when they walk on two feet or four feet (knuckle walking). When on two feet, they expend extra energy to maintain stability by swaying back and forth sideways and to prevent their very bent hips, knees, and ankles from collapsing due to gravity [132]. Chimps are optimized for speed (when galloping on their knuckles), power, agility and stability in motion, and climbing. They and other quadrupeds can run much faster and, at these very fast speeds, for longer times than humans [87].

We will see that there are similarities and differences in the cycles, kinematics, forces, and energetics of walking and running. One similarity is that in both the foot pushes back to get a forward reaction force. It is difficult to walk in sand because of the small reaction force on the feet until the sand is compressed; this compression is seen by your footprints. Footprints in sand are deeper in the front that at the heel because the foot is pushing off (backward) in the sand. At the start of a race, sprinters push off from starting blocks for forward propulsion of the reaction force.

3.3.1 Kinematics

The photographs in Fig. 3.9 show the stages of walking during a step by the right foot. The stages of walking are diagrammed in Fig. 3.10. The hip, knee, and ankle sagittal plane angles are usually referenced to the time the foot touches the ground (foot strike, FS—or heel strike/contact) and when the toe leaves the ground (toe off, TO or foot off) (Fig. 3.10). From foot strike to toe off is the stance phase (foot flat or loading, midstance, terminal stance, and then preswing). From toe off to foot strike

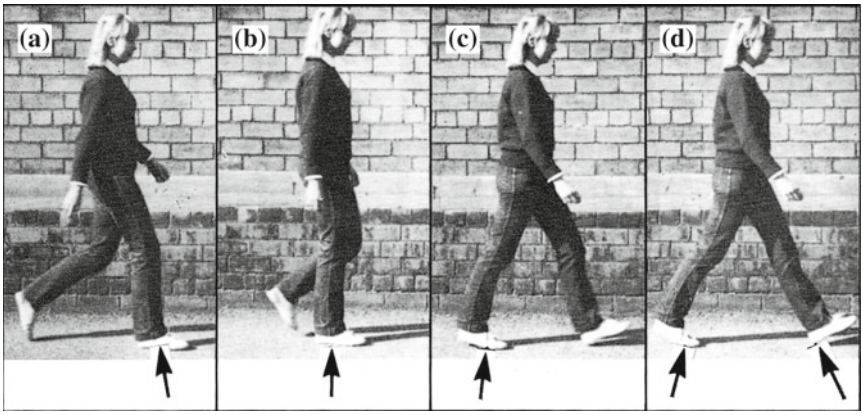


Fig. 3.9 Walking, with *arrows* showing the directions of the forces on the feet (from [5]. Copyright 1992 Columbia University Press. Reprinted with the permission of the press)

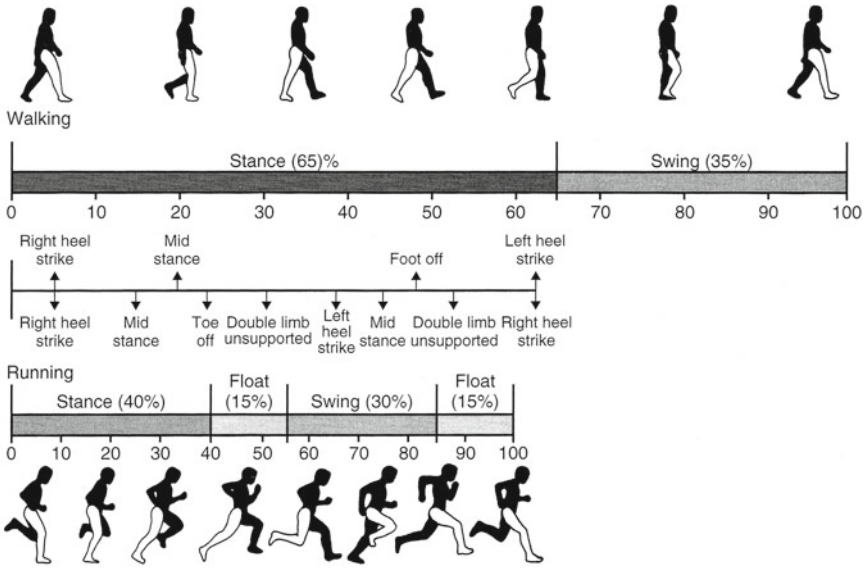


Fig. 3.10 Walking and running gait sequences. For walking, from foot strike (or heel strike) to toe off (or foot off) is the stance phase, with subdivisions sometimes called: foot flat or loading, midstance, terminal stance, and then preswing. From toe off to foot strike is the swing phase, with subdivisions called: initial swing, midswing, and then terminal swing (from [125]. Used with permission)

is the swing phase (initial swing, midswing, and then terminal swing). For walking, the stance phase occurs during the first 60–65% of the cycle for each leg and the swing phase occurs during the last 35–40% of this gait cycle. There are two feet on the ground for the first and last ∼10% of the stance phase for walking, just after foot strike and just before toe off. Both feet are never off the ground at the same time. The evolution of the three motion angles during a walking cycle is shown in the top row of Fig. 3.11.

At contact the upper leg (hip) is flexed by about 30° (relative to the vertical in this analysis), then extends to about 10° extension, and during preswing and then most of swing it flexes to 35° (Fig. 3.11). At contact the lower leg (knee) is nearly extended (∼5°), flexes to 20° during midstance, extends again, flexes to 40° during preswing, peaks at 60–70° flexion at midswing, and then extends again. At contact, the foot (ankle) is slightly plantarflexed (0–5°), extends a bit more to ∼7°, dorsiflexes as the leg rotates forward over the planted foot to 15°, dorsiflexes to 15° for propulsion near the end of stance; after toe off, it dorsiflexes to get toe clearance and plantarflexes to prepare for contact again.

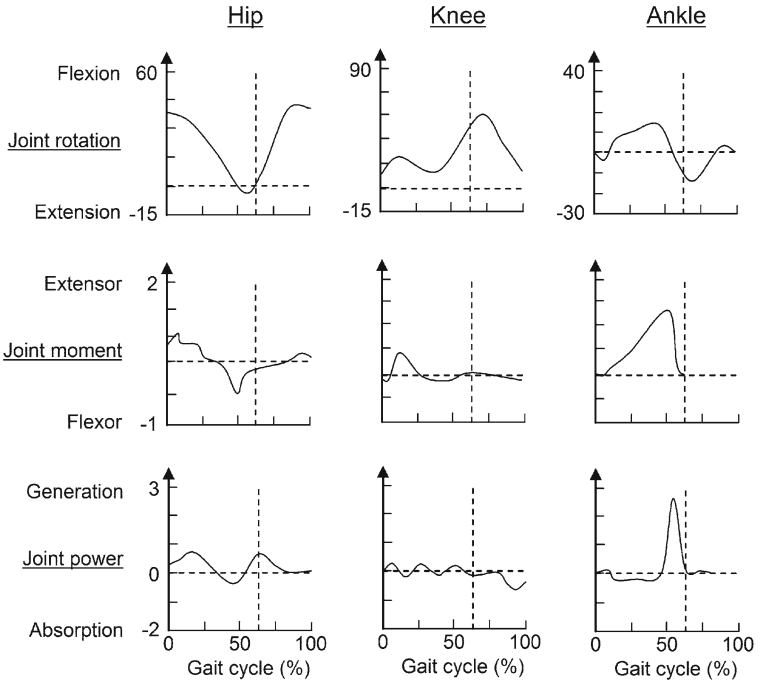


Fig. 3.11 Average values for sagittal plane joint rotation (in °), moment (torque) per mass (in N-m/kg body mass), and power per mass (in W/kg body mass) for the hip, knee, and ankle during a step in walking (based on [55]. Also see [11])

3.3.2 Muscular Action

Each flexion/extension action described in the kinematics is caused by the several flexor/extensor muscles for that joint. The muscular activity at each phase of the walking cycle is shown for the three sagittal joints in Figs. 3.12 and 3.13. Tables 3.1, 3.2 and 3.3 describe these muscle groups [102, 122].

Let us discuss which muscle groups are important during the stance phase and then the swing phase during walking. The ankle dorsiflexors help to lower the leg to the ground at heel strike (the initial contact, and the first part of stance) by eccentric muscle contraction (see Chap. 5). The hip extensors act after contact in early stance to continue this deceleration by reversing the forward swing of the leg, while the intrinsic muscles and long tendons of the foot preserve the shape of the arch of the foot. The knee extensors help the person accept the body weight during loading and then stabilize the knee during midstance. The ankle plantarflexors decelerate mass and control dorsiflexion up to mid-stance, while the hip abductors stabilize the pelvis. In terminal stance (heel off) and then preswing (toe off), the body is accelerated by the ankle plantarflexors by concentric muscle contraction (see Chap. 5) and then by the

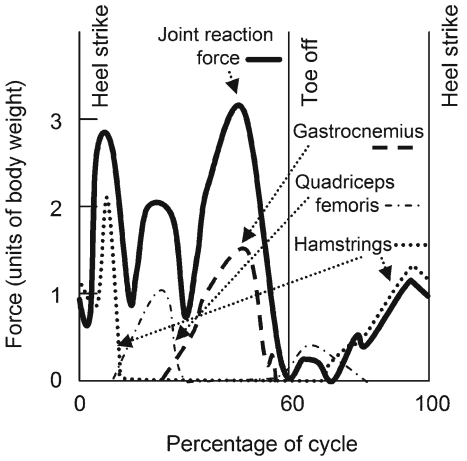


Fig. 3.12 Joint reaction forces transmitted through tibial plateau during one gait normal walking (*solid line*), along with the muscles forces—averaged for 12 subjects (based on [103, 112])

long flexors of the toes, while the pelvis is still stabilized, and the thigh is decelerated by eccentric contraction in preswing.

In initial or preswing to midswing, the hip flexors advance the thigh by concentric contraction, the ankle dorsiflexors enable foot clearance from initial swing to midswing, and in terminal swing the hip extensors decelerate the thigh by eccentric contraction, the knee-flexors decelerate the lower leg by eccentric contraction, the ankle dorsiflexors help position the foot, and the knee extensors extend the knee to prepare for foot contact.

Alternatively, we could analyze this motion by describing when each muscle group is activated. The hip extensors act after contact in early stance and the hip flexors then advance the limb during preswing to midswing; from stance to swing the hip adductors and hamstrings are activated. The quadriceps muscles are activated before contact, then resist extensive knee flexion (by eccentric muscle contractions) during stance to avoid buckling due to gravity, and then extend the knee (by concentric muscle contractions) through early midstance as the leg rotates over the planted foot; they also control hip flexion prior to stance. The dorsiflexors fire during the swing phase to achieve foot clearance and during contact to control the foot placement. The plantar flexing soleus and gastrocnemius muscles fire during stance to control the motion of the leg over the foot and then to propel the foot and body forward. Figure 3.11 shows the joint moment and power generated during the gait cycle.

3.3.3 Friction

Is friction good or bad for walking and other locomotion? It is both.

First, what is friction? On a microscopic scale it is a complex set of interactions between two surfaces and any medium in between them and it is not easily modeled.

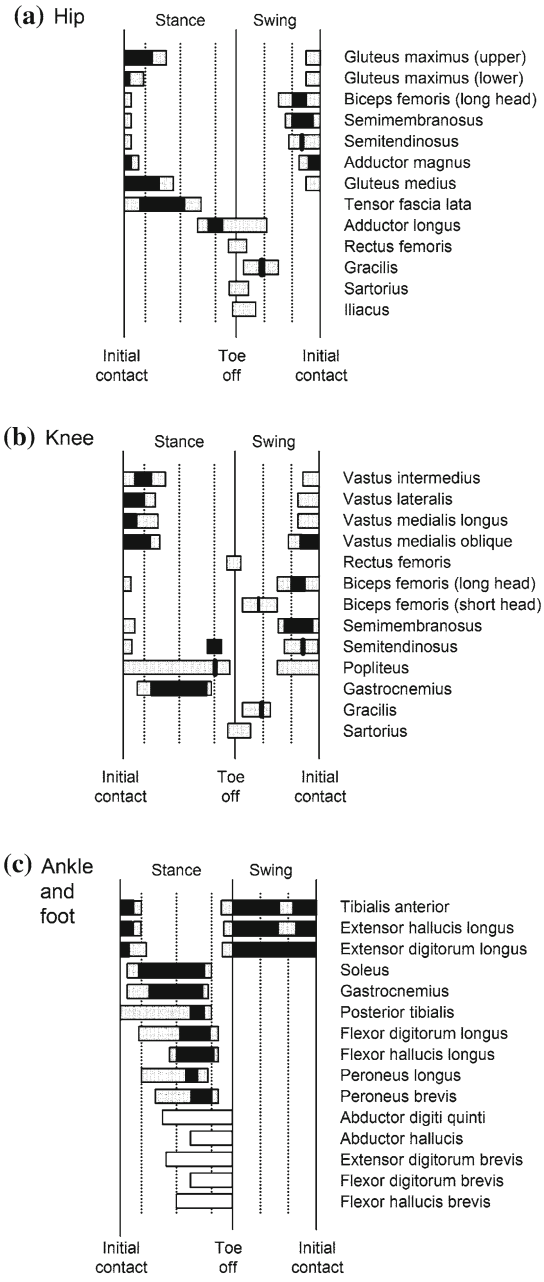


Fig. 3.13 Electromyographic (EMG) activity of the muscles of the **a** hip, **b** knee, and **c** ankle and foot during walking in a healthy person. *Dotted regions* represent activation <20% of maximum voluntary contractions, while *black regions* represent >20% activation. *White regions* in (**c**) for the intrinsic muscles of the foot show at least some level of activation. The four phases of stance are, in time sequence, loading, midstance, terminal stance, and preswing; the three phases of swing are initial swing, midswing, and terminal swing (Also see Fig. 3.10.) (based on [11, 116])

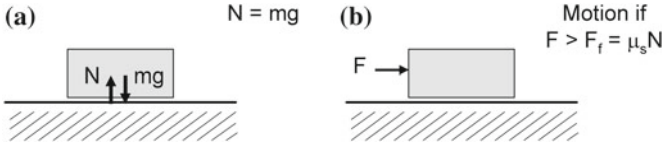


Fig. 3.14 **a** Static equilibrium of an object on a table and **b** lateral force required to overcome static friction to enable motion

Table 3.5 Static and kinetic (when noted) coefficient of friction, μ

Common objects and the body	
Rubber (tire) on dry (wet) concrete road, static	1.0
Brake material on cast ion, dry; static	0.4
Brake material on cast ion, with mineral oil; static	0.1
Graphite on steel, static	0.1
Steel on unlubricated steel	0.7
Steel on lubricated steel or ice	0.15
Teflon on teflon (or steel)	0.04
Ice on ice	0.1
Ice on ice, 4 m/s, 0 °C, kinetic	0.02
Wood on wood	0.25–0.50
Articular cartilage in the human knee, kinetic	0.005–0.02
Articular cartilage in the human hip, kinetic	0.01–0.04
Athletic equipment (kinetic)	
Skates on ice	0.003–0.007
Skis on snow	0.05–0.20
Tennis balls on wood	0.25
Basketball shoes on clean (dusty) wooden floor	1.0–1.2 (0.3–0.6)
Cleated shoes on astroturf	1.2–1.7
Jogging shoes on felt carpet, clay, asphalt	0.9–1.1, 0.3–0.5, 0.6–0.8

Using data from [19, 20, 110]

On a macroscopic scale it is modeled using a coefficient of friction μ. An object of weight $W_\mathrm{b} = m_\mathrm{b} g$ feels a normal force $N = m_\mathrm{b} g$ (see Fig. 3.14). Initially the object is static and the friction interaction is characterized by a static coefficient of friction μ_s (Table 3.5). Consider a lateral force on that object F. For low forces, the frictional force is equal in magnitude and opposite in direction to this force and the object does not move. The object will move only if this force exceeds $F_\mathrm{f} = \mu_\mathrm{s} N$. With the object moving, the net lateral force on the object is $F - F_\mathrm{f}$, where now $F_\mathrm{f} = \mu_\mathrm{k} N$. μ_k is the kinetic, dynamic, or translational coefficient of friction, which can be different from μ_s.

Internal friction is usually troublesome. Friction in the knee joints during walking is bad. In fact, it is terrible. The knee is a synovial joint where the cartilage on the femur and tibia, and the surrounding synovial fluid contained in a sac, provide a very

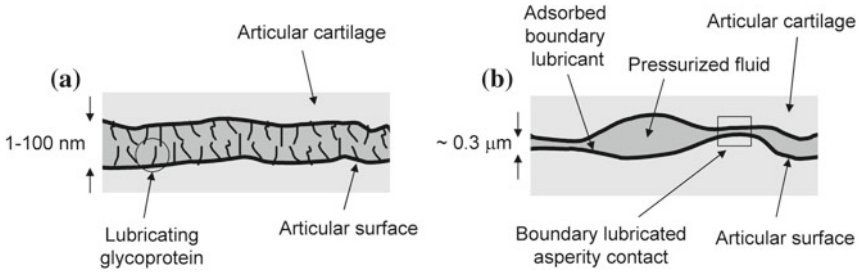


Fig. 3.15 In synovial joints **a** boundary layer lubrication of the articular cartilage surfaces for relatively flat and nearby surfaces and **b** mixed lubrication at articular cartilage, showing boundary lubrication when the separation is on the order of the surface roughness and fluid-film lubrication in areas of more widely separated surfaces (based on [105])

low coefficient of kinetic friction $\mu_k = 0.003 - 0.01$. (This is smaller than most—if not all—man-made systems. See Table 3.5.) Such synovial joints are found in the knees and hips, which have relatively thick $\geq$5 mm cartilage, and in the finger joints, which have relatively thin cartilage layers, $\sim 1 - 2$ mm. Lubrication at such articulating surfaces can occur by fluid-film lubrication, in which the surfaces are relatively far apart and are separated by synovial fluid; boundary lubrication, in which the surfaces are very close together (1–100 nm) and the lubrication is provided by the lubricating glycoprotein on the surfaces (Fig. 3.15a); or by a mixed lubrication, in which there is a mixture of both (Fig. 3.15b). Synovial joints with fluid-film lubrication have an extremely low coefficient of friction of $\simeq$0.02, whereas boundary-lubricated surfaces typically have coefficients of friction two orders of magnitude higher than those that are fluid-film lubricated. (Articular cartilage is a fairly complex and dynamic material, containing synovial fluid, etc. (Fig. 4.51).)

Loss of the synovial fluid from traumatic injury, loss of cartilage from long-term wear or injury, and excessive bone growth in the joint region—as in osteoarthritis—produce major pain during attempted motion. (The synovial fluid about the author's knee left the containing sac during a pantomime in junior high school, as a result of a collision of his knee with a door. Until his leg was stabilized by a cast, his severe pain was lessened during locomotion by his walking backward, with no rotation in the afflicted joint.)

External friction can be necessary. Without friction we could not walk or run. We would merely slip. When the heel of the foot touches the ground during walking, friction from the ground must slow it and then stop the forward motion of the foot (Fig. 3.16). The fraction of a second or so later when the ball of the foot is about to leave the ground, friction from the ground helps propel the body forward. This initial deceleration and subsequent acceleration of the foot typically requires forces of about $0.15\,mg$ and so a static coefficient of friction $\mu_s > 0.15$ is necessary to walk. Figure 3.17 shows how this required minimum coefficient of friction increases with walking speed; it is about 0.17 for a typical walking speed of 3 mph (miles per hour) (a 20 min mile). Figure 3.18 shows how it varies during the course of a step, and how

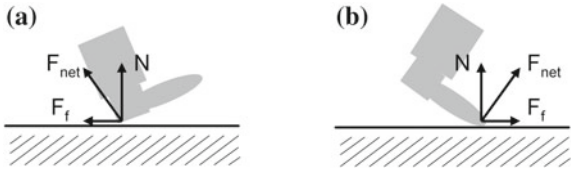


Fig. 3.16 **a** Heel contact stage and decelerating the foot and **b** toe-off stage and accelerating the foot

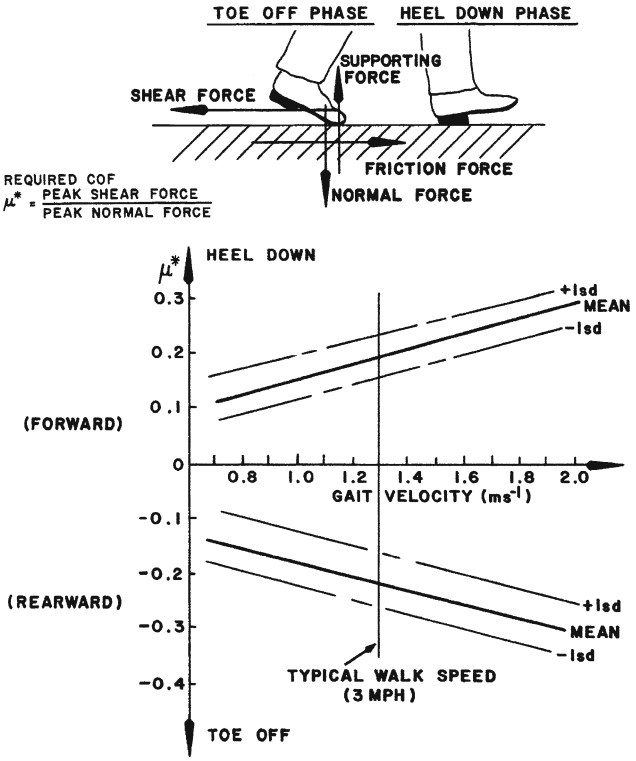


Fig. 3.17 Coefficient of friction required during walking at different speeds (from [24]. Reprinted with permission of Wiley)

it increases when pushing different loads [25]. Table 3.5 lists the kinetic coefficients of friction for several cases. Obviously, walking on ice is difficult. (For other reasons, walking on thin ice is also ill-advised.)

More generally, in every type of forward locomotion, the reaction force to pushing back on the ground is what propels you forward (Newton's Third Law of Motion) [16], and maintaining sufficient friction between the foot and ground prevents slipping. Depending on the activity, this can be aided by wearing shoes with rubber soles, which are often contoured to maximize friction and surface contact, or, on softer

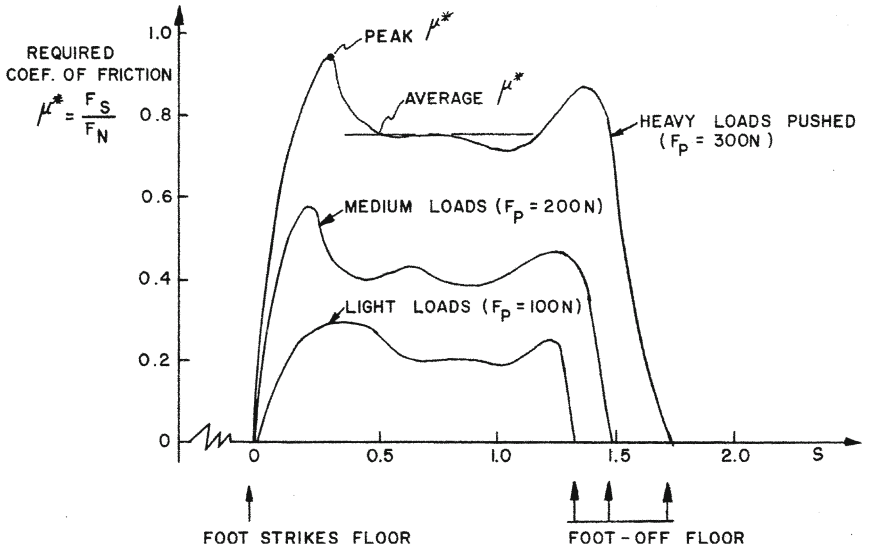


Fig. 3.18 Dynamic coefficient of friction required to push a weighted cart with different loads (from [23]. Reprinted with permission of Wiley)

surfaces (grass), by wearing cleats and spikes that dig into the ground. Minimizing surface friction is the goal when you want to move fast on ice (skating in ice skating and hockey) or snow (skiing), but creating friction is sometimes important. When ice skaters need to accelerate or push off, they use the edge or tip of the skate blade to dig into the ice and push off, and use opposite motions to decelerate. Downhill skiers accelerate due to gravity, and use the ski edges and ski poles to change direction and tweak speed. In swimming, swimmers propel forward by pushing water backward by using their arms and legs.

3.3.4 Energetics

Our motion during walking is surprisingly and deceptively simple when we analyze the kinetic and potential energy of the body during a walking cycle. Let us choose the horizontal direction of walking as the x direction and the vertical direction as the z direction. The kinetic energy KE of an object is $m_b v^2/2$, where m_b is the body mass and v is the magnitude of the velocity, the speed, of the center of mass. This total KE can be separated into that due to motion in the horizontal and vertical directions, as

$$\mathrm{KE} = \mathrm{KE_H} + \mathrm{KE_V}, \tag{3.1}$$

$$\mathrm{KE_H} = \frac{1}{2} m_\mathrm{b} v_x^2 = \frac{1}{2} m_\mathrm{b} \left(\frac{\mathrm{d}x}{\mathrm{d}t} \right)^2, \tag{3.2}$$

$$\mathrm{KE_V} = \frac{1}{2} m_\mathrm{b} v_z^2 = \frac{1}{2} m_\mathrm{b} \left(\frac{\mathrm{d}z}{\mathrm{d}t} \right)^2. \tag{3.3}$$

The potential energy is

$$\mathrm{PE} = m_\mathrm{b} g z_\mathrm{CM}, \tag{3.4}$$

where z_CM is the vertical position of the center of mass of the body.

When we describe a system where energy is not supplied, say by an external driving force, or lost, such as to friction, the total energy E of this system is constant, where

$$E = \mathrm{KE} + \mathrm{PE}. \tag{3.5}$$

We would not expect KE + PE to be constant during walking, but it is almost constant.

The photographs in Fig. 3.9 show the stages of walking during a step by the right foot. In stage (a), the right foot has just decelerated, much of the heel of the foot is on the ground, and the foot is in front of the body's center of mass. In stage (b), the right foot is squarely on the ground and under the center of mass. In stage (c), the right foot is behind the center of mass, mostly the ball is in contact with the ground, and forward propulsion is beginning. In stage (d), the front of the right foot is propelling the body forward, while the heel of the left foot has made contact with the ground and has begun to decelerate. We will examine the horizontal and vertical forces (F_H and F_V) and energies at each of these stages (Fig. 3.19).

The horizontal forces are due to friction. During stage (a) in Fig. 3.19, F_H slows the right foot, and during stages (c) and (d) it accelerates it. Because the horizontal speed of the center of mass of the whole body decreases when the right foot decelerates and it increases when the foot accelerates, during stage (a) $\mathrm{KE_H}$ decreases and during stages (c) and (d) it increases.

The center of mass (CM) of the body is changing during this step. The right leg is becoming more vertical and straighter from stage (a) to stage (b), so the center of mass (z_CM) is rising and PE is increasing. From stages (b) to (c) to (d), the right leg is becoming less vertical and is also bending, so the center of mass is falling and PE is decreasing. After stage (d) the left leg gets straighter and PE starts to increase again.

Because changes in the height of the center of mass must be the result of vertical forces, we see that from stages (a) to (b), z_CM begins to rise, which must mean that there is a net positive vertical force and $F_\mathrm{V} > m_\mathrm{b} g$. From stages (b) to (c), z_CM begins to fall, which must mean that there is a net negative vertical force and $F_\mathrm{V} < m_\mathrm{b} g$.

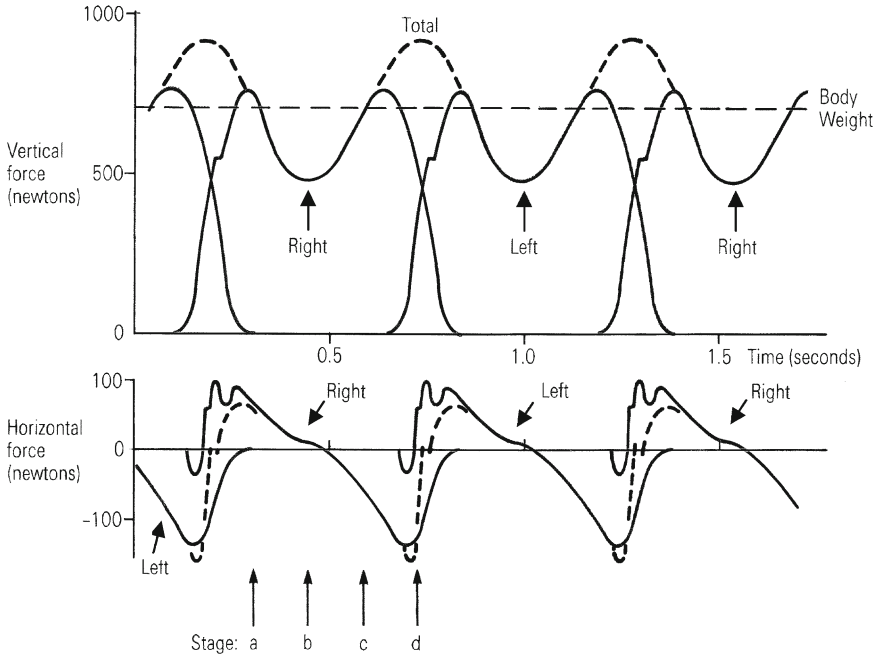


Fig. 3.19 Vertical and horizontal forces on feet during walking (from [5]. Copyright 1992 Columbia University Press. Reprinted with the permission of the press)

So we see that from stage (a) to (b), the center of mass is rising and PE is increasing, while the right foot is decelerating and KE_H is decreasing. During stage (c), the center of mass is falling and PE is decreasing, while the right foot is accelerating and KE_H is increasing. We see that PE and KE_H are out of phase, one is decreasing, while the other in increasing, and vice versa. This also occurs in harmonic motion, such as that for a mass on a spring or a pendulum.

Figure 3.20 shows that during a step PE + KE_V and KE_H are out of phase. (KE_V also changes during these stages, as a result of the motion of the center of mass z_{CM}; however, it changes relatively little.) More surprisingly, their sum PE + KE_V + KE_H is fairly constant during the step. Again, these two observations are reminiscent of harmonic motion, during which the PE and KE are out of phase and their sum does not change during a cycle. This exchange of kinetic and potential energy also occurs in a pendulum and for a rolling egg, as in Fig. 3.21, and this observation has led some to compare the leg during walking to a pendulum, by what is known as the ballistic or pendulum model of walking. Before pursuing this analogy further, we will review harmonic and pendulum motion.

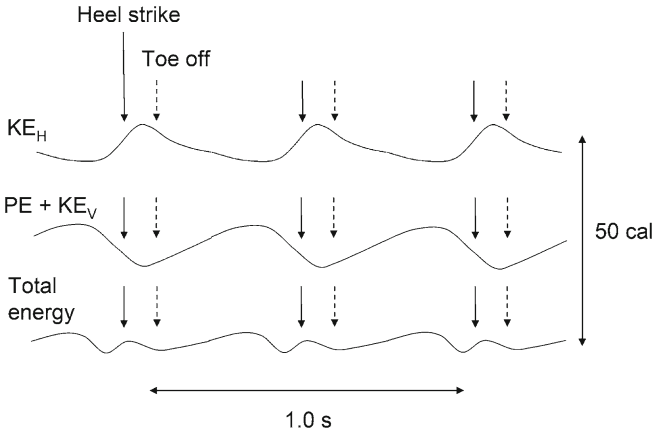


Fig. 3.20 Changes in mechanical energy during walking from force plates, with heel strike and toe off shown, for (from *top* to *bottom curve*) forward kinetic energy, gravitational potential energy plus vertical kinetic energy, and total energy (based on [22, 93])

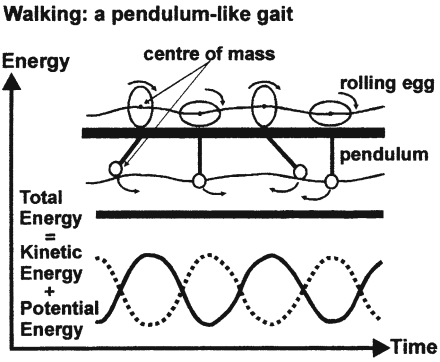


Fig. 3.21 The cyclic exchange of kinetic and potential energy in a pendulum and a rolling egg, with constant total energy, is similar to that in walking (from [99]. Used with permission)

3.3.5 Review of Harmonic Motion, Pendulums, and Moments of Inertia

Simple Harmonic Oscillator

The position x from equilibrium for a body of mass m attached by a spring with spring constant k (Fig. 3.22) is determined from Newton's Second Law of Motion $F = ma$, where F is the restoring force on the mass due to the spring and $a = \mathrm{d}^2x/\mathrm{d}t^2$ is the acceleration of the mass. So

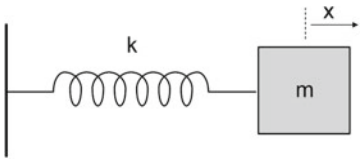


Fig. 3.22 Simple harmonic oscillator, with an object with mass m attached to a spring with spring constant k

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx \tag{3.6}$$

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x = -\omega^2 x, \tag{3.7}$$

where $\omega = (k/m)^{1/2}$ is the resonant frequency of this harmonic oscillator. The solution can be written as

$$x = A\cos(\omega t + \phi), \tag{3.8}$$

where A is an arbitrary amplitude and ϕ is a phase (which is not very significant here) (see Appendix C for more information about the solution).

The kinetic energy during this motion is

$$\mathrm{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = \frac{1}{2}mA^2\omega^2\sin^2(\omega t + \phi), \tag{3.9}$$

because $\mathrm{d}x/\mathrm{d}t = -A\omega\sin(\omega t + \phi)$. The potential energy is

$$\mathrm{PE} = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \phi) = \frac{1}{2}mA^2\omega^2\cos^2(\omega t + \phi) \tag{3.10}$$

using $\omega = (k/m)^{1/2}$. The PE and KE are clearly 90° out of phase during this oscillatory motion. The total energy,

$$E = \mathrm{KE} + \mathrm{PE} = \frac{1}{2}mA^2\omega^2\sin^2(\omega t + \phi) + \frac{1}{2}mA^2\omega^2\cos^2(\omega t + \phi) = \frac{1}{2}mA^2\omega^2, \tag{3.11}$$

is constant during this motion.

The radial frequency ω has units of radians per second. The cycle frequency f is $\omega/2\pi$, which has units of cycles per second (cps) or Hz (Hertz). This means that a complete cycle (period) occurs in a time $T_{\mathrm{period}} = 1/f = 2\pi/\omega$, during which time ωt changes by 2π rad.

Mass on a Pendulum

We will use the pendulum to model walking. In a simple pendulum, a ball of mass m is at the end of a plumb (string) of length L (Fig. 3.23a). For now we will assume that the plumb is massless. Assume that the plumb is at an angle θ from the vertical. The

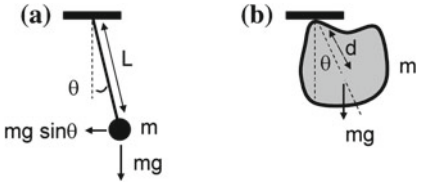


Fig. 3.23 **a** Mass on a simple pendulum and **b** complex pendulum

position of the ball along the arc of the plumb is $L\theta$. The speed of the ball normal to the plumb is $v = L\mathrm{d}\theta/\mathrm{d}t$, and the acceleration of the ball normal to the plumb is $L\mathrm{d}^2\theta/\mathrm{d}t^2$. (For small angles, $|\theta| \ll 1$, the lateral coordinate is $x \simeq L\theta$, and so $\mathrm{d}^2x/\mathrm{d}t^2 \simeq L\mathrm{d}^2\theta/\mathrm{d}t^2$.) From the diagram, the tension T in the string is balanced by $mg\cos\theta$, and there is a net acceleration of the ball $mg\sin\theta$ that tends to decrease the magnitude of θ. Newton's Second Law $F = ma$, can be written as (with $ma = F$)

$$mL\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -mg\sin\theta. \tag{3.12}$$

For small angle motion ($|\theta| \ll 1$), so $\sin\theta \approx \theta$ and

$$mL\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -mg\theta \tag{3.13}$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{g}{L}\theta = -\omega^2\theta. \tag{3.14}$$

Physically and mathematically this is similar to the simple harmonic oscillator, now with a resonant frequency $\omega = (g/L)^{1/2}$ and solution

$$\theta(t) = B\cos(\omega t + \beta), \tag{3.15}$$

where B is an arbitrary amplitude and β is a phase (which again is not very significant here) (see Appendix C for more information about the solution).

The kinetic energy during this motion is

$$\mathrm{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\left(L\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{1}{2}mB^2L^2\omega^2\sin^2(\omega t + \beta), \tag{3.16}$$

because $v = L\mathrm{d}\theta/\mathrm{d}t = -BL\omega\sin(\omega t + \beta)$. The potential energy is

$$\mathrm{PE} = mgL(1 - \cos\theta) \approx \frac{1}{2}mgL\theta^2, \tag{3.17}$$

because for small θ ($|\theta| \ll 1$), $1 - \cos\theta \approx 1 - \theta^2/2$, and so

$$\text{PE} \approx \frac{1}{2} mgLB \cos^2(\omega t + \beta) = \frac{1}{2} mB^2 L^2 \omega^2 \cos^2(\omega t + \beta) \tag{3.18}$$

using $\omega = (g/L)^{1/2}$. Again, the PE and KE are out of phase during this oscillatory motion. As with the harmonic oscillator, the total energy

$$\begin{aligned} E &= \text{KE} + \text{PE} \\ &= \frac{1}{2} mB^2 L^2 \omega^2 \sin^2(\omega t + \beta) + \frac{1}{2} mB^2 L^2 \omega^2 \cos^2(\omega t + \beta) \qquad (3.19) \\ &= \frac{1}{2} mB^2 L^2 \omega^2 \qquad (3.20) \end{aligned}$$

is constant during the motion of the pendulum.

During walking the leg can be modeled as a pendulum because of this interplay between the KE and PE. We will use this pendulum (ballistic) model to determine the oscillation frequency of the leg $\omega = (g/L)^{1/2}$ and from this we will obtain the walking speed. Before doing this, we need to modify this simple pendulum model because the leg cannot be approximated by a mass (foot) at the end of a massless string (upper and lower legs). (We will also use this refined model in our analysis of throwing a ball.)

More Complex Pendulum

If we multiply both sides of (3.12) by L, we get

$$mL^2 \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -mgL \sin\theta. \tag{3.21}$$

The right side is actually the net torque τ acting on the ball of the simple pendulum about an axis at the top of the string, where the magnitude of the distance vector $r = L$, the force on the ball is mg, and the angle from this $\mathbf{r}$ to $\mathbf{F}$ is $360° - \theta$, according to our convention (Fig. 3.23b). With $\sin(360° - \theta) = -\sin\theta$,

$$\tau = mgL \sin(360° - \theta) = -mgL \sin\theta = mL^2 \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}. \tag{3.22}$$

The mL^2 term is the *moment of inertia* I of the ball about the pivot axis, which is at the hip. More generally, the moment of inertia of a complex object can be written as the sum of contributions from distinct masses m_i each a distance R_i from the axis

$$I = \sum_i m_i R_i^2 \tag{3.23}$$

or as an integral of mass density $\rho(\mathbf{r})$ at positions $\mathbf{r}$ that are a distance R from the axis, integrated over the volume V

$$I = \int \rho(\mathbf{r}) R^2 \mathrm{d}V. \tag{3.24}$$

For these complex objects with moment of inertia I, total mass m, and center of mass a distance d from the axis

$$\tau = -mgd \sin\theta = I \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} \tag{3.25}$$

and with the small angle approximation $\sin\theta \approx \theta$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{mgd}{I} \sin\theta \approx -\frac{mgd}{I}\,\theta. \tag{3.26}$$

The resonant frequency is now $\omega = (mgd/I)^{1/2}$ (see Appendix C for more information about the solution).

Also, the angular momentum of the pendulum $L = I\Omega$ with $\Omega = \mathrm{d}\theta/\mathrm{d}t$, and $\tau = \mathrm{d}L/\mathrm{d}t$.

More on Moments of Inertia

Consider an object of mass m with moment of inertia I about a given axis through its center of mass. You can obtain its moment of inertia about any axis laterally displaced a distance q from this axis by using the parallel axis theorem

$$I_{\text{about displaced axis}} = I_{\text{about cm}} + mq^2. \tag{3.27}$$

This *parallel axis theorem* is illustrated in Fig. 3.24a, and can be proved from (3.24).

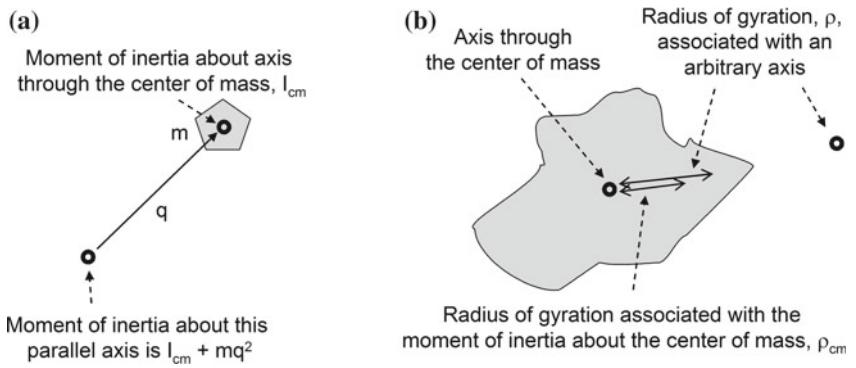


Fig. 3.24 **a** The parallel axis theorem is illustrated for an object of mass m for an axis about the center of mass normal to the page and about an arbitrary parallel axis, and **b** the radius of gyration is schematically illustrated for moments of inertia about the center of mass and an arbitrary axis

One simple way of providing the moment or inertia of any object is by giving its *radius of gyration*, ρ, which is defined by

$$I = m\rho^2. \tag{3.28}$$

This means the moment of inertia of the object would be the same if all of the mass were distributed a distance ρ from the axis. One example of this is the mass m distributed on a circle of radius ρ about the axis. In another example, two points with mass $m/2$ are placed on either side of the axis. Of course, the radius of gyration of a ball on a string is the length of the string, but in this case the center of mass is not on the rotation axis. Figure 3.24b illustrates the radius of gyration for an axis through the center of mass and about an arbitrary axis. The average radii of gyration for body segments are given in Table 1.9.

3.3.6 Ballistic (or Pendulum) Model of Walking

Figure 3.10 shows that during the stance phase the center of mass moves similar to an inverted pendulum, while in the swing phase the swinging leg moves like a pendulum. A simple form of the ballistic model says that during the swing cycle your leg is like a pendulum with radial frequency ω. This means that a complete cycle (period) occurs in a time $T_{\text{period}} = 2\pi/\omega$. One forward swing of the leg corresponds to a half cycle, which takes $T_{\text{half period}} = \pi/\omega$. For a leg of length L_{leg} swinging through an arc of $\Delta\theta$, the step length is $\sim L_{\text{leg}}(\Delta\theta)$ for small $\Delta\theta$ (Fig. 3.25a), so we might say that the stepping or walking speed v is this step length divided by $T_{\text{half period}}$ or

$$v = \frac{L_{\text{leg}}(\Delta\theta)}{T_{\text{half period}}} = \frac{\omega}{\pi} L_{\text{leg}}(\Delta\theta). \tag{3.29}$$

(We will revisit this equation later.) For ball on a string $\omega = (g/L_{\text{leg}})^{1/2}$, so $T_{\text{half period}} = \pi(L_{\text{leg}}/g)^{1/2}$. For a body height $H = 1.8\,\text{m}$, Fig. 1.15 and Table 1.6

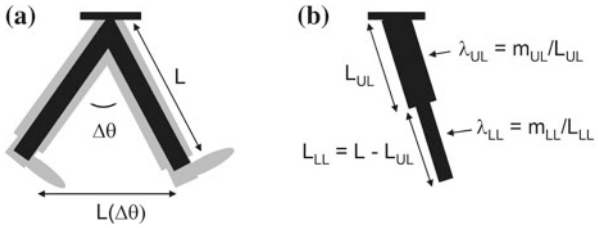


Fig. 3.25 **a** Ballistic (pendulum) model of walking, with simple leg of uniform linear mass density, and with **b** complex leg with upper and lower legs with different linear mass densities

show that the leg length is $L_{\text{leg}} = 0.53\,H = 0.95\,\text{m}$. With $g = 9.8\,\text{m/s}^2$, $T_{\text{half period}} = 0.98\,\text{s}$.

How does this compare to real walking? The speed for a slow walk is about $0.5\,\text{m/s} = 1.1\,\text{mph}$. (Test this!) With a step length of 0.3 m, the time for a step is $T_{\text{half period}} \simeq 0.60\,\text{s}$. For a fast walk the speed is about $2.0\,\text{m/s} = 4.5\,\text{mph}$ and the step length is larger, 0.7 m, and so the step time is about 0.35 s. The model predictions are fair for the slow walk with this ball on a string model (0.98 s model result compared to 0.6 s actual time). Fast walking cannot be modeled as a free pendulum; it is clearly a forced pendulum. You can feel this in your legs as you change from a slow, leisurely pace to a fast, vigorous pace.

Can we improve the model of slow walking by using a more refined model for the mass distribution of the leg? For a leg of length L with a uniform linear density (and the same thickness throughout, which is not exactly true) and mass m_{leg}, as suggested by Fig. 3.25a, we need to use the expression from the more general, refined model (3.26) with $\omega = (m_{\text{leg}}gd/I)^{1/2}$ and $T_{\text{half period}} = \pi(I/m_{\text{leg}}gd)^{1/2}$. The distance of the center of mass from the axis is $d = L_{\text{leg}}/2$. The moment of inertia $I = m_{\text{leg}}L_{\text{leg}}^2/3$. This is obtained by integrating over a uniform linear mass density (mass per unit length) $\lambda = m_{\text{leg}}/L_{\text{leg}}$ from $R = 0$ to L_{leg} in (3.24):

$$I = \int \rho(\mathbf{r})R^2\mathrm{d}V = \int_0^{L_{\text{leg}}} \lambda R^2\mathrm{d}R = \frac{m_{\text{leg}}}{L_{\text{leg}}}\frac{L_{\text{leg}}^3}{3} = \frac{1}{3}m_{\text{leg}}L_{\text{leg}}^2. \tag{3.30}$$

(With a uniform cross sectional area A, $\rho(\mathbf{r}) = \lambda/A$ and $dV = A\ \mathrm{d}R$, so $\rho(\mathbf{r})\mathrm{d}V = \lambda\ \mathrm{d}R$.) Then $\omega = (m_{\text{leg}}gd/I)^{1/2} = (m_{\text{leg}}g(L_{\text{leg}}/2)/(m_{\text{leg}}L_{\text{leg}}^2/3))^{1/2} = (3g/2L_{\text{leg}})^{1/2}$ and $T_{\text{half period}} = \pi(2L_{\text{leg}}/3g)^{1/2} = 0.80$ s, again for $L_{\text{leg}} = 0.95\,\text{m}$. This prediction is fairly close to the estimated 0.60 s step time, and agreement is much better than for the ball on string model. We should remember that this is just a model; energy recovery is not perfect, and ~70%, for walking (Fig. 3.26).

What happens if we refine the model a bit more? The leg does not have a uniform linear mass density; the upper leg (thigh) is heavier per unit length than the lower leg (with the foot) (Fig. 3.25b). The entire leg has a mass $m_{\text{leg}} = 0.161\,m_{\text{b}}$ and length $L_{\text{leg}} = 0.530\,H$. Tables 1.6 and 1.7 and Fig. 1.15 show that the upper leg has a mass $m_{\text{u,leg}} = 0.10\,m_{\text{b}} = 0.621\,m_{\text{leg}}$ and has a length $L_{\text{u,leg}} = 0.245\,H = 0.462\,L_{\text{leg}}$ (H = body height), while the lower leg and foot have a mass $m_{\text{l,leg}} = 0.061\,m_{\text{b}} = 0.379\,m_{\text{leg}}$ and a length $L_{\text{l,leg}} = 0.285\,H = 0.538\,L_{\text{leg}}$.

The center of mass distance d from the hip joint and the moment of inertia about the hip are needed. The center of mass of two objects is in general given by

$$r_{\text{CM}} = \frac{m_1r_1 + m_2r_2}{m_1 + m_2}, \tag{3.31}$$

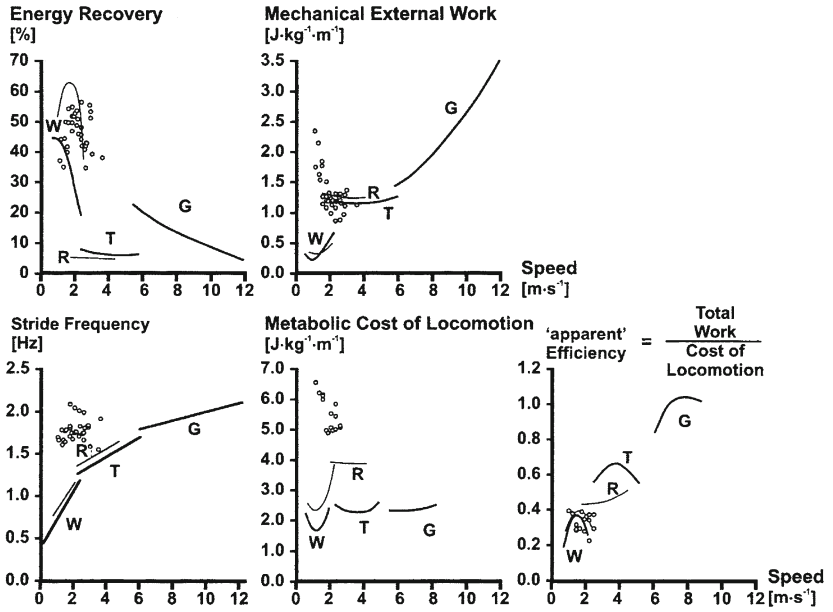


Fig. 3.26 Gait parameters as a function of speed for humans (*thin curves*) and horses (*thick curves*), for different gaits: walking (*W*), running (*R*), trotting (*T*), galloping (*G*), and human skipping (*open circles*) (from [99]. Used with permission. Also see [59])

which here is

$$d = \frac{m_{\mathrm{u,leg}}(L_{\mathrm{u,leg}}/2) + m_{\mathrm{l,leg}}(L_{\mathrm{u,leg}} + L_{\mathrm{l,leg}}/2)}{m_{\mathrm{u,leg}} + m_{\mathrm{l,leg}}} \tag{3.32}$$

$$= \frac{(0.621\, m_{\mathrm{leg}})(0.231\, L_{\mathrm{leg}}) + (0.379\, m_{\mathrm{leg}})(0.731\, L_{\mathrm{leg}})}{m_{\mathrm{leg}}} = 0.421\, L_{\mathrm{leg}}. \tag{3.33}$$

The moment of inertia is (Fig. 3.25b)

$$I = \int_0^{L_{\mathrm{u,leg}}} \lambda_{\mathrm{u,leg}} R^2 \mathrm{d}R + \int_{L_{\mathrm{u,leg}}}^{L_{\mathrm{leg}}} \lambda_{\mathrm{l,leg}} R^2 \mathrm{d}R = \frac{1}{3}\lambda_{\mathrm{u,leg}} L_{\mathrm{u,leg}}^3 + \frac{1}{3}\lambda_{\mathrm{l,leg}}(L_{\mathrm{leg}}^3 - L_{\mathrm{u,leg}}^3), \tag{3.34}$$

where $\lambda_{\mathrm{u,leg}} = m_{\mathrm{u,leg}}/L_{\mathrm{u,leg}} = 0.62\, m_{\mathrm{leg}}/0.46 L_{\mathrm{leg}} = 1.34\, m_{\mathrm{leg}}/L_{\mathrm{leg}}$ and $\lambda_{\mathrm{l,leg}} = m_{\mathrm{l,leg}}/L_{\mathrm{l,leg}} = 0.38\, m_{\mathrm{leg}}/0.54 L_{\mathrm{leg}} = 0.70\, m_{\mathrm{leg}}/L_{\mathrm{leg}}$. With $L_{\mathrm{u,leg}} = 0.46\, L_{\mathrm{leg}}$, we see that $I = 0.256\, m_{\mathrm{leg}} L_{\mathrm{leg}}^2$. (Problem 3.27 compares these models of the moments on inertia with measured data, using the parallel axis theorem and the radii of gyration in Table 1.9.)

Then $\omega = (m_{\text{leg}} g d / I)^{1/2} = (m_{\text{leg}} g (0.421\, L_{\text{leg}}) / (0.256\, m_{\text{leg}} L_{\text{leg}}^2))^{1/2} = (1.64\, g / L_{\text{leg}})^{1/2}$. We see that $T_{\text{half period}} = \pi (0.61\, L_{\text{leg}} / g)^{1/2} = 0.76\,\text{s}$, which is 4.5% faster than the uniform density model, and not a great improvement over it.

What is wrong with this model? We have made some dubious assumptions. The swing phase of each leg occurs only over 35% of the gait cycle, so the legs are swinging only during 70% of the cycle and not the whole cycle. Equation (3.29) actually describes how fast the swinging foot moves and not the center of mass of the body. Figure 3.10 suggests that during the swing phase the center of mass moves only about half of the step length, and so if the center of mass is assumed to move at a constant speed during the gait cycle the speed v would be about half that given in (3.29). More important, the assumptions that the leg just passively swings and that it remains straight are far too simple. We should not expect this model to be quantitatively accurate [100, 129].

3.3.7 Inverted Pendulum Model

During stance the leg is like an inverted pendulum, with the center of mass traveling on a circular arc of radius R (Fig. 3.27). To maintain the circular motion, a centripetal force with magnitude $m_{\text{b}} v^2 / R$ is needed, where v is approximately the walking speed. (This is the same as an upward "centrifugal force" in a frame that is rotating with the leg.) When this exceeds $m_{\text{b}} g$, the center of mass will not come down. This sets a maximum walking speed of $v_{\text{max}} = (gR)^{1/2} \sim 3.0\,\text{m/s}$ for $R = 0.9$ m; at faster speeds people have to move in a different way: jogging, running, etc. On the moon ($g_{\text{Moon}} = 0.17\, g_{\text{Earth}}$) and Mars ($g_{\text{Mars}} = 0.4\, g_{\text{Earth}}$) this limit is smaller, 1.2 m/s and 1.9 m/s, respectively. This low upper limit for the walking speed on the moon explains why astronauts had to resort to hopping on the moon to move fast. (This is the real "moonwalking," for those who also recall Michael Jackson's "moonwalk.") Children

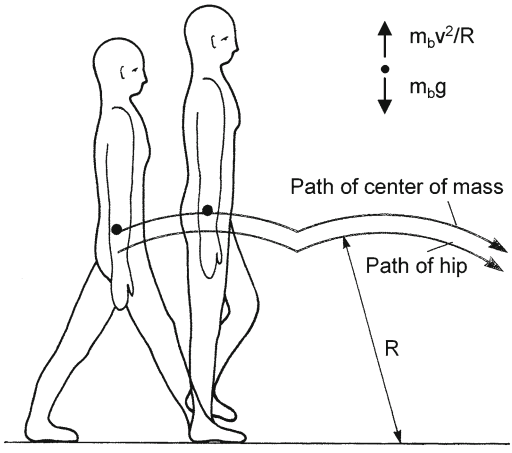


Fig. 3.27 Takeoff during fast walking, in the inverted pendulum model (from [5]. Copyright 1992 Columbia University Press. Reprinted with the permission of the press)

(on Earth) have smaller R and consequently have to start running at lower speeds than adults (and sometimes need to run to keep up with adults). Adults usually start running at speeds even slower than the 3.0 m/s estimate, about 2.0 m/s, possibly in an effort by the body to minimize the metabolic energy cost per unit distance traveled (see Fig. 3.26). Speeds near 4 m/s are attainable in race walking because the racers distort their hips at midstance in a way that flattens the trajectory of the center of mass—thereby increasing R.

3.4 Running

Running is not just fast walking; it is qualitatively different. During walking, each foot is on the ground for more than half the time, and sometimes both are on the ground at the same time. Both feet are never off the ground at the same time. During running, each foot is on the ground for less than half the time, and—unlike walking—sometimes neither foot is on the ground. This distinction can have serious implications. Jane Saville, an Australian race walker, was in first place 200 m from the finish line in the 20 km race walking competition in the 2000 Summer Olympics. As photographs later confirmed, both of her feet were off the ground at the same time and she was disqualified. (Following standard practice, she had received two prior cautions before her red card disqualification.) She later won the bronze medal in this event in the 2004 Summer Olympics.

In running a sprint (up to 400 m), the runner starts in a four-point stance and pushes off the starting block with both legs to get forward acceleration, with the body fairly horizontal at first to keep pushing forward. The sprinter then accelerates as fast as possible with the body becoming vertical to increase stride length, to attain and maintain maximum speed over the short time to the finish line [16]. In running a long-distance race, fast acceleration at the beginning is less important. The goal instead is to maintain a maximum speed with the needed endurance. (For world record running speeds see Table 6.34 below.)

3.4.1 Kinematics

Figure 3.28 shows snapshots of different stages of running. One difference between walking and running is that the leg becomes almost straight during walking at midstance, but it never becomes straight during running. Another difference is that during walking the stance is longer than the swing, while the reverse is true for running. As seen in Fig. 3.10, during running the stance occurs during the first 40% of the cycle after foot strike, and the foot is off the ground for the next 60% of this cycle, from toe off to the next foot strike. The leg floats for the first 15% of the cycle after toe off, swings forward during the next 30%, and then floats for 15% until foot strike.

The changes of sagittal joint angles are qualitatively the same as in walking (Fig. 3.11 top), but with differences, such as with the knee angle. Figure 3.29 shows

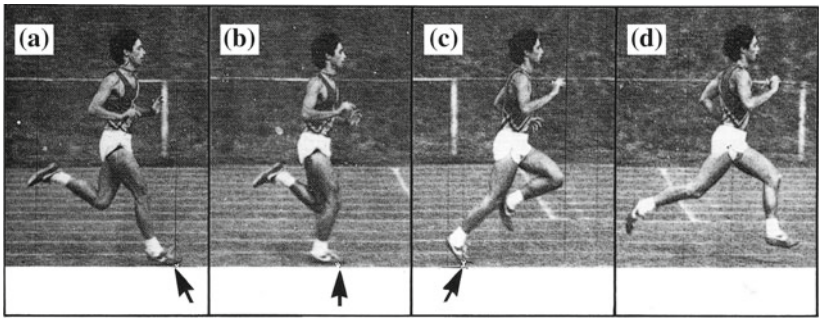


Fig. 3.28 Stages of a running stride, with the *arrows* indicating the directions of the forces on the feet (from [5]. Copyright 1992 Columbia University Press. Reprinted with the permission of the press)

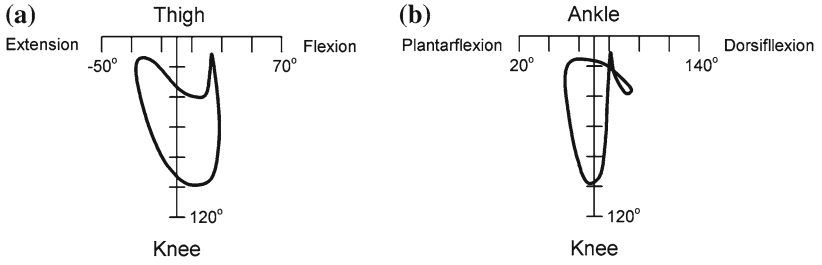


Fig. 3.29 Mean locus of **a** thigh angle versus knee angle and **b** ankle angle versus knee angle during running at 3.57 m/s, using joint angles as defined in Fig. 3.1 (based on [98, 141])

a representative example of how the locus of these angles varies with respect to each other during a gait cycle. For a running speed of 3.4 m/s (∼8 min mile), the thigh angle varies from 39° (flexion) to −21° (extension) for a range of 60°. At footstrike the thigh is about 25° from the vertical because the hip is flexed; any initial angle change after FS is relatively small. During the extension prior to footstrike the knee angle is 13° and increases to 41° during cushioning flexion (for a range of 27° during cushioning flexion), and then decreases to 15° during extension during the propulsive phase. At swing phase flexion the angle is 104°, for a total knee range of 90°. For the ankle, the dorsiflexion prior to footstrike is 91°, that at the stance phase is 112°, and that at plantar flexion is 64°, for a total range of ankle motion of 48°. These values change with running speed and with grade for uphill and downhill running. Table 3.6 shows typical loci of the thigh and ankle angles versus the knee angle during a stride.

Your legs are not the only things that move during running and fast walking (along with your center of mass). Your arms swing back and forth and your torso rotates, both 180° out of phase with your legs. Why? Consider what would happen if this did not happen. With every stride by your right leg you would need to gain much positive angular momentum about the vertical axis, while with every stride with your left leg you would need to acquire negative angular momentum, as is evident in Fig. 3.30. This is just like the rotor in a washing machine. The motion in your arms and torso

Table 3.6 Maximum thigh, knee, and ankles angles (in degrees) for running at different speeds and grades on a treadmill

	Level			Ramp		
Speed (m/s)	3.4	4.2	5.0	3.4	3.4	3.4
Grade	0%	0%	0%	−20%	0%	20%
Thigh						
Flexion	39.0	41.0	46.6	25.8	39.0	54.5
Extension	−20.6	−23.1	−28.0	−18.4	−20.6	−18.7
Knee						
Extension before footstrike	13.4	16.7	15.5	4.0	13.4	37.7
Cushioning flexion	40.6	39.4	42.9	41.2	40.6	46.3
Propulsion phase extension	15.1	16.8	14.2	32.1	15.1	13.0
Swing phase flexion	103.9	108.1	117.3	99.6	103.9	113.2
Ankle						
Dorsiflexion before footstrike	91.3	91.8	92.5	–	91.3	97.6
Stance phase dorsiflexion	112.4	108.7	110.8	103.5	112.4	116.4
Plantar flexion	64.2	58.7	57.1	76.4	64.2	59.4

Using data from [98]

cancels ∼90% of these changes, as seen in this figure. Your swinging arms serve as passive dampers and are not driven by shoulder muscles [119].

3.4.2 Muscular Action

Each flexion/extension action described by kinematics is caused by the flexor/extensor muscles of that joint. The muscular activity at each phase of the cycle is qualitatively similar to that shown for the three sagittal joints for walking in Figs. 3.11, 3.12 and 3.13. Tables 3.1, 3.2 and 3.3 describe these muscle groups. Figure 3.31 shows muscle activity during running.

With advancing age people generally walk and run slower because they take shorter strides. In walking, there is less activation of knee extensor and ankle plantar flexor muscles, and some increased hip extensor activation to partly compensate for

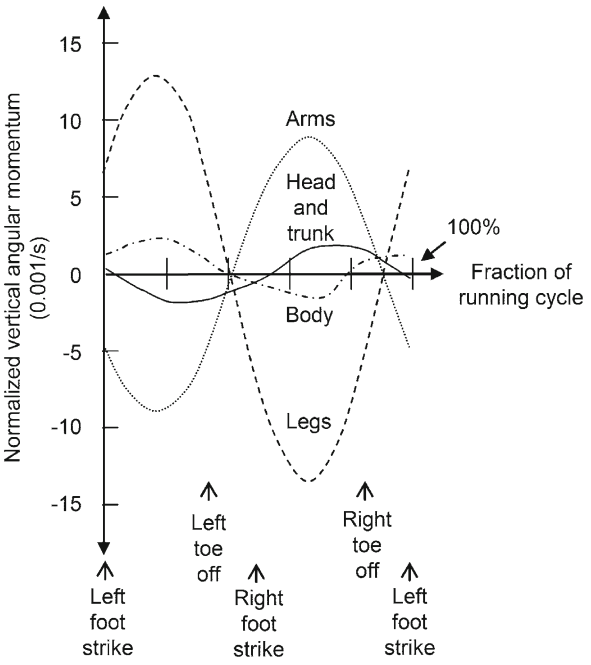


Fig. 3.30 Normalized mean vertical angular momenta of the arms, head-plus-trunk, legs, and whole body during a running cycle, at 4.5 m/s. The normalized vertical angular momentum is obtained by dividing the vertical angular momentum (in kg-m^2/s) by body mass (in kg) and the square of the runner's standing height (height in m) (based on [69, 70])

this [38]. In running, stride length and running speed typically decrease by ~0.08 m and ~0.10 m/s per decade from 20 to 60 years of age, due to lessened ankle muscle activation, without changes in the activation of the hip and knee muscles [39]. (Also see the discussions in Chaps. 6 and 9 on aerobic capacity and oxygen in the blood with advancing age and Fig. 6.12 on how world-class marathon running times increase with age.)

3.4.3 Energetics

Figure 3.28 shows that the body slows down during stage (a) when the right foot is planted on the ground, and it accelerates during stage (c) as the right foot propels the body forward, as in walking. This means that the horizontal kinetic energy

$$\mathrm{KE_H}(t = \text{stage b}) < \mathrm{KE_H}(t = \text{stage d}), \tag{3.35}$$

because stage (b) is right after (a) and stage (d) is right after (c). During stage (d) both feet are off the ground, the body is in the air, and the body center of mass and potential energy are relatively high. In contrast, during stage (b) the right foot is on the ground, the right leg is bent (while it is straight during walking) and so the center

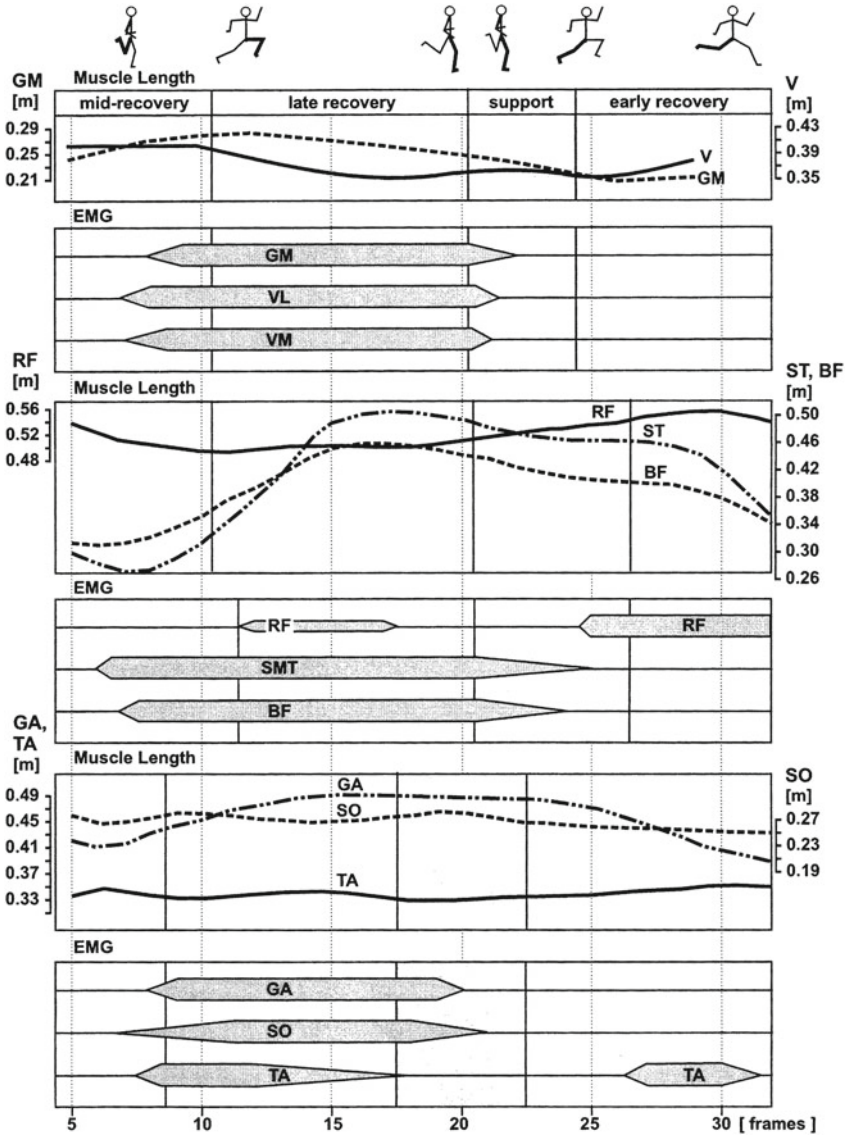


Fig. 3.31 Muscle length and electromyographic (*EMG*) activity versus time during one running cycle of the right leg, for the gluteus maximus (*GM*), vastus lateralis (*VL*), biceps femoris (*BF*), vastus medialis (*VM*), combined VL and VM (*V*), rectus femoris (*RF*), semitendinosus (*ST*), combined semimembranosus (*SM*) and ST (*SMT*), gastrocnemius (*GA*), soleus (*SO*), and tibialis anterior (*TA*) (from [65]. Used with permission). Also see [91]

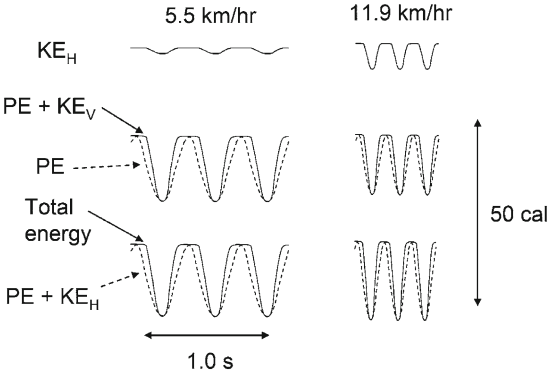


Fig. 3.32 Changes in mechanical energy during running at two different speeds from force plates, for (from *top* to *bottom curve*) horizontal kinetic energy; gravitational potential energy plus vertical kinetic energy and gravitational potential energy; and total energy and gravitational potential energy plus horizontal kinetic energy (based on [22, 93])

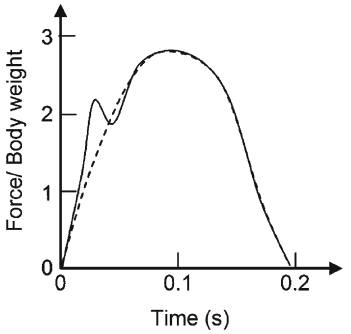


Fig. 3.33 Vertical force exerted on the ground during a running stride (*solid line*, with *dashed line* approximation). This was calculated from force plates and was averaged for five-runners running at 4.5 m/s (based on [5])

of mass and potential energy are relatively low. Consequently,

$$\mathrm{PE}(t = \text{stage b}) < \mathrm{PE}(t = \text{stage d}) \tag{3.36}$$

and so

$$\mathrm{KE_H}(t = \text{stage b}) + \mathrm{PE}(t = \text{stage b}) \ll \mathrm{KE_H}(t = \text{stage d}) + \mathrm{PE}(t = \text{stage d}). \tag{3.37}$$

We see that during running $\mathrm{KE_H}$ and PE (or really PE + $\mathrm{KE_V}$, with $\mathrm{KE_V}$ being relatively small) are not out of phase and their sum changes much during each stride (see Fig. 3.32). The pendulum model cannot be applied to running at all [5].

Not all of the lessened KE and PE during stage (b) is lost; some is saved within the body as stretched tendons and ligaments, as in a pogo stick. We will examine this now and revisit it again later after we have learned a bit more about the elastic properties of parts of the body in Chap. 4.

Figure 3.33 shows a force plot of the vertical normal forces on the foot on the ground during a run at 4.5 m/s, which corresponds to a competitive speed during a marathon (2 h 37 min over 42.2 km or 26 miles, 385 yd). The peak force in this example is $2.7\, m_b g = 2.7\ W_b$. For a 70 kg person (700 N, 160 lb), this is ~1,900 N.

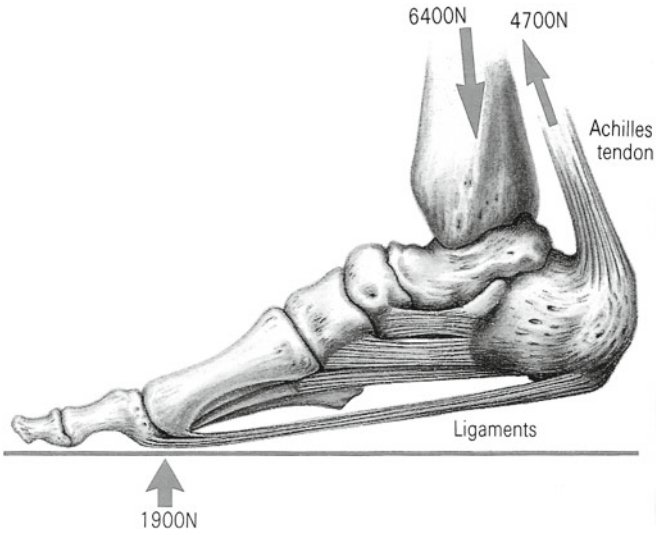


Fig. 3.34 Peak forces on foot during running stride (from [5]. Copyright 1992 Columbia University Press. Reprinted with the permission of the press)

During sprinting, this force can increase to $3.6\,m_b g$. (These forces are estimated later in this chapter in the discussion of collisions.)

The torque balance in the quasistatic condition when this normal force peaks at 1,900 N is depicted in Fig. 3.34, which is similar to the second class lever discussed in Chap. 2. In equilibrium, the Achilles tendon exerts an upward force of 4,700 N and there is a 6,400 N reaction force down on the foot from the rest of the body, because $4{,}700 \sim 1{,}900$ N (120/47 mm).

The Achilles tendon is also known as the calcaneal (kal-kane'-ee-ul) tendon. It attaches the triceps surae (sur-ee'), which includes the soleus and the medial and lateral heads of the gastrocnemius (gas-trok-nee'-mee-us) muscles (Figs. 1.8 and 3.5, Table 3.3), to the calcaneus (kal-cane'-ee-us), also known as the heel.

During this marathon-type run, there is a loss of about 100 J of kinetic energy each time the foot touches the ground and decelerates to a stop. Not all of it is lost. Some of it is stored in the body, and a portion of that is recoverable, mostly from body components that store the energy like springs. To maintain a steady average running speed, the portion that cannot be recovered must be supplied by the body in the acceleration phase of the stride. *How much does the body slow down during this step?* For given mass and speed, just calculate the change in speed when 100 J of kinetic energy is lost (see Problem 3.34).

Approximately 35 J of this 100 J is stored in stretching the Achilles tendon. We will see in the discussion about mechanical properties in Chap. 4 how this value is consistent with the forces exerted on the Achilles tendon. Of this 35 J, about 93% is recoverable during the acceleration phase, so this tendon acts almost like a

friction-free spring. During the step, the foot arch flattens and it takes about 17 J to do this. About 80% of this is recoverable. Some energy is also lost to the quadriceps tendon over the kneecap. Much of the remainder is lost to muscles acting as brakes, and this is not recoverable. Overall, about 50 J is recoverable from the body's springs and about 50 J must be supplied by the muscles during the acceleration phase. The notion that 50% of the energy is recoverable energy is not universally accepted; Fig. 3.26 suggests that only about 5% is recoverable. The materials properties of the body are clearly significant—for analyzing the recoverable energy storage and the needed braking-action of muscles. We will examine both soon. The long bones in the leg can also store energy, but we will see that they store little energy. There are also questions about the efficiency and timing of the energy return, and potential confusions between the loss of kinetic energy during a step with the much larger ($\sim 5\times$) metabolic energy needed to replace this lost kinetic energy.

Good running shoes lessen the impact in each step. *Can they also help by storing energy as springs, thereby lessening the amount of energy needed in each step?* It is not clear that this occurs. Good running shoes are squeezed up to at most 10 mm in the shoe sole under the forefoot upon impact. They can store about 7 J, of which about 54–66% could be returned, so it would seem that they can help a bit. However, by cushioning the forces felt by the feet, significantly less recoverable energy is stored by the body. Even more cushioning would produce instability during stance and take off. Furthermore, even if much energy could be stored elastically in shoes, it is not clear if much of it would be recoverable. Stored energy has to be released at the right time, over the right duration (frequency), and at the right position [110]. The resonance frequencies of unloaded shoes are ~100–200 Hz and loaded shoes likely 20–30 Hz. Because any stored energy in the heel needs to be released 100–200 ms after foot contact and it is really released 50–100 ms (~1/20 Hz) after contact, the timing does not seem right. Also, if energy is stored in one part of the shoe because of impact (say the heel) it does little good if the extra "kick" is needed in the front of the foot for forward propulsion. Sprinters use running shoes with essentially no padding during races because their body returns more energy without the extra padding. During practice, they wear shoes with padding to limit the wear and tear on their legs, i.e., the stress on their tendons and so on. Later in this chapter we will discuss how foot injuries during running might be controlled by how we land with our feet and the running shoes we use (Fig. 3.33) [85, 87].

A good running track can return up ~12 J per step (Table 3.11, later). Such a track has a stiffness (spring constant) about 3× the lower leg stiffness (80,000 N/m), and can increase running speeds by ~2%. Such an increase in speed decreases 100 m sprint times by ~0.2 s and could decrease marathon times by ~80 s. (Of course, marathons are not run on such a track.) These tracks can be tuned to have resonant frequencies of ~2 Hz, which matches the times needed for the recovery of the stored energy. This impact of the step is a type of collision that will be discussed later in this chapter. This is also analyzed in a problem at the end of Chap. 4 (Problem 4.39).

Still, more of the energy lost per step is recoverable in walking than during running, perhaps somewhat over 70% (Fig. 3.26). Kangaroos recover about 50% during each hop.

Accelerating to Sprint Speed

The average speed for world class runners at the 100 and 200 m distances is about the same, $\simeq$10.4 m/s (in 2014) (see Table 6.34). It is slower at longer distances, because the stores of energy that activate muscles at the fastest rate are depleted after $\sim$200 m (Chap. 6). It is also slower at shorter distances, because it takes several strides to achieve peak speeds, starting from a dead stop. We now model this acceleration process by examining the energetics after each stride [67, 68, 78, 90]. The stride is step (c) in Fig. 3.28, with one foot on the ground propelling the body forward. The body consists of the other leg, which has mass $m_{\rm leg}$, and the body above the legs (upper body), which has a mass $m_{\rm b} - 2m_{\rm leg}$, where $m_{\rm b}$ is the total body mass. (In this model we are ignoring the other stages: free flight, the slowing down of the body during touchdown, etc., and we are effectively incorporating all into this step (c).)

During the first stride, the foot on the ground, say the right foot, propels the upper body and the left leg to a speed with a force F as they move a distance L, and they attain a speed of v_1 at the end of the stride. The work done FL equals the kinetic energy in the upper body, $(m_{\rm b} - 2m_{\rm leg})v_1^2/2$, and that in the left leg, $m_{\rm leg}v_1^2/2$; the right foot is still on the ground, so the kinetic energy of the right leg is small and can be neglected. Therefore, after the first stride

$$FL = \frac{1}{2}(m_{\rm b} - 2m_{\rm leg})v_1^2 + \frac{1}{2}m_{\rm leg}v_1^2 = \frac{1}{2}(m_{\rm b} - m_{\rm leg})v_1^2 \tag{3.38}$$

and

$$v_1^2 = \frac{2FL}{m_{\rm b} - m_{\rm leg}}. \tag{3.39}$$

In the second stride the left foot is on the ground, does work FL, and accelerates the upper body from v_1 to v_2 and the right leg from 0 to v_2, so

$$\begin{aligned} FL &= \frac{1}{2}(m_{\rm b} - 2m_{\rm leg})(v_2^2 - v_1^2) + \frac{1}{2}m_{\rm leg}v_2^2 \\ &= \frac{1}{2}(m_{\rm b} - m_{\rm leg})v_2^2 - \frac{1}{2}(m_{\rm b} - 2m_{\rm leg})v_1^2 \end{aligned} \tag{3.40}$$

and

$$v_2^2 = \frac{1}{m_{\rm b} - m_{\rm leg}}(2FL + (m_{\rm b} - 2m_{\rm leg})v_1^2). \tag{3.41}$$

Using (3.39) for v_1^2

$$v_2^2 = \frac{2FL}{m_{\rm b} - m_{\rm leg}}\left(1 + \frac{m_{\rm b} - 2m_{\rm leg}}{m_{\rm b} - m_{\rm leg}}\right). \tag{3.42}$$

Similarly, after the third stride, with the right foot on the ground,

$$v_3^2 = \frac{1}{m_\mathrm{b} - m_\mathrm{leg}}(2FL + (m_\mathrm{b} - 2m_\mathrm{leg})v_2^2) \tag{3.43}$$

and using (3.42)

$$v_3^2 = \frac{2FL}{m_\mathrm{b} - m_\mathrm{leg}}\left(1 + \frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}} + \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)^2\right). \tag{3.44}$$

After n strides, v_n is obviously given by

$$v_n^2 = \frac{2FL}{m_\mathrm{b} - m_\mathrm{leg}} \times \left(1 + \frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}} + \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)^2 + \cdots + \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)^n\right). \tag{3.45}$$

The terms in the brackets form a geometric series $1 + x + x^2 + \cdots + x^n = (1 - x^n)/(1 - x)$ for $0 < x < 1$, so

$$v_n^2 = \frac{2FL}{m_\mathrm{b} - m_\mathrm{leg}} \frac{1 - \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)^n}{1 - \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)} = \frac{2FL}{m_\mathrm{leg}}\left(1 - \left(\frac{m_\mathrm{b} - 2m_\mathrm{leg}}{m_\mathrm{b} - m_\mathrm{leg}}\right)^n\right). \tag{3.46}$$

The final speed is (with $n \to \infty$)

$$v_\mathrm{final} = \sqrt{\frac{2FL}{m_\mathrm{leg}}}. \tag{3.47}$$

Table 1.7 shows that $m_\mathrm{leg} = 0.161\, m_\mathrm{b}$, so a 70 kg sprinter, with $F = 560\,\mathrm{N}$ and $L = 1.0\,\mathrm{m}$ stride, attains a final speed of 10.0 m/s.

3.4.4 *Bouncing Ball/Pogo Stick Model*

Much as walking can be modeled as a pendulum, running can be modeled using mechanical analogies. Figure 3.35 shows the bouncing ball model. This is fairly equivalent to the even simpler pogo stick model, which does not have the complication of rotational ball motion. When the ball or stick is at its highest point, the gravitational PE is maximized, the KE is zero, and there is no stored *elastic energy* EE in the ball due to compression of the ball or in the pogo stick due to compression of the spring (which is also a form of potential energy). When it is about to touch the ground,

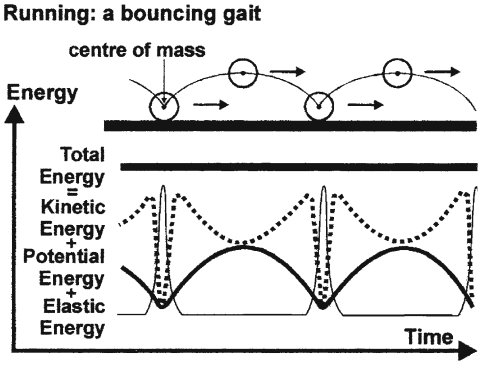


Fig. 3.35 Bouncing ball model of running, showing that the sum of the ball kinetic energy (*dotted line*), gravitational potential energy (*thick line*), and elastic potential energy (*thin line*) is constant (*very thick line*) during bounces (from [99]. Used with permission)

the PE is almost minimized, the KE is almost maximized and the EE is still zero. During the elastic collision with the ground, the EE increases while the KE goes to zero, and then all of the EE is recovered, initially mostly as KE. The total energy TE = KE + PE + EE is always constant [99]. There is exchange between the sum of the kinetic and gravitational potential energies with the elastic energy in a bouncing ball.

There is an analogy between running and this model, but it is not very strong. The PE and KE are both maximized during stage (d) in Fig. 3.26, while EE is zero. During stage (b), PE and KE are both minimized, and presumably there is EE stored in the tendons, foot arch, etc. Consequently, the phase of KE is different from that in the model. However, the analogy is fine for the phases of PE + KE = TE − EE and of EE. Unlike walking, there is relatively little stored energy returned in running (Fig. 3.26); it is not very elastic.

Figure 3.36 shows the pogo stick model for running and a double pogo stick model for skipping. Skipping is a third, and fairly uncommon gait for bipeds. It differs from walking in that it has a significant flight phase, and from running in that both feet can be on the ground (Table 3.7). Skipping is energetically very efficient: about 50% of the energy is recoverable (Fig. 3.26). These models for biped locomotion can also be used to model quadruped locomotion (Fig. 3.36). Whereas there are two main types of locomotion in bipeds (walking, running), there are three for quadrupeds (walking, trotting, galloping). The types of quadruped motion are described in Table 3.7.

3.5 Jumping

We will first examine the vertical jump and pole vault using an energetics approach and then the high jump and long jump by analyzing their equations of motion. In the last three athletic endeavors, kinetic energy from running is converted into potential energy. The vertical jump is a half-collision with the floor. (Here the half-collision being the second half of a collision with the floor.) Other types of jumping, as

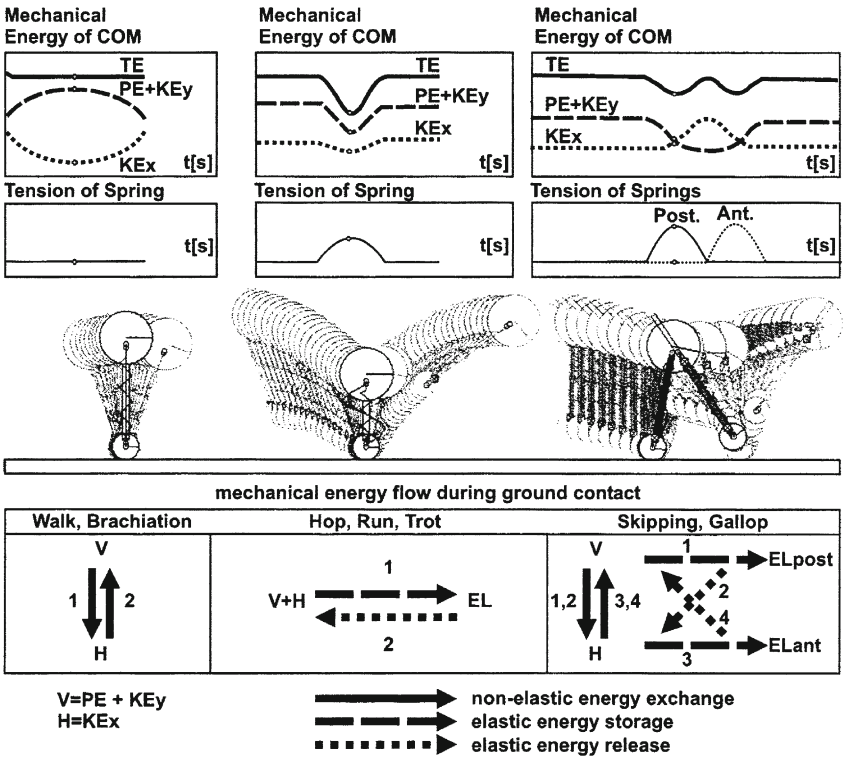


Fig. 3.36 Three mechanical models of locomotion, the (inverted) pendulum (*left*, for walking), single pogo stick (*middle*, for hopping and running), and double pogo stick (*right*, for skipping), with tension in pogo stick spring and the modes of energy exchange (from [99]. Used with permission)

on trampolines and in others common in gymnastics, involve full collisions with surfaces; how much energy these surfaces return to the jumper becomes important. This will be addressed in the discussion of collisions.

3.5.1 Vertical Jump

How high can you jump? It depends on how fast you can take off. The four stick diagrams in Fig. 3.37 show the four stages of the vertical jump.

First the person stands upright. The center of mass is about 1.0 m from the ground, 5 cm above the hip joints. Figure 3.37b shows the body after the *crouch*, after which the center of mass is about 0.65 m above the ground. The *extension* phase is next, which ends in *takeoff* shown in the third diagram. At takeoff (Fig. 3.37c) the center of mass is about 1.05 m above the ground, a bit higher than in Fig. 3.37a since the person is on his or her toes. The center of mass rises by a distance s during extension,

Table 3.7 Comparison of motion in bipeds and quadrupeds

Biped type	Foot sequence[a]	Features
Walk	R, L, R, L, etc.	No flight phase
Run	R, flight, L, flight, etc.	
Skip		
Right unilateral skipping	L, R, flight; L, R, flight; etc.	Left unilateral is R, L, flight, etc.
Bilateral skipping	L, R, flight; R, L, flight; etc.	
Quadruped type (biped analog)	Foot sequence[b]	Features
Walk (walk)	FL, HR, FR, HL	Stride: up to 3 hooves on ground
Trot (run)	Two running bipeds w/50% phase lag	Rack is trot with 0% phase lag
Gallop (none)		
Slow gallop or canter	HL, HR, FL simul.; FR flight	Right cantor; left switches R and L
Transverse gallop (fast run)	HL, HR flight; FL, FR flight	Right transverse
Rotary gallop (fast run)	HL, HR flight; FR, FL flight	Clockwise or counterclockwise

[a] *R* right foot, *L* left foot
[b] *FL* fore left, *HR* hind right, *FR* fore right, *HL* hind left (from [99])

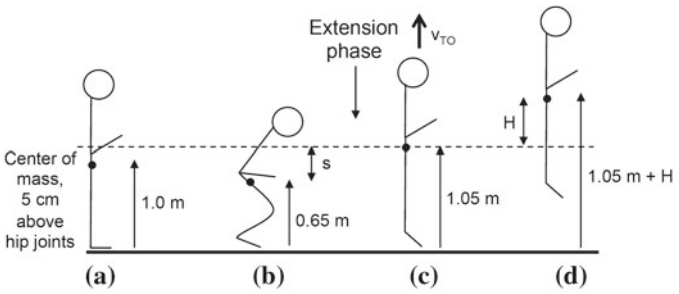


Fig. 3.37 Stick diagram of a vertical jump, including initial standing in (**a**), crouching in (**b**) and then extension to take-off in (**c**) and free flight in (**d**) (based on [113])

and attains a vertical speed of v_{TO} at takeoff. After takeoff the person is in *free flight* for a distance H and then the vertical speed v is zero, as shown in Fig. 3.37d.

The kinetic energy at takeoff is all converted into potential energy at the apex of free flight. Therefore

$$\frac{1}{2}m_{\mathrm{b}}v_{\mathrm{TO}}^2 + m_{\mathrm{b}}g(1.05\ \mathrm{m}) = 0 + m_{\mathrm{b}}g(1.05\ \mathrm{m} + H) \tag{3.48}$$

$$\frac{1}{2}m_{\mathrm{b}}v_{\mathrm{TO}}^2 = m_{\mathrm{b}}gH, \tag{3.49}$$

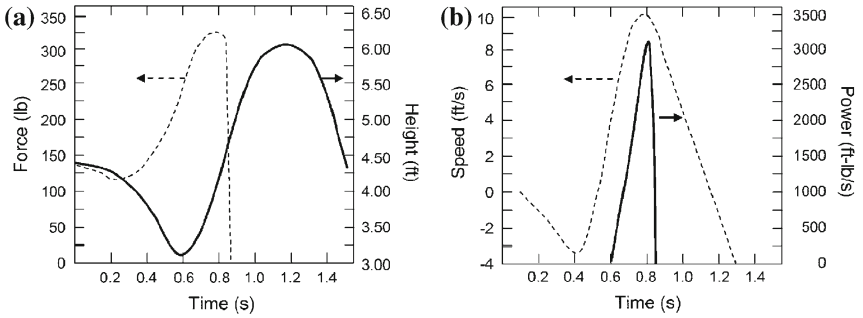


Fig. 3.38 Kinematic and dynamic data during a vertical jump, including **a** reaction force from the ground (*left scale*) and height of the center of mass (*right scale*), and **b** vertical speed of the center of mass (*left scale*) and applied power (*right scale*, this power is the product of the force in (**a**) and speed in (**b**)) (based on [113])

so $H = v_{\mathrm{TO}}^2/2g$ and the faster the takeoff speed the higher the jump.

During extension both legs generate a normal force $N(t)$ that leads to a net vertical force $F_{\mathrm{V}}(t) = N(t) - m_{\mathrm{b}}g$. Figure 3.38 shows this normal force in measurements made on several men in the early 1930s at Columbia University [113]. This vertical force increases and then decreases, and while it is applied the center of mass accelerates upward and travels a distance s. The work W done on the center of mass during this phase is

$$W = \int_0^s F_{\mathrm{V}}(t)\mathrm{d}z. \tag{3.50}$$

(Note that to perform this integration $F_{\mathrm{V}}(t)$ needs to be converted to a function of z.) This gets converted into the vertical kinetic energy so

$$\int_0^s F_{\mathrm{V}}(t)\mathrm{d}z = \frac{1}{2}m_{\mathrm{b}}v_{\mathrm{TO}}^2 = m_{\mathrm{b}}gH \tag{3.51}$$

and

$$H = \frac{\int_0^s F_{\mathrm{V}}(t)\mathrm{d}z}{m_{\mathrm{b}}g}. \tag{3.52}$$

For $W_{\mathrm{b}} = m_{\mathrm{b}}g = 140\,\mathrm{lb}$ (64 kg, 620 N), an average normal force during extension $\langle N(t)\rangle$ of 300 lb (Fig. 3.38), and $s = 1.4\,\mathrm{ft}$ (0.43 m), we find

$$H = \frac{\int_0^s F_{\mathrm{V}}(t)\mathrm{d}z}{m_{\mathrm{b}}g} = \frac{(N(t) - m_{\mathrm{b}}g)s}{m_{\mathrm{b}}g} \tag{3.53}$$

$$= \frac{(300\,\mathrm{lb} - 140\,\mathrm{lb})1.4\,\mathrm{ft}}{140\,\mathrm{lb}} = 1.6\ \mathrm{ft}\ (0.49\ \mathrm{m}). \tag{3.54}$$

How long is takeoff? Let us assume that the vertical acceleration is constant and equal to a during extension. (This would be true only if the vertical normal force were constant—and it is not.) From $F_V = m_b a$

$$a = \frac{N(t) - m_b g}{m_b}. \tag{3.55}$$

If the duration of takeoff is τ

$$v_{TO} = a\tau \tag{3.56}$$

and using

$$s = \frac{1}{2} a\tau^2 \tag{3.57}$$

we see

$$\tau^2 = 2\left(\frac{m_b g}{N(t) - m_b g}\right)\left(\frac{s}{g}\right) = 2\left(\frac{140\text{ lb}}{160\text{ lb}}\right)\left(\frac{1.4\text{ ft}}{32.2\text{ ft/s}^2}\right) \tag{3.58}$$

because $\langle N(t)\rangle - m_b g = 300\,\text{lb} - 140\,\text{lb} = 160\,\text{lb}\,(710\,\text{N})$. The acceleration occurs over a time $\tau = 0.28\,\text{s}$.

How are the jump height and takeoff time related? Dividing (3.56) by (3.57) eliminates a to give

$$v_{TO} = \frac{2s}{\tau}, \tag{3.59}$$

so

$$H = \frac{v_{TO}^2}{2g} = \frac{(2s/\tau)^2}{2g} = \frac{2s^2}{g\tau^2}. \tag{3.60}$$

If you can decrease the extension time, you can jump much higher! We will see that extension times are shorter in people with larger fractions of fast-twitch muscles in their legs. However, there can be a trade-off with shorter extension times. When muscles achieve steady-state motion, they develop less tension the faster they contract, which is much less than the isometric (fixed length) force that they can generate. (In Chap. 5 we will consider these and other features of muscle activation.) Measurements of the force on the floor using force plates have shown that while the peak force on a representative runner's foot might be as high as 1,710 N, the force on each foot during his high jump turns out to be much less, 715 N [5]. This surprising observation is explained by the faster muscle contraction during jumping. Average forces are also expected to be lower in the high jump because muscles develop less tension when they are much longer or shorter than their resting length. Also, the momentum arm about the knee joint is smaller for the bent knee in the squatting position of a vertical jump.

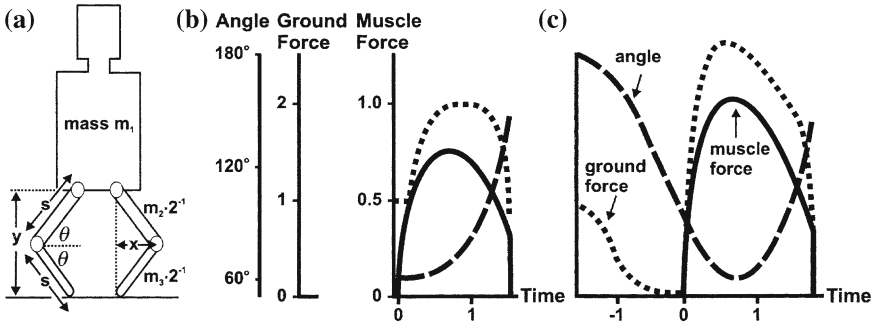


Fig. 3.39 **a** A mechanical model of vertical jumping, with **b** a simulated jump and **c** a simulated countermovement jump, with muscle force, ground reaction force, and angle plotted versus time in (**b**) and (**c**) (from [7], as from [6]. Used with permission)

Figure 3.39 shows a model of two variations of the vertical jump, which includes muscle activation, tendon stretching, and body response [7]. The squat jump starts in a squatting position. Initially, as the muscles contract there is only slow movement of the knees and consequently much tendon stretching. Due to fast contraction, muscle forces in the squat jump are 0.2× the isometric level. Just before liftoff these muscle forces and tendon recoil produce forces 0.4× the isometric level. The countermovement jump starts with an upright person first falling freely to the starting position of the squat and then jumping upward. The muscle forces initially generated are larger than the isometric values because the muscles are initially braking (eccentric contraction, Chap. 5), and this leads to even more tendon stretching and upward force generated, and a higher elevation jump than the squat jump. This countermovement jump is higher also because muscles are stretched immediately before they are contracted, leading to greater forces during this contraction [145] (as in the later phases of this jump). (This effect is not included in the Fig. 3.39 model.)

3.5.2 Pole Vault

The pole vault is not really a jump, but a propulsion. Still, it is fairly similar to the long jump in that horizontal kinetic energy is converted to a propulsion. More specifically, the pole vaulter's horizontal kinetic energy is mostly converted into elastic potential energy stored in the pole, which is then mostly converted to gravitational potential energy of the pole vaulter (Fig. 3.40). A fast running pole vaulter can propel above and beyond the bar if the two "mostly"s in the previous sentence are really "essentially all"s. This happens with good techniques and good poles. To propel over the bar there has to be some remnant horizontal kinetic energy, but this necessary amount is minimal.

Figure 3.40 shows the relevant heights: the height of the center of mass at takeoff (the takeoff height, H_1), the maximum increase in the height of the pole vaulter's

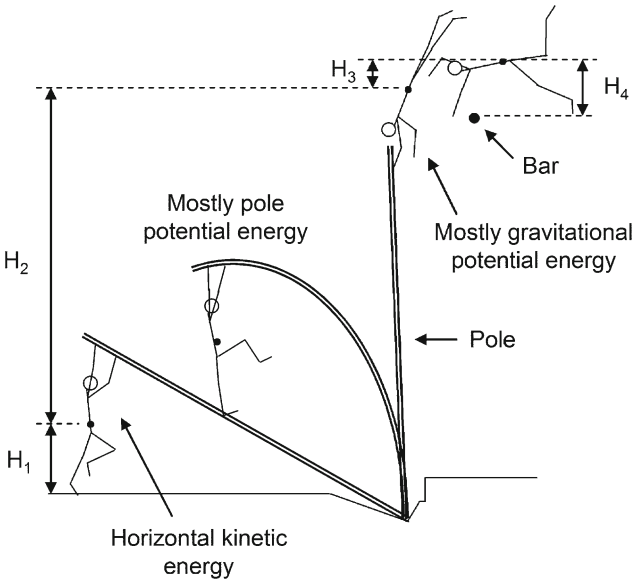


Fig. 3.40 Schematic of the heights involved in the pole vault (based on [66])

center of mass while on the pole (the swing height, H_2), the further increase in height due to vertical kinetic energy after release of the pole (the flight height, H_3), and maximum height of the center of mass above the cross bar (the clearance height, H_4—which could be negative). The takeoff angle of world-class pole vaulters using fiberglass poles is $\simeq$13–15° (relative to the vertical).

The ideal case of total conversion of one type of energy to another gives a surprisingly good prediction of actual pole vaulting performance. Initially the pole vaulter accelerates to a speed $v_{\mathrm{H,i}}$ and has a center of mass h_{CM} above the ground (H_1 in Fig. 3.40). During running h_{CM} is approximately 0.9 m, corresponding to the slightly crouched position during running. The goal is to convert the initial $E_{\mathrm{H,i}} = mv_{\mathrm{H,i}}^2/2$ into the potential energy. Over the bar, the pole vaulter has a center of mass $h_{\mathrm{bar}}+h_{\mathrm{min}}$, where h_{bar} is the height of the bar and h_{min} (H_4 in Fig. 3.40) is the distance the center of mass of the vaulter needs to be above the bar without touching it, which we will take as $\sim$0.1 m. This corresponds to an increase in the height of the center of mass by $h_{\mathrm{bar}} + h_{\mathrm{min}} - h_{\mathrm{CM}}$. A good pole vaulter can achieve a speed $v_{\mathrm{H,i}}$ $\sim$9.5 m/s. (World-class sprinters in a 100 m race, who do not carry poles while running, can attain speeds of 10.4 m/s.)

If the initial kinetic energy is converted into potential energy over the bar:

$$\frac{1}{2}mv_{\mathrm{H,i}}^2 = mg(h_{\mathrm{bar}} + h_{\mathrm{min}} - h_{\mathrm{CM}}) \tag{3.61}$$

$$h_{\mathrm{bar}} = h_{\mathrm{CM}} - h_{\mathrm{min}} + \frac{v_{\mathrm{H,i}}^2}{2g}. \tag{3.62}$$

With $g = 9.8\,\text{m/s}^2$, we see that $h_{\text{bar}} = 5.4\,\text{m}$. In a good pole vault $h_{\text{bar}} \sim 5.4\,\text{m}$; the world record is 6.16 m (in 2014) (see Problem 3.50). Of course, the pole vaulter is still moving forward over the bar and so v_{H} does not actually become zero. Also, the maximum height of a pole vault jump can be higher than predicted here because the vaulter can extend his or her arms to push up from the pole when it is vertical, just before releasing it.

The pole is very elastic. It stores and returns energy to the pole vaulter much more efficiently than do human body components during jumping and running. There is an optimal pole stiffness that maximizes performance [134]. If the pole is too stiff, it straightens before the athlete is at the maximum height and the pole pushes the athlete horizontally away from the bar. If the pole is not stiff enough, it straightens too slowly and the athlete passes the bar before attaining the highest position.

3.5.3 High Jump and Long Jump

We analyzed the vertical jump and pole vault by considering the energetics at different stages of the motion. Analyzing jumping by following the equations of motion is more complex and requires the general multisegment model presented later this chapter. We now examine a model of jumping from a running start that is simpler than this to learn about the differences in the techniques needed to be successful high and long jumpers [3]. The high jump using the now-standard Fosbury flop method is depicted in Fig. 3.41 and the long jump is illustrated in Fig. 3.42. This model [3] focuses on one joint (the knee) and two segments (the upper and lower legs, which are assumed to be of equal length). Though this model is conceptually simple, it still needs to be solved numerically. The energetics of the high jump and the long jump are analyzed in the problems at the end of this chapter (Problems 3.51–3.54, and 3.55–3.57, respectively).

The jumper runs and plants one foot, which is considered a point at the end of the lower leg and the origin of a two-dimensional system (Fig. 3.43); the center of mass of the body is at (x, y) and at the time of implant it is moving horizontally at the approach speed of v. The line from the center of mass to the origin makes an angle

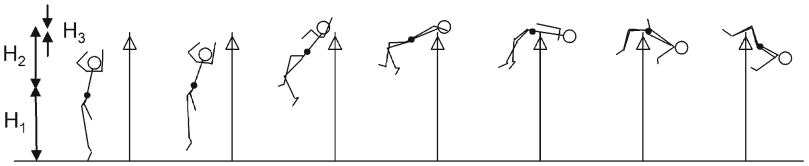


Fig. 3.41 Multisegment model of a high jump using the Fosbury flop method, showing that the center of mass (the *closed circle*) of the jumper is always below the bar (in this example). Different heights during a high jump are also shown, with H_1 being the initial height of the center of mass, H_2 the maximum elevation of the center of mass, and H_3 the distance that the final center of mass is above the bar. The *arrows* are shown assuming the highest center of mass is above the bar, so H_3 would really be negative for a successful Fosbury flop (based on [66])

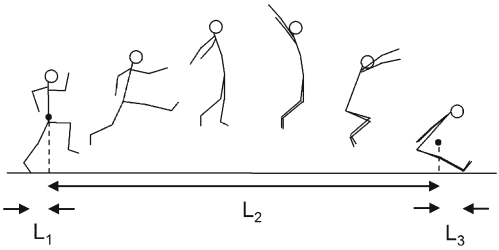


Fig. 3.42 Different lengths during a hang-style long jump (based on [66])

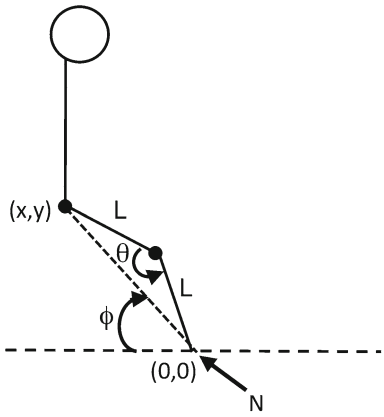


Fig. 3.43 Schematic of a force model of the high jump and long jump, with torques applied about the knee joint by muscles (based on [3])

$\phi = \arctan(-y/x)$ with the horizontal. In this simple model the center of mass is assumed to be at the height of the hip joints, although it is really 5 cm above them as noted in Chap. 2. The angle of the knee joint is θ, with $(x^2 + y^2)^{1/2} = 2L\sin(\theta/2)$, where L is the length of the upper and lower legs. The center of mass is followed from the time the foot is planted until takeoff and then after takeoff. There is a normal or reaction force, N, on the foot until takeoff, which, along with gravity, are the only forces exerted on the center of mass.

Lift is provided by the torque τ that is developed by the knee extensor muscles, which power the upper and lower legs (see Fig. 3.2); it is assumed that there is no torque generated about the hip joints. This knee torque produces a force at the ground with magnitude N, where $\tau = NL\sin(\pi/2 - \theta/2)) = NL\cos(\theta/2))$, so the reaction force from the ground has magnitude $N = \tau/(L\cos(\theta/2))$; it needs to exceed the body weight to enable liftoff. Because there are no torques about the hip joints in this model, which is also the center of mass here, this force is parallel to the vector from the planted foot to the hips and not actually normal to the surface, as is shown in Fig. 3.43.

In Chap. 5 we will see that the maximum forces developed by muscles depend both on their length and contraction speed (Fig. 5.29). If they contract slowly during the motion, we can use a model in which the muscle force depends only on its length (Fig. 5.19). We will soon use this to model throwing a ball (even though this assumption might not be perfectly valid), and will, in fact, pretty much ignore the

effects of small changes in muscle length. If the muscle lengths vary little and remain near those needed for maximum force during the motion, we can use a model in which the muscle force depends only on the muscle contraction speed, which is given by the Hill force-velocity curve (5.21) (Fig. 5.26). We will use this in this model of jumping [3].

As is often done, we will recast the Hill force-velocity curve by changing from muscle forces to joint torques and from muscle contraction speeds to the rates of change of joint angles ($\omega = -d\theta/dt$, so ω is positive when the knee angle is becoming smaller). (This is described in Problem 5.20.) For joint flexion ($\omega > 0$)

$$\tau = \tau_{\max}\left(\frac{\omega_{\max} - \omega}{\omega_{\max} + G}\right) \tag{3.63}$$

and in resisting joint extension (for $\omega < 0$)

$$\tau = \tau_{\max}, \tag{3.64}$$

where $\omega_{\max}$ and $G \sim 3$ are constants. (In the classic Hill force-velocity formulation, (3.64) would be $\tau = 1.8\tau_{\max}$.) At the time of foot implant, ω is zero and the torque can be at its maximum value, and during knee flexion ω increases and the torque decreases until it becomes zero when $\omega = \omega_{\max}$.

When the foot is still on the ground, the equations of motion of the center of mass of the body are

$$\frac{d^2x}{dt^2} = \frac{N}{m}\cos\theta \tag{3.65}$$

and

$$\frac{d^2y}{dt^2} = \frac{N}{m}\sin\theta - g. \tag{3.66}$$

The equations in this model need to be numerically integrated from the time of foot implant to takeoff for typical muscle properties, with takeoff beginning when the torque falls to zero. (Numerical analysis is beyond the scope of this book.)

After takeoff the center of mass moves in free flight, with these same equations of motion, but now with $N = 0$; consequently, from takeoff to landing they are easily solved analytically. Free flight depends on the center of mass position (x_{TO}, y_{TO}) and velocity ($v_{\mathrm{x,TO}}$, $v_{\mathrm{y,TO}}$) at time of takeoff. By usual kinematics, at time t after takeoff $v_{\mathrm{x}}(t) = v_{\mathrm{x,TO}}$ and $v_{\mathrm{y}}(t) = v_{\mathrm{y,TO}} - gt$. The maximum height of the center of mass is $y_{\mathrm{TO}} + v_{\mathrm{y,TO}}^2/2g$ (which is significant for the high jump) and the distance travelled by the center of mass during flight is $(v_{\mathrm{x,TO}}/g)(v_{\mathrm{y,TO}} + (v_{\mathrm{y,TO}}^2 + 2gY)^{1/2})$ (which is significant for the long jump), where Y is how much lower the center of mass is at landing than at takeoff (Fig. 3.42). See Problem 3.96 for a derivation of this result, which is also important in throwing and hitting a ball at one height and having it land or caught at a different height. $Y = y_{\mathrm{TO}}$ if the center of mass is at ground level at

landing. This distance traveled is a bit different from the distance the implanted foot travels (from its takeoff to landing) (Fig. 3.42).

The most interesting variables in this jump model are the approach speed and the angle of the lower leg at foot implant. The most significant results are the maximum increase in the height H_2 of the center of mass and the height itself H_1+H_2 (Fig. 3.41), and the distance it travels until landing L_2 (Fig. 3.42). In the high jump the goal is to maximize H_2 for a given H_1, and this occurs when the leg angle at implant is $\sim$45–50° and the approach speed is $\sim$7 m/s. It is a bit surprising that smaller H_2 is predicted (and also seen) for even faster approach speeds; such faster speeds are easily attainable by runners, but are not desirable in the high jump.

In the long jump the goal is to maximize landing L_2, and the model shows that this occurs with steeper leg angles at implant of $\sim$60–70° and the fastest possible approach speeds. In contrast to the high jump, long jumps are always better with faster speed, to the $\sim$11 m/s speed of sprinters.

For both jumps, the model analysis of best strategies is relatively insensitive to the specific choice of muscle parameters and (our) ignoring of drag, and agrees with the guidance jumpers are given: High jumpers should not try to attain speeds as fast as sprinters, and should have relatively shallow leg angles at foot implant (which occurs when they have low centers of mass and their implant feet are well in front their centers of mass). Long jumpers should run as fast as possible, with their implant lower legs at a steeper angle.

3.6 Throwing a Ball

Sports involving balls and other projectiles involve several of the same basic components: (1) human motion to throw the ball or swing a bat or racket, (2) hitting the ball (collision of the bat with the ball), (3) the flight of the ball, and (4) the bouncing and skidding or rolling of the ball after it contacts the ground. Throwing a baseball or football involves 1, 3, and sometimes 4, while hitting a baseball or golf ball involves all four. We will address the throwing part of component 1 now and then the swinging motion part, contact with the ball during swinging, and ball flight, bouncing and rolling later in this chapter. We will then tie up this discussion in Appendix G, which gives an overview of the physics of sports, where we will also provide general references on the physics of various sports.

The biomechanics of throwing a ball and, of course, of pitching in baseball are quite complex. They involve the coordinated motion of many muscles about several body joints (Table 3.8), as is also true for running, jogging, etc. We will model throwing a ball by first greatly simplifying the problem and subsequently reassessing many of our initial assumptions. The first assumption is that only the arm is involved in throwing. This is a major assumption.

The diagrams in Fig. 3.44 shows four models of throwing a ball overhand. One can consider throwing by:

Table 3.8 Joints and muscles involved in the acceleration phase of the overarm throw

Joint	Joint action moved	Segment	Major muscles
Shoulder joint	Horizontal abduction	Upper arm	Posterior deltoid
			Infraspinatus
			Teres minor
	Medial rotation		Subscapularis
			Latissmus dorsi
			Pectoralis major
Shoulder girdle	Abduction	Scapula	Serratus anterior
			Pectoralis minor
Elbows	Extension	Forearm	Triceps brachii
Radioulnar	Pronation	Forearm	Pronator teres
			Pronator quadratus
Wrists	Flexion	Hand	Flexor carpi radialis
			Flexor carpi ulnaris
			Flexor carpi digitorum
			Superficialis

Each joint and its action are shown, along with the segment being moved and the muscles undergoing concentric action
Using data from [64]

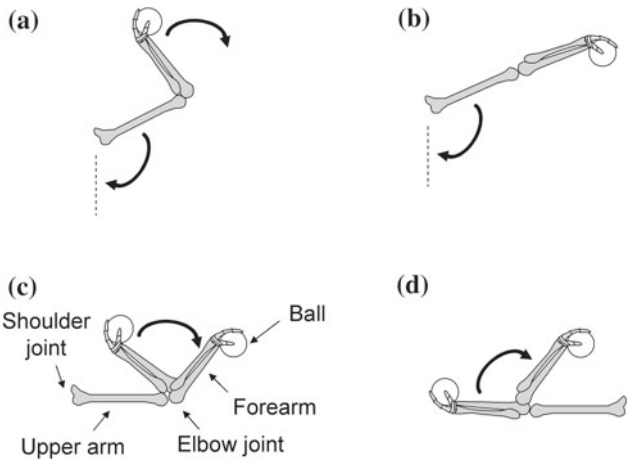


Fig. 3.44 Models of throwing a ball with **a** motion of the shoulder and elbow (extension), **b** motion of the shoulder only with a fixed arm, **c** extension of the elbow and a stationary shoulder, and **d** flexion of the elbow and a stationary shoulder

(a) The entire motion with motion in the shoulder and elbow joints (Fig. 3.44a).
(b) Motion of a stiff arm (elbow fixed)—motion only in the shoulder joint, using the deltoid muscles (Fig. 3.44b).
(c) Motion of only the forearm (lower arm) (shoulder fixed)—motion only in the elbow joint using the extensor triceps brachii (Fig. 3.44c).
(d) Motion of only the forearm (shoulder fixed)—motion only in the elbow joint using the flexor biceps brachii (Fig. 3.44d).

Similar diagrams can be drawn showing several analogous ways of throwing a ball underhand, which involve these same (a), (b), and (d) models.

Each of these models represents a fairly unnatural motion for throwing (try them!), but they represent good first-order models. Model (a) involves two-body joints, and consequently is a two-segment model (because two segments can move). Models (b)–(d) involve one-body joint and are one-segment models. More sophisticated models require a multisegment model.

We will analyze throwing with the single-segment model (d) where the forearm moves due to the flexor biceps brachii contraction, which causes the forearm to rotate about the elbow joint. This can be a model of either overhand or underhand throwing. Although it is very simple, this model has several complexities and we will analyze each one. Our goal is to see how fast we can throw a ball, specifically a baseball. We will first assume that the biceps brachii are the only muscles used to flex the forearms; later we will examine other muscles that can contribute to form an "effective biceps" muscle.

We have analyzed the torque in this system already, in Chap. 2. As seen in Fig. 2.9d (Case 3)

$$\begin{aligned}\sum \tau_z &= Md_{\mathrm{M}} \sin\theta - W_{\mathrm{F}} d_{\mathrm{F}} \sin\theta - W_{\mathrm{B}} d_{\mathrm{B}} \sin\theta \\ &= (Md_{\mathrm{M}} - W_{\mathrm{F}} d_{\mathrm{F}} - W_{\mathrm{B}} d_{\mathrm{B}}) \sin\theta, \end{aligned} \tag{3.67}$$

with the weight in the hand now called a ball (B). For simplicity, we first assume that the humerus is vertical. In a static situation the torques add up to zero. Here they do not. For our desired arm motion, the sum must be positive.

Figure 3.45 explains a small subtlety concerning the defined angle θ, which is the angle the moving forearm makes with the fixed humerus (upper arm). The forearm makes an angle θ' with the fixed x-axis. In relating the net torques to the second derivative of the angle, as in (3.67), the dynamical variable is actually what we have called θ' here, so

$$\sum \tau_z = I \frac{\mathrm{d}^2\theta'}{\mathrm{d}t^2}. \tag{3.68}$$

Because $\theta + \theta' = \pi/2$ (=90°), a fixed quantity, we see that $\mathrm{d}\theta/\mathrm{d}t + \mathrm{d}\theta'/\mathrm{d}t = 0$ and $\mathrm{d}^2\theta/\mathrm{d}t^2 + \mathrm{d}^2\theta'/\mathrm{d}t^2 = 0$, so

$$\frac{\mathrm{d}^2\theta'}{\mathrm{d}t^2} = -\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} \tag{3.69}$$

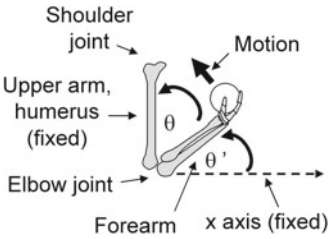


Fig. 3.45 Angles involved in throwing a ball, include the angles of the rotating forearm relative to the x-axis (θ') and the fixed upper arm (θ)

and from (3.25)

$$\sum \tau_z = -I \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} \tag{3.70}$$

or

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\frac{\sum \tau_z}{I}. \tag{3.71}$$

Now we need to evaluate the moment of inertia I and the torques. The moment of inertia about the elbow pivot equals the sum of the components due to the arm and the ball. Using (3.23) and (3.30) this is

$$I = \frac{1}{3} m_\mathrm{F} L^2 + m_\mathrm{B} L^2, \tag{3.72}$$

where L is the length of the forearm ($0.146\,H$) plus half the hand ($0.108\,H/2$), which is $L \sim 0.2\,H \sim 36$ cm. The weight of the forearm (with the whole hand) $\sim 0.022\,W_\mathrm{b}$, so $m_\mathrm{F} \sim 2$ kg for a 90 kg player. (We are modeling a 6 ft tall (1.8 m), 200 lb (90 kg) major leaguer.) A regulation major league baseball must weigh between 5 and $5\frac{1}{4}$ oz (Official Baseball Rules (Major League Baseball), Rule 1.09), which corresponds to a mass of 0.146 kg. This gives $I = 1{,}053$ kg-cm^2. (These are mixed MKS/CGS units. This is often not the best practice, but is fine if we are careful.)

Evaluating the components of torque due to the forearm and ball is straightforward. Using $W = mg$,

$$W_\mathrm{F} d_\mathrm{F} = (2\text{ kg})(9.8\text{ m/s}^2)(18\text{ cm}) = 353\text{ N-cm} \tag{3.73}$$

$$W_\mathrm{B} d_\mathrm{B} = (0.146\text{ kg})(9.8\text{ m/s}^2)(36\text{ cm}) = 51.5\text{ N-cm}, \tag{3.74}$$

for a total of 404 N-cm.

In evaluating the torque due to the muscle, we will initially make some assumptions. (1) We assume that the distance from the point of biceps brachii insertion to the pivot axis, d_M, is 4 cm, independent of the arm angle θ. Figure 3.46 shows that $d_\mathrm{M} = d_\mathrm{M}(\theta)$ and so including this variation would change our final answer a bit, but it will not change our overall conclusions. (2) We assume a fixed value of the maximum muscle force M, to see how fast we can throw a ball. As we will see in Chap. 5, this is an assumption because (a) M depends on muscle length (l_M), which depends

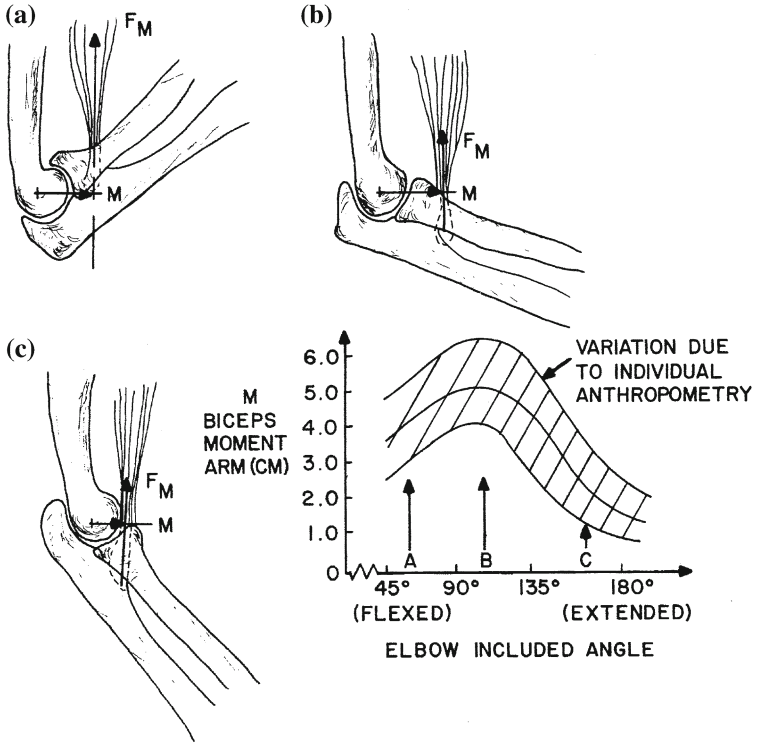


Fig. 3.46 Variation of the moment arm of biceps brachii versus elbow angle (from [23]. Reprinted with permission of Wiley)

on θ, and (b) M depends on the muscle contraction speed $v_M = dl_M/dt$, which depends on $d\theta/dt$. We will revisit several of these assumptions in Problems 5.25 and 5.26. For now we use the fact that most muscles exert a maximum force per unit cross-sectional area, which we will initially take as $\sim$20 N/cm^2, a relatively modest value.

For biceps brachii with a 2 in diameter, the cross-sectional area is $\pi(1\,\text{in})^2 = 20.3\,\text{cm}^2$, so $M = 405\,\text{N}$ and $Md_M = 1{,}620\,\text{N-cm}$. So (3.67) becomes

$$\sum \tau_z = (1{,}620\,\text{N-cm} - 404\,\text{N-cm})\sin\theta = (1{,}216\,\text{N-cm})\sin\theta \tag{3.75}$$

and

$$\frac{d^2\theta}{dt^2} = -\frac{\sum \tau_z}{I} = -\frac{(1{,}216\,\text{N-cm})\sin\theta}{1{,}053\,\text{kg-cm}^2}. \tag{3.76}$$

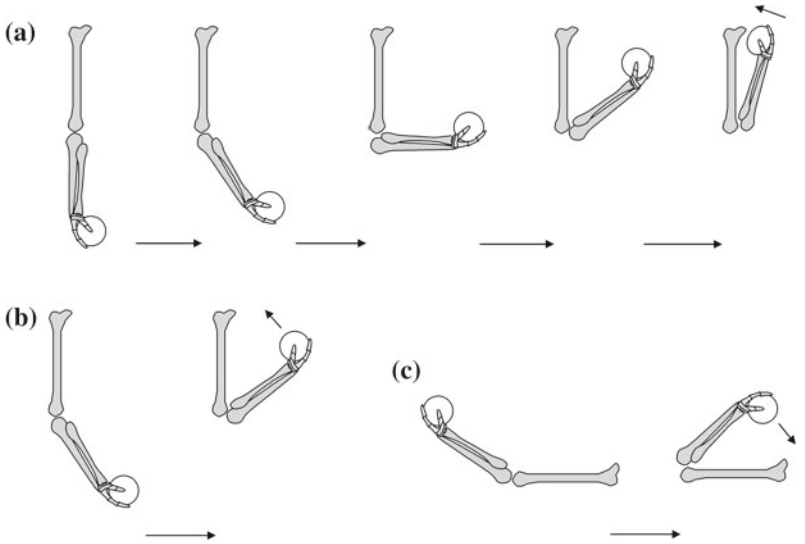


Fig. 3.47 Throwing models with **a** 180° rotation, **b** 90° rotation for vertical upper arm, and **c** 90° rotation for horizontal upper arm. The direction of the velocity vector of the released ball is shown in the last segment of each. Clearly, in this simple model a different position of the upper arm is needed to release the ball horizontally

Because $1\,\mathrm{N} = 1\,\mathrm{kg\text{-}m/s^2} = 100\,\mathrm{kg\text{-}cm/s^2}$,

$$\frac{\mathrm{d}^2\theta(t)}{\mathrm{d}t^2} = -116/\mathrm{s}^2 \sin\theta(t), \tag{3.77}$$

where we have expressed $\theta(t)$ as an explicit a function of time.

We want to solve this equation to find the speed of the baseball at the end of the throwing motion. The speed of the ball at any time t is $v_\mathrm{B} = |L\mathrm{d}\theta(t)/\mathrm{d}t|$. Before examining the end of the motion, let us examine the conditions at the beginning of the throw, the so-called initial conditions. From the diagram in Fig. 3.47a, we see that at the beginning, $t = 0$, the arm is straight, so $\theta(t = 0) = \pi(=180°)$. Because the ball is still, $\mathrm{d}\theta(t = 0)/\mathrm{d}t = 0$. At the end of the throwing motion the ball is released, say at t_final. This could occur at $\theta = 0$, so $\theta(t_\mathrm{final}) = 0$. In a normal throwing motion the release occurs before $\theta = 0$. In any case, if the release time is t_final, the ball leaves the hand with a speed $v_\mathrm{B} = |L\mathrm{d}\theta(t_\mathrm{final})/\mathrm{d}t|$.

We will restrict the range of motion for θ from $3\pi/4$ to $\pi/4$ (135° to 45°) (Fig. 3.47b) because this range involves the most efficient region for the muscles. So we see that

$$\theta(t = 0) = 3\pi/4 \qquad \mathrm{d}\theta(t = 0)/\mathrm{d}t = 0 \tag{3.78}$$

and

$$\theta(t = t_{\text{final}}) = \pi/4 \qquad \mathrm{d}\theta(t = t_{\text{final}})/\mathrm{d}t = ?, \tag{3.79}$$

where the last quantity will give us the final answer.

We can numerically integrate (3.77) to find the final speed; however, we will make some further approximations that will simplify the solution. We approximate $\sin\theta(t)$ by a constant. Over the restricted angle range of interest, we replace $\sin\theta(t)$ by $\langle\sin\theta\rangle = 0.707$ ($= \sin\pi/4 = \sin 3\pi/4$). This will lead to a slight underestimate of the speed because $\sin\theta(t)$ varies from this value to 1.0. Now (3.77) becomes

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -116/\mathrm{s}^2\langle\sin\theta\rangle = -81.7/\mathrm{s}^2 = -\alpha. \tag{3.80}$$

This is easily solved (see Appendix C for more information about the solution). After integrating both sides from time 0 to t

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\alpha t + c_1. \tag{3.81}$$

c_1 is determined from the initial condition that $\mathrm{d}\theta(t = 0)/\mathrm{d}t = 0$, which gives $c_1 = 0$ and

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\alpha t. \tag{3.82}$$

This equation is integrated again to give

$$\theta(t) = -\frac{1}{2}\alpha t^2 + c_2. \tag{3.83}$$

Because $\theta(t = 0) = 3\pi/4$, we find $c_2 = 3\pi/4$ and $\theta(t) = -\alpha t^2/2 + 3\pi/4$. At t_{final}, $\theta(t = t_{\text{final}}) = \pi/4$, so

$$\pi/4 = -\frac{1}{2}\alpha t_{\text{final}}^2 + \frac{3\pi}{4}, \tag{3.84}$$

which gives $t_{\text{final}} = (\pi/\alpha)^{1/2}$. Using (3.82), $|\mathrm{d}\theta(t = t_{\text{final}})/\mathrm{d}t| = |-\alpha t_{\text{final}}| = (\pi\alpha)^{1/2}$. The speed of the ball when it is released by the hand is

$$v_{\text{B}} = \left| L\frac{\mathrm{d}\theta(t = t_{\text{final}})}{\mathrm{d}t} \right| = L\sqrt{\pi\alpha}, \tag{3.85}$$

with arm angular speed $|\mathrm{d}\theta(t_{\text{final}})/\mathrm{d}t| = \sqrt{\pi\alpha}$. For $L = 36\,\text{cm}$ this is $v_{\text{B}} = 63.8(\alpha)^{1/2}$, in units of cm/s with α in units of $1/\text{s}^2$.

Now we can evaluate several specific cases:

Case a: For these 2 in diameter biceps brachii with $\langle\sin\theta\rangle = 0.707$, we see that $\alpha = 81.7/\text{s}^2$, $t_{\text{final}} = 0.2\,\text{s}$ and $v_{\text{B}} = 577\,\text{cm/s} = 5.77\,\text{m/s} = 18.9\,\text{ft/s} = 12.9\,\text{mph}$.

(Remember, 60 mph = 88 ft/s = 26.8 m/s = 96.6 km/h.) Because power major league baseball pitchers can throw a fastball at speeds approaching (or reaching) 100 mph, our calculated speed is very slow. Maybe we should expect a more modest speed of say ∼70 mph, but our result is still much too slow. Let us reconsider our assumptions one by one.

Case b: Let us choose a larger value for $\langle \sin\theta \rangle$ say 1.0 (because of the smaller range of θ), so now we see that $\alpha = 115.5/\text{s}^2$ and $t_{\text{final}} = 0.165\,\text{s}$. Now $v_{\text{B}} = 6.86\,\text{m/s} = 22.5\,\text{ft/s} = 15.3\,\text{mph}$. This is still too slow.

Case c: Now let us make our muscles bigger, so they have a 3 in diameter. Because the maximum muscle force is proportional to area, the torque provided by the muscles is now 1,620 N-cm $\sin\theta \times 9/4 = 3{,}645$ N-cm $\sin\theta$. The total torque is (3,645−404 N-cm) $\sin\theta = (3{,}241$ N-cm) $\sin\theta$. Still using $\langle \sin\theta \rangle = 1.0$ now, we find $\alpha = 308/\text{s}^2$ and $t_{\text{final}} = 0.10\,\text{s}$. Now $v_{\text{B}} = 11.2\,\text{m/s} = 36.7\,\text{ft/s} = 25.0\,\text{mph}$. This is getting better, but the speed is still too slow.

Case d: The last thing we will do right now is to remove the effect of gravity. This amounts to ignoring the mass of the arm and ball. We have really been examining the situation in the diagram like Fig. 3.47b. If we were to throw it as in Fig. 3.47c, then gravity would slow the motion in the first half of the operation and would increase it in the second half, so to first-order let us ignore it. This means that we ignore the -404 N-cm $\sin\theta$ torque term in Case (c). Now $\alpha = 346/\text{s}^2$, with arm angular speed 33.0 rad/s = 1,890 °/s, and we see that $t_{\text{final}} = 0.094\,\text{s}$ and $v_{\text{B}} = 11.9\,\text{m/s} = 38.9\,\text{ft/s} = 26.5\,\text{mph}$. This does not help much.

Something is clearly wrong. What? Let us examine some of our remaining assumptions:

1. Is the muscle force per unit area really ∼20 N/cm^2? No. The maximum force per unit area is closer to 30–40 N/cm^2. However, since the maximum occurs only for part of the throwing cycle (Fig. 5.19), this assumption is not that bad. Using 30 N/cm^2, would increase the speed by $(1.5)^{1/2}$ giving ≃32.5 mph for Case (d).
2. The muscle tension depends on both the muscle length and contraction speed. We will address limitations of a model that ignores contraction speed in Chap. 5 (in Fig. 5.29, and Problems 5.25 and 5.26). In any case, including the contraction speed would only decrease the speed of the ball.
3. The biceps brachii are not the only muscles used to flex the elbow (Fig. 2.10, Case 4). Inclusion of the other muscles increases the torque by a factor of 2.6, which should increase the speed by $(2.6)^{1/2}$ to 52 mph. This is likely an overestimate given that a physiological 3 in diameter for the biceps brachii (only) is an overestimate. Let us estimate that this brings the speed up to 45 mph.
4. The underhand throwing assumption with elbow flexing does not limit us within the confines of the model. Still, in overhand throwing the elbow is extended by the triceps brachii. For similar musculature and moments, elbow extension should still give the same upper throwing speed.
5. We assumed a simple one-segment model with motion of the elbow joint. We have ignored (a) the motion of the whole arm, with both the elbow and shoulder joints moving (two-segment model, Fig. 3.44a) and possibly also the wrist and (b) the use



Fig. 3.48 Sandy Koufax pitching in Game 1 of the 1963 World Series in Yankee Stadium on October 2, 1963. He and the Los Angeles Dodgers won Games 1 and 4, by the scores of 5-2 and 2-1, in a four-game sweep of the New York Yankees. A home run by Mickey Mantle was the sole run he gave up in his second complete-game victory over Whitey Ford (Photo reprinted with the permission of the Los Angeles Dodgers)

of back and leg muscles for propulsion (like a spring) in throwing (multisegment model). This is clearly seen for Sandy Koufax pitching in Fig. 3.48, and is why baseball pitchers do a lot of running and have very strong legs, especially power pitchers.

This last assumption is the real reason for the slower-than-expected calculated speed. The physics is right. Our answer is the final speed for throwing a baseball in the manner described. The model is just much too simple to describe a real throwing motion. The effective muscle cross-section area is really much larger because many more muscles are involved and the moment arms of these additional torques are large.

Sandy Koufax has said, "Everybody who performs an athletic event of any kind is a system of levers. …" He continued his description of the most critical point of the pitcher's windup, "The front leg is charged with stopping the torso. When the torso stops, the arm catches. … So the arm now develops more speed than it had when it was simply moving with the body." He continued his description of the pitching motion, "It's a two-armed catapult. You try to get the front half out as far as you can. All power pitchers do that. When the front leg stops, the upper body catches, the arm straightens, and you see a straight line. Basically you try to work as much as you can with leverage and weight and energy transfer. …" [81].

A more general multisegment model is obviously needed. Motion of six joints is possible: ankle, knee, hip, shoulder, elbow, and wrist. If each leads to the same forward motion of the ball at the end of the pitch (they do), the speed from each can be added to get the final speed of the ball. The ankle, knee, and hip all lead to forward propulsion, such as the first step of a sprint. Running 100 yd in 10 s translates to 30 fps or ≈20 mph. It takes several steps to reach this steady-state speed. We estimate that the ankle, knee, and hip can add 10 mph. The time required for the upper arm to rotate about the shoulder and this rotation angle are about the same as those for the

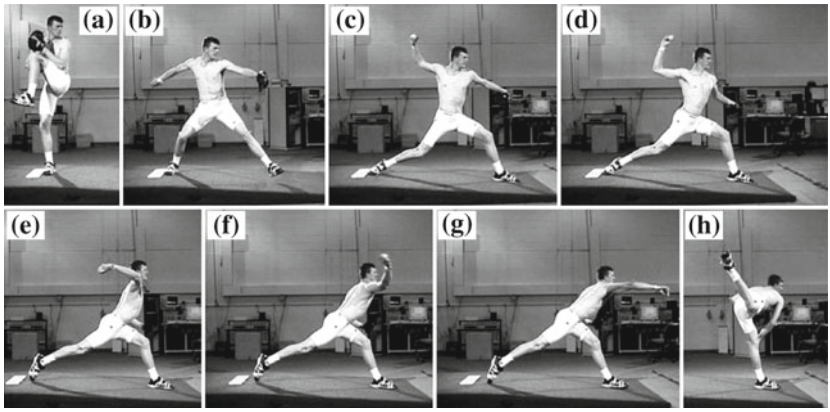


Fig. 3.49 The mechanics of the pitching motion. The six phases of baseball pitching are shown from windup (**a**) to stride (**b**), arm cocking (**c**), arm acceleration (**d**–**f**), arm deceleration (**g**), and follow-through (**h**) (this photo was provided by the American Sports Medicine Institute)

lower arm about the elbow, so the elbow and shoulder joints both contribute about 45 mph. The wrist may contribute a little to the speed (but much to ball rotation, as is described in the problems). Altogether, the final speed is ∼100 mph, which makes sense for a power pitcher.

The mechanics of the pitching motion are clearer when viewing the six phases of baseball pitching in Fig. 3.49 [109, 146]. After the windup, the pitcher is at the balance point, which we will say occurs at $t = 0$ s. The pitcher strides and makes foot contact at ∼0.50 s. The pelvis rotates transferring energy from the legs to the upper body (maximum rotation speed of ∼550–700°/s) and this causes the upper trunk to rotate (∼1,100–1,350°/s). The arm straightens as the elbow extends (∼2,200–2,800°/s) and the shoulder rotates (∼6,900–9,800°/s), and then the ball is released at ∼0.64 s, which is ∼0.14 s after foot contact. Then the front leg straightens to help decelerate the body. We see that the measured acceleration times are consistent with our simple calculations, ∼0.1 s. We should not be surprised that if we included the contribution of the even faster rotation of the shoulder to that of our calculated elbow extension, we could have predicted a ∼100 mph fastball. The torques applied to the forearm from the elbow and the upper arm from the shoulder during a pitch are shown in Figs. 3.50 and 3.51 (along with the range of variation). These values are consistent with those in our simple model.

Later this chapter we will present a more general multisegment model that can be used to improve our one-segment model of pitching and throwing. As one step toward this, we now overview three models of throwing [4], each with two joint angles as shown in Fig. 3.52, which can be used to learn how the timing of muscle activation in different joints strongly affects complex motions such as throwing. In each case there is a proximal angle and a distal angle (the one nearer the release of the projectile), which are both powered by joint muscles. Figure 3.52a can be thought

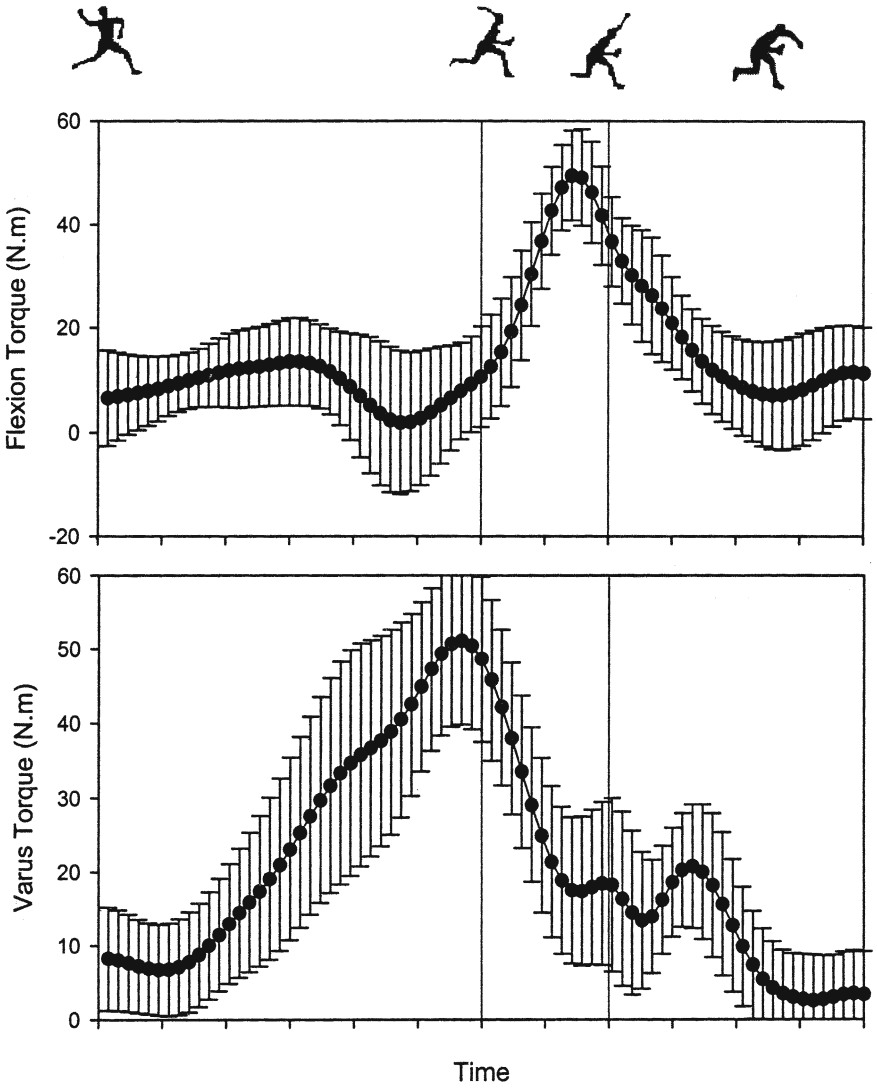


Fig. 3.50 Torques applied to the forearm at the elbow during a baseball pitch, with the time scale relative to the duration of the pitch (from [146]. With the kind permission of Springer Science and Business Media)

of as a model of shot putting, however one with no horizontal motion, with the knee and elbow being the proximal and distal joints. Figure 3.52b models underarm motion, as in bowling (but with a stationary trunk) and underarm throwing (but including only part of the actual motion in underarm pitching in softball of baseball (submarine pitching)), with the sequential activation of the shoulder and elbow joints.

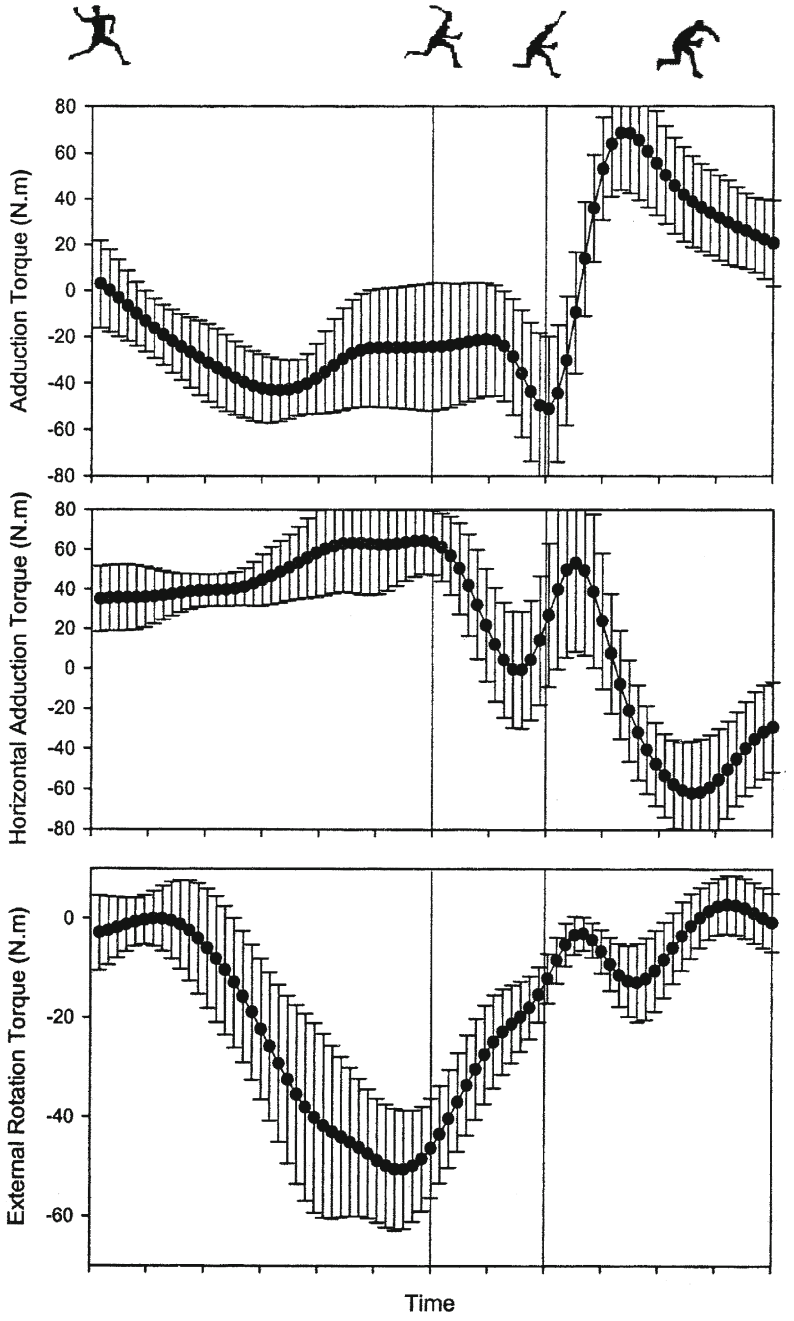


Fig. 3.51 Torques applied to the upper arm at the shoulder during a baseball pitch, with the time scale relative to the duration of the pitch (from [146]. With the kind permission of Springer Science and Business Media)

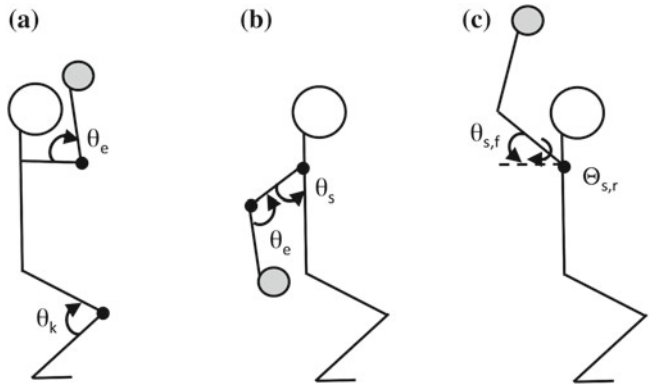


Fig. 3.52 Schematic of three models of throwing a projectile, with torques applied about two rotational degrees of freedom: about the knee and elbow joints for shot putting in (**a**), the shoulder and elbow joints for bowling and underarm throwing in (**b**), and the shoulder flexion and (humerus) rotation motions for overarm pitching and passing a football in (**c**) (based on [3])

Figure 3.52c shows a model for overarm throwing, as used in baseball pitching and passing a football, with both shoulder flexion and rotation, but with a fixed trunk and the elbow fixed at 90°. (In our one-segment model of pitching we followed only the elbow joint.) The equations that describe these segments and the joint muscles are similar to those we used in our model of high and long jumping earlier in this chapter. It is assumed that each of the two effective joint muscles is either inactive or fully activated at a give time. The proximal one is activated initially and the distal one is activated after a delay, and both remain activated until the projectile is thrown. Numerical simulation of each model [4] shows that there is a delay time between firing the proximal and distal muscles that optimizes projectile speed; this is in accord with the actual delay observed in progressively more distal muscles in throwing and similar motions. It may be surprising that for the first two models these optimal times are longer than those that maximize the total work done, but this is so because such additional delays increase the fractions of the total work that become the projectile kinetic energies.

These models help reinforce the view that throwing entails sequential activations of muscles, in order through the legs, hips, torso, shoulder, elbow and wrist, leading to rapid rotation of the last three joints [26, 40, 46, 71, 114, 121]. In our upcoming discussion of multisegment modeling we will include the bones, muscles and joints in the different moving segments and see that some forces are transmitted from one segment to the next. While muscle contractions (Chap. 5) are ultimately responsible for joint rotations, there may not be a one-to-one relation between muscle force and rotational acceleration in each joint. Such models usually do not include the possibility that the elastic energy in stretched tendons (which attach muscles to bones, Chap. 4) and ligaments (which connect bones to each other, Chap. 4) that lead to rotation about a given joint could have been stored during an earlier stage

of the motion [121]. The rotation of the humerus makes the largest contribution to the projectile speed [72]; it can be over 9,000°/s over a few milliseconds [114] and is the fastest that can occur in the body. However, the shoulder muscles themselves may not be able to provide enough power to account for all of the humerus rotational acceleration, as is addressed in Problem 3.87. In fact, perhaps ∼90% of the ∼200 J stored in shoulder ligaments and tendons in the arm during the earlier cocking phase of pitching is transmitted to the shoulder motion, which would account for ∼54% of the ∼346 J of the rotational work done in humerus rotation by quality baseball pitchers. Elastic energy recovery could also assist attaining high joint rotational speeds and power during the elbow extension phase, beyond that powered by the triceps brachii alone (Problems 5.11–5.13) [121]. Although elastic energy recovery during pitching increases the pitching speed, and this is good, these tendons and ligaments can become damaged after repeated use (Chap. 4) and so this energy storage might be one origin of pitching injuries [46].

3.6.1 Comparing Throwing Motions

The overarm throwing motion in baseball pitching is similar to the throwing and striking motions in several sports, including throwing footballs [47], javelins, and spears, long-distance (fast break) basketball passes, hitting volleyballs, and tennis and badminton serving [8, 121]. The sequence and timing of the phases are similar to those Fig. 3.49, but differ due to the details of the sport. The modeling analysis is similar; for throwing, performance and model predictions depend on the object mass and shape, and whether the thrower is stationary (aside from the usual forward propulsion of the body center of mass during the throwing motion, as in baseball and football) or is moving (due to running, as in javelin throwing).

Naturally, heavier objects cannot be thrown as fast as lighter ones (Problem 3.78) [47]. Moreover, you need to keep the projectile closer to the shoulder joint when throwing heavier objects, possibly to reduce ligament and tendon damage in the shoulder and elbow [121]. This is done by decreasing the arm abduction angle and increasing elbow flexion, thereby reducing the moment of inertia of the arm about this joint. When throwing elongated objects, such as footballs, javelins and spears, the palm of the hand is alongside the elongation direction, while when throwing spherical objects, such as baseballs, the palm is behind the ball. The forearm is more supinated in throwing the former than the latter [121].

In particular, both baseball and football throwing can be divided into the six phases (Fig. 3.49): windup, stride, arm cocking, arm acceleration, arm deceleration, and follow-through. Football quarterbacks rotate their arms ∼9° less than baseball pitchers [47], release the ball when they are more erect, and take shorter strides than pitchers. Aside from differences in the ball, it is hardly surprising that the pitcher and quarterback windup and follow-through phases are quite different because the quarterback does not have the luxury of a windup with a knee lift or an extended follow-through due to onrushing massive and quick defensive players. Major-league

pitchers can throw up to ∼90–100 mph, while strong-armed quarterbacks can throw footballs only as fast as ∼50–60 mph, because the ball is heavier (0.42 vs. 0.14 kg) so elbow rotation is slower (see Problem 3.78). The ball needs to be grasped well, but differently, in the two sports; baseball pitchers use different finger arrangements for different types of pitches, while quarterbacks need to clutch the larger and somewhat compressible football with more force. (Some quarterbacks seem to prefer throwing footballs that are relatively less inflated. The physics of deflated footballs is addressed in Problems 3.88–3.90.) Pitchers seem to develop more frequent arm injuries, in part because they develop greater compressive forces at the elbow and shoulder and adduction torques at the shoulder during arm deceleration, and their greater frequency of throwing (a starting pitching may throw ∼120 pitches competitively every five days and a quarterback ∼50 passes every seventh day) [47]. Of course, collisions between quarterbacks and 350 lb defensive lineman and lighter but even faster linebackers frequently cause other very serious injuries for quarterbacks, which we will discuss in the discussion of collisions of the human body.

3.6.2 Power Generated During a Throw

How much power is exerted during a throw, even the quite slow throw using elbow motion only? In linear motion with a constant force exerted F, the work done in moving the amount a distance Δx is

$$W = F\Delta x \tag{3.86}$$

and the power needed to do this in a time Δt is

$$P_{\text{power}} = Fv, \tag{3.87}$$

where the linear speed $v = \Delta x/\Delta t$.

For rotational motion the analogous expressions are

$$W = \tau\Delta\theta \tag{3.88}$$

$$P_{\text{power}} = \tau\Omega, \tag{3.89}$$

for an angular motion of $\Delta\theta$ in a time Δt with rotational speed $\Omega = \Delta\theta/\Delta t$. This is consistent with the linear results, as is seen by examining the power needed to exert a torque in a circular motion with a radius r. Because $\tau = rF$ and $v = \Omega r$,

$$P_{\text{power}} = \tau\Omega = (rF)\Omega = (\Omega r)F = Fv. \tag{3.90}$$

For multijoint (multisegment) models, the net torque and angular speed about each joint need to be considered and so

$$P_{\text{power}} = \sum_k \tau_k \Omega_k. \tag{3.91}$$

The torque generated in the case with the 3 in diameter muscles is 3,645 N-cm (Case c, 25.0 mph), and it has been assumed to be constant during the throw. The average angular speed is $\Omega_{\text{av}} = \Delta\theta/\Delta t$ where $\Delta\theta = 3\pi/4 - \pi/4 = \pi/2$ and $\Delta t = t_{\text{final}} = 0.10$ s, so $\Omega_{\text{av}} = 15.7$ rad/s. The angular speed increases linearly with time during this motion (from (3.82)) from 0 at the beginning to the peak value at the end $\Omega_{\text{peak}} = 31.4$ rad/s.

Using (3.89), the average power generated is:

$$P_{\text{power, av}} = (3{,}645\,\text{N-cm})(15.7\,\text{rad/s}) = 57{,}000\,\text{N-cm/s} = 570\,\text{W} = 0.76\,\text{hp}. \tag{3.92}$$

We have made use of 1 W=1 N-m/s and 1 horsepower (hp) = 746 W. The peak power

$$P_{\text{power peak}} = (3{,}645\,\text{N-cm})(31.4\,\text{rad/s}) = 1{,}140\,\text{W} = 1.5\,\text{hp}, \tag{3.93}$$

which seems large, even for this slow throw. (For the 2 in diameter biceps, the average and peak powers are 178 and 356 W, respectively.)

3.7 Batting and Swinging Motions

When we bat in baseball and swing rackets our bodies undergo complex multisegment motions. The motion in serving in tennis is very similar to that in overhand throwing (Fig. 3.49). There are similarities and significant differences between overhand throwing and many other motions, such as swinging at a golf ball, hitting a ping pong (or table tennis) ball, taking a slap shot in ice hockey, swinging a bowling ball, and hitting a pool (or billiard) ball with a cue. All of these motions (and throwing) need to be analyzed by multisegment modeling. We will examine some simple models here and more complex models later in this chapter.

Several swinging motions, such as swinging a golf club, can be modeled as a double, hinged pendulum, as in Fig. 3.53 [36, 140]. In the first pendulum, from A to B, the hands and wrist undergo circular motion about a fixed (really, an approximately fixed) point near the shoulders. In the second pendulum, from B to C, the golf club and head undergo circular motion about the pivot of this pendulum, which are the hands and wrists undergoing motion in the first pendulum and which serve as a hinge at B. In the golf drive, the body generates a large torque about the pivot in this first pendulum, and later the wrists provide a smaller torque about the pivot of the second pendulum. If the golfer were to apply no torque at the hands and wrists and let them serve as a free hinge, the club head would fall due to gravity after the initial

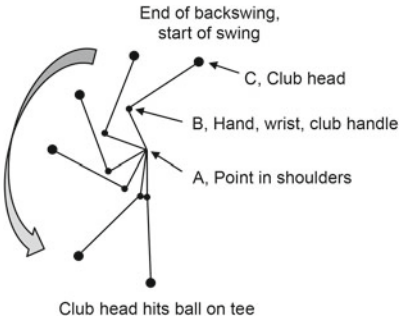


Fig. 3.53 Double pendulum model of a golf swing (based on [36])

backswing (decreasing the angle between AB and BC of this second pendulum) at the beginning of the swing. As the first pendulum begins to swing, the club head tries to move in a straight line (increasing this angle) and exerts a centrifugal pull on the hinge due to its circular motion about its pivot (the hinge), as is analyzed in Problem 3.178. This slows the first pendulum, and lets the second pendulum catch up and the arms and golf club then tend to line up; the speed of the club (lower) pendulum is very important! The motion is qualitatively the same even when the golfer applies a torque to the second pendulum, especially way before impact. The solution to the equations of motion for this hinged pendulum is beyond the level of this book and can be found in [36].

In each pendulum, joints act as hinges through which energy is efficiently transferred due to motion [36, 87]. If the wrists were fixed during the golf swing, so the arms and club could be viewed as a single pendulum, only about one quarter of the energy of the swing would be fed to the golf clubhead. Because of wrist cocking and uncocking during the swing, about two thirds can be transferred from the heavier upper (arm) pendulum through the hinge at B to the clubhead at C in the lower pendulum. This transfer from the upper to lower pendulum can occur even without torque applied at the wrists. The club can be accelerated further when torque is applied in phase to add to, and not subtract from, the transfer from the upper to lower pendulum. These considerations apply to motions in hockey, baseball [27], and tennis [34] and other racket games and the use of any club, bat or racket to transfer energy through the wrist hinge to the swung object. The knee acts as such a hinge to transfer energy from the leg muscles to the foot in kicking balls, as in football kickoffs and punts and in soccer. The elbow acts as such a hinge in throwing and in bowling, when ball speed can be increased by allowing the arm to straighten just before releasing the ball.

What fraction of our muscle mass goes into swinging a golf club? About 1.3 kW is needed to accelerate the club head, ∼0.5 kW for the club shaft during a golf drive, and ∼0.7 kW for the hands and arms of the player, for a total of ∼2.5 kW. (The metabolic rate need to sustain this activity is ∼18× this value.) This 2.5 kW power is almost as much as an Olympic sprinter develops the first sec of a sprint, greater than that developed by a weightlifter, and comparable to that developed in the high

jump [36]. We will see in Chap. 5 that ~250 W is generated per kg of active muscle, so ~10 kg of active muscle is needed to generate 2.5 kW. If these are half of the opposing, antagonist pairs of muscles, then 20 kg of muscles are involved. If a 73 kg person has 30 kg of muscle, we see that most of it must go into the swing. Since ~4/7 of human muscle, ~17 kg, is below the waist and ~3/7 above the waist, ~13 kg, most of the power in golf and other swings must come from the massive leg and thigh muscles [36].

3.8 Other Types of Motion

References [5, 66, 110] describe many other types of motion. One such motion is bicycling. Figures 3.53, 3.54, 3.55, and 3.56 show the joint angles during a cycling crank cycle, along with changes in muscle-tendon length and positive (contracting muscles) and negative (extending muscles) work by muscles during this cycle [66].

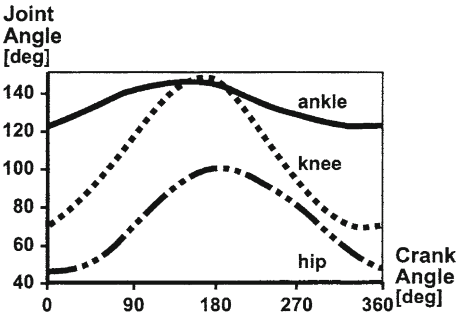


Fig. 3.54 Joint angles in the right leg during a crank cycle in bicycling. The hip angle is that between the pelvis and the upper leg (thigh) (180° with full extension), the knee angle is that between the lower leg (shank, calf) and upper leg (thigh) (180° with full extension), and the ankle angle is that between the lower leg and a line from the ankle joint axis to the pedal spindle axis (from [65], as from [76]. Used with permission)

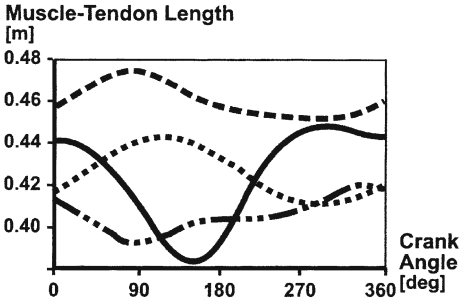


Fig. 3.55 Changes in the muscle/tendon lengths in biarticulate muscles (that span two joints) during a crank cycle in bicycling, for the rectus femoris (*dashes*), biceps femoris (*dots*), semimembranosus (*solid line*), and gastrocnemius (*dashes* and *dots*) (from [65], as from [76]. Used with permission)

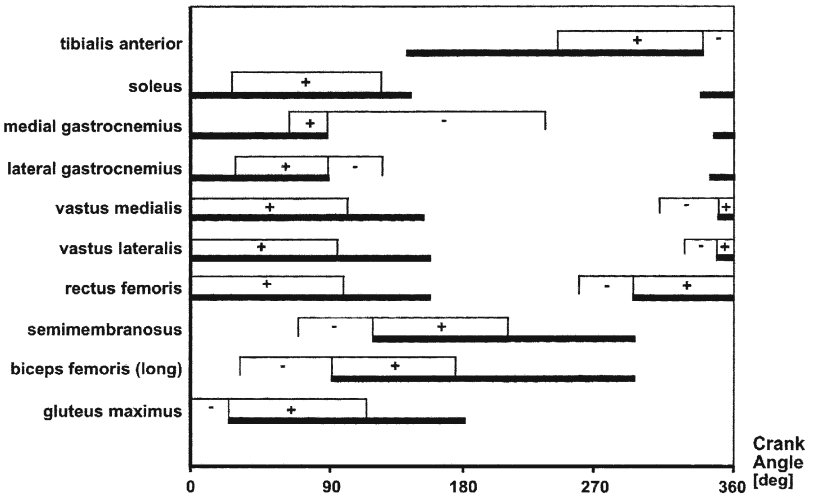


Fig. 3.56 Crank angles corresponding to positive work (contracting muscles) and negative work (extending muscles) for ten muscles during one crank cycle in bicycling (from [65], as from [76]. Used with permission)

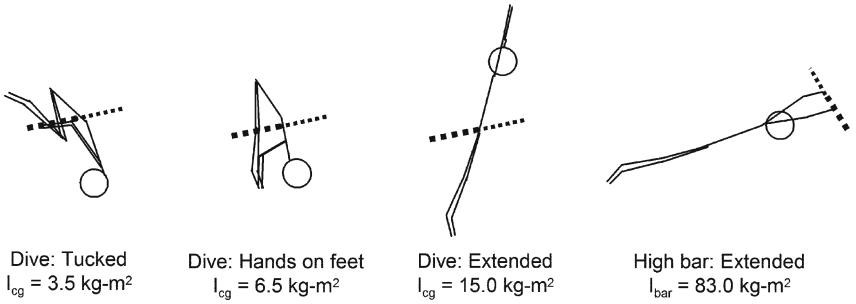


Fig. 3.57 Moments of inertia during diving positions (about the respective rotation axes) and in the high bar gymnastic event (about the bar) (based on [66])

We will analyze swimming and potential human flight in Chap. 7 when we discuss drag and lift in fluids.

Angular kinematics and kinetics of the entire body are very important in several sports, such as several gymnastics events, acrobatics, diving, and ice skating (see [66]). (In several activities, such as in throwing a ball, the angular motion of a few body segments dominate.) Figure 3.57 shows the moment of inertia about the rotation axis in diving and high-bar events. You can also determine the moment of inertia about the three orthogonal axes, with origin at the center of mass: the *somersault* (from side-to-side, the y axis in Fig. 1.1), *cartwheel* (front-to-back, x axis), and *twist* (longitudinal, z axis) axes (Table 3.9). You can clearly see that you can decrease your moment of inertia greatly during a dive and concomitantly increase your rotation

Table 3.9 Moments of inertia of the body during diving about three perpendicular axes, in kg-m^2

Position	Somersault	Cartwheel	Twist
Layout	12.55	15.09	3.83
Open pike (arms out to side)	8.38	8.98	4.79
Closed pike (fingers touching toes)	8.65	6.60	3.58
Tuck	4.07	4.42	2.97

Using data from [43, 97]

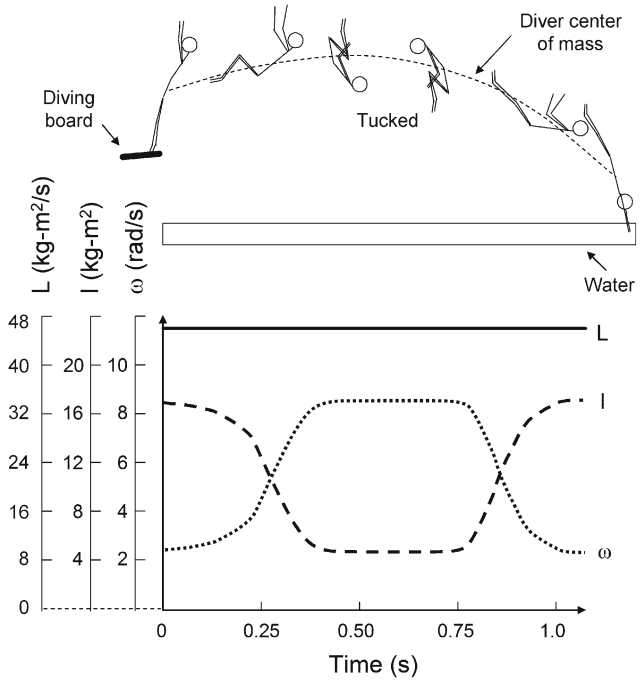


Fig. 3.58 Interplay between angular momentum, moment of inertia, and angular velocity during a tucked backward one-and-one half dive. In this backward dive the body is initially straight, curls up with arms around the tucked and bent legs by the knees, and becomes straight again before impact. The diver gains enough vertical momentum from the springboard to have enough time to complete this maneuver before impact. A sketch of the motion of the diver's center of mass is also shown (Based on [66])

rate, as seen in Fig. 3.58, because the angular moment is constant when no external torques are applied. This control of rotation rate is also important in ice skating, and is analyzed in Problem 3.60. During an athletic maneuver in flight, total angular momentum of the body is constant but the angular momentum of different body segments can change, as is seen by the transfer of the angular momentum from the upper body to the legs during a piked front dive in Fig. 3.59. By appropriate motion of your arm, you replace some of this somersaulting angular to twisting angular during

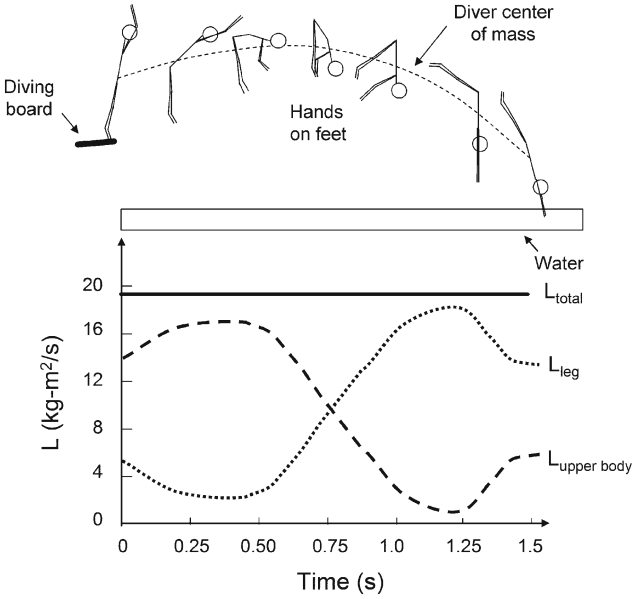


Fig. 3.59 Interplay between angular momentum localized first in the upper body and then in the legs during a piked front dive. In this forward dive the body is initially straight, bends over at the waist with the hands holding on to the ankles of the straight legs, and then becomes straight again before impact. The diver gains sufficient vertical momentum from the springboard to have enough time to complete this maneuver before impact. A sketch of the motion of the diver's center of mass is also shown (based on [66])

a dive. For example, if your arms are initially extended above your head during a front dive and you swing one arm sideways bringing it alongside your body, your body will rotate in the opposite direction to conserve total angular momentum, in a side-somersaulting rotation.

3.9 Multisegment Modeling (Advanced Topic)

More sophisticated modeling of throwing and most motions requires multisegment modeling, which we first visited in the discussion of statics in Chap. 2. Our models of jumping and throwing involved one or two moving segments. They provided insight but no quantitative accuracy. Including all the segments and joints involved leads to a more accurate, but complex model that requires numerical solution, which is beyond our scope. We now look at how to set up a general multisegment model by considering the two-segment model.

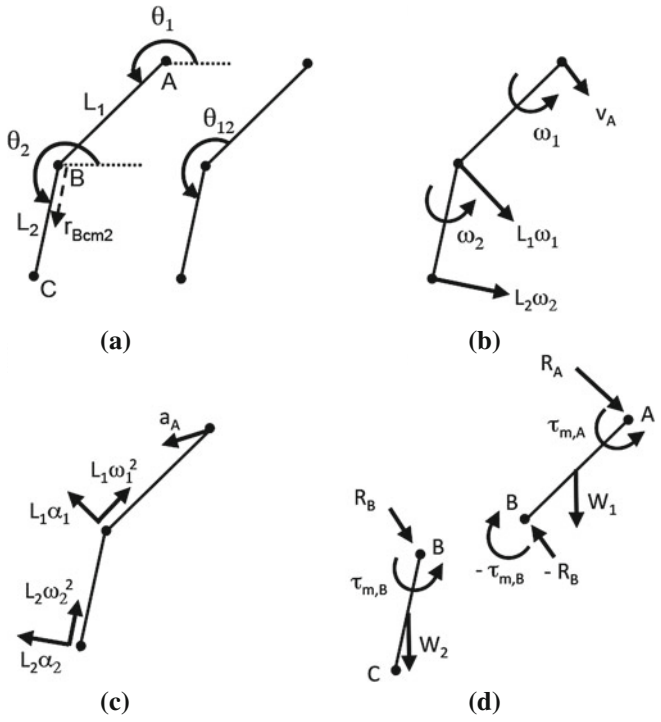


Fig. 3.60 Schematic of two-segment modeling, showing **a** distances and angles, **b** angular speeds about joints and linear speeds of joints, **c** accelerations, and **d** the weight forces, reaction forces at the joints, and muscle torques for both segments (based on [43])

Figure 3.60 shows a two-segment model with proximal segment 1 of length L_1 and mass m_1 rotating about joint A and distal segment 2 of length L_2 and mass m_2 rotating about joint B. We are interested in tracking the distal end of segment 2 at position C. In throwing a ball, this could represent the upper arm rotating about the shoulder joint and the forearm or lower arm rotating about the elbow, with position C being the hand and ball, with the wrist fixed. In kicking a ball, it could represent the upper leg rotating about the hip joint and the lower leg rotating about the knee, with position C being the foot and ball, with the ankle fixed. These models can be upgraded to three-segment models by including a hand or foot that can rotate about the wrist or ankle [42, 43, 120].

The position of point C, $\mathbf{r}_C$, depends on the position of point A, $\mathbf{r}_A$, and the evolution of the vectors describing each segment from its proximal to distal end, $\mathbf{r}_{AB}$ and $\mathbf{r}_{BC}$:

$$\mathbf{r}_C = \mathbf{r}_A + \mathbf{r}_{AB} + \mathbf{r}_{BC}. \tag{3.94}$$

The linear velocity of point C, $\mathbf{v}_C$, gives the speed of the ball in the nonrotating hand in throwing and the nonrotating foot in kicking. It depends on the linear velocity

of point A, $\mathbf{v}_A$, which gives the speed of the shoulder in throwing and the hip in kicking; the rotational speed of segment 1 about joint A, ω_1; and the rotational speed of segment 2 about joint B, ω_2, so:

$$\mathbf{v}_C = \mathbf{v}_A + (\omega_1 \times \mathbf{r}_{AB}) + (\omega_2 \times \mathbf{r}_{BC}). \tag{3.95}$$

Both rotations are in the tangential direction, as seen in Fig. 3.60b. The linear acceleration of point C, $\mathbf{a}_C$, depends on the linear acceleration of point A, $\mathbf{a}_A$, and the rotational speed and acceleration, $\boldsymbol{\alpha}_1$, of segment 1 about joint A and the speed and acceleration, $\boldsymbol{\alpha}_2$, of segment 2 about joint B:

$$\mathbf{a}_C = \mathbf{a}_A + (\alpha_1 \times \mathbf{r}_{AB}) + (\omega_1 \times \omega_1 \times \mathbf{r}_{AB}) + (\alpha_2 \times \mathbf{r}_{BC}) + (\omega_2 \times \omega_2 \times \mathbf{r}_{BC}), \tag{3.96}$$

with the ω^2 terms pointing radially inward and the α terms pointing tangentially (Fig. 3.60c). (This α is different from the parameters used in our discussion of throwing.)

We now set up the dynamical equations describing the forces and torques acting on the centers of mass of each segment, treating the more distal segment first. The equation of motion for the linear acceleration of the center of mass of segment 2 is

$$\mathbf{R}_{2B1} + \mathbf{W}_2 = m_2 \mathbf{a}_2, \tag{3.97}$$

where $\mathbf{R}_{2B1}$ is the reaction force on segment 2 at B due to segment 1, W_2 is the weight of segment 2 (m_2g), and $\mathbf{a}_2$ is the linear acceleration of the center of mass of segment 2. It is

$$\mathbf{a}_2 = \mathbf{a}_A + (\alpha_1 \times \mathbf{r}_{AB}) + (\omega_1 \times \omega_1 \times \mathbf{r}_{AB}) + (\alpha_2 \times \mathbf{r}_{Bcm2}) + (\omega_2 \times \omega_2 \times \mathbf{r}_{Bcm2}), \tag{3.98}$$

which is (3.96), with $\mathbf{r}_{BC}$ replaced by the vector from point B to the center of mass of segment 2, which is $\mathbf{r}_{Bcm2}$ with length r_{Bcm2} (Fig. 3.60a).

The equation of motion for the rotational acceleration of segment 2 about its center of mass is

$$\tau_{m,B} + (\mathbf{r}_{cm2B} \times \mathbf{R}_{2B1}) = I_{2cm2}, \tag{3.99}$$

where $\boldsymbol{\tau}_{m,B}$ is the torque about joint B due to muscles, $\mathbf{r}_{cm2B}$ is the vector from the center of mass of segment to joint B ($= -\mathbf{r}_{Bcm2}$), and I_{2cm} is the moment of inertia of segment 2 about its center of mass. The joint reaction force $\mathbf{R}_{2B1}$ from (3.97) can now be inserted into (3.98) to give the dynamical equation for the rotation of segment 2 about joint B. In two dimensions in scalar variables this not so simply reduces to

$$\tau_{m,B} = I_{2p}\alpha_{21} + (I_{2p} + m_2 L_1 r_{Bcm2} \cos\theta_{21})\alpha_1 + (m_2 L_1 r_{Bcm2} \sin\theta_{21})\omega_1^2, \tag{3.100}$$

where I_{2p} is the moment of inertia of segment 2 about joint B, $\theta_{21} = \theta_2 - \theta_1$, which corresponds to the elbow angle for throwing and the knee angle for kicking, and $\alpha_{21} = \alpha_2 - \alpha_1$ [42, 43, 120]. For arbitrary θ_{21}, the total torque has contributions

from the reaction forces from the accelerations both along and normal to this segment (Fig. 3.60d).

The analogous equation of motion for segment 1 is [42, 43, 120]

$$\begin{aligned}\tau_{\mathrm{m,A}} =& I_{1\mathrm{p}}\alpha_1 + (I_{2\mathrm{p}} + m_2(L_1^2 + 2L_1 r_{\mathrm{Bcm2}}\cos\theta_{21}))\alpha_1 + (I_{2\mathrm{p}} + m_2 L_1 r_{\mathrm{Bcm2}}\cos\theta_{21})\alpha_{21} \\ & - (m_2 L_1 r_{\mathrm{Bcm2}}\sin\theta_{21})\omega_{21}^2 - (2m_2 L_1 r_{\mathrm{Bcm2}}\sin\theta_{21})\omega_1\omega_{21},\end{aligned} \tag{3.101}$$

where $I_{1\mathrm{p}}$ is the moment of inertia of segment 1 about joint A and $\omega_{21} = \omega_2 - \omega_1$. Obtaining this is a bit complicated.

Problems 3.81–3.83 show that under appropriate conditions (3.100) and (3.101) reduce to our torque equation for the one-segment model of throwing (3.68).

3.10 Collisions of the Human Body

So far we have examined how people begin to move or continue to move. Now we will investigate what happens when motion is stopped very fast, in a collision. We humans tend to break down when we collide with objects. We can see the reasons for this by determining how fast given body parts decelerate during such collisions. We will also use this analysis to estimate the forces on a foot upon impact during running and the return of energy during jumping. In these examples the person is directly involved in the collision, for example in the collision of a person's head with the car dashboard in a car crash or with the opponent's fist in boxing. Football is the epitome of a collision "ball" sports. A player tries to advance the football. The goal of the defensive player is to collide with the ball carrier to stop his progress. The goal of most of the offensive players, other than the ball carrier, is to collide with the defensive players to stop them from colliding with the ball carrier. In addition to collisions between people, in some instances a person causes a collision with an object, such as by hitting a baseball with a bat. In others, there is a collision within the body, as during chewing food.

3.10.1 Kinematics of a Collision

We need to learn how humans react to rapid decelerations. During a constant acceleration a in a time t, the velocity of an object changes from the initial value v_{i} to the final value

$$v_{\mathrm{f}} = v_{\mathrm{i}} + at \tag{3.102}$$

and the distance traveled by the object during that time is

$$\Delta x = v_{\mathrm{i}}t + \frac{1}{2}at^2. \tag{3.103}$$

Say a mass m is traveling at a speed v, which we will assume is positive. If it decelerates at a uniform rate to speed $v_f = 0$ in a time $t = t_{coll}$, during the collision the acceleration $a_{coll} = \Delta v/\Delta t = -v_i/t_{coll}$. The analysis will be clearer if we define the positive quantity $a_{decel} = -a_{coll} = v_i/t_{coll} > 0$. The distance the mass travels during that collision is

$$\Delta x = v_i t_{coll} + \frac{1}{2} a_{coll} t_{coll}^2 = v_i t_{coll} + \frac{1}{2}\left(\frac{-v_i}{t_{coll}}\right) t_{coll}^2 \tag{3.104}$$

$$= \frac{1}{2} v_i t_{coll} = \frac{1}{2} a_{decel} t_{coll}^2 = \frac{v_i^2}{2a_{decel}}, \tag{3.105}$$

where $v_i = a_{decel} t_{coll}$ has been used.

The magnitude of the force felt by the object is:

$$|F| = \left| m \frac{\Delta v}{\Delta t} \right| = \frac{m v_i}{t_{coll}}. \tag{3.106}$$

The force per unit area A on the collision region is

$$P = \frac{|F|}{A} = \frac{m|\Delta v/\Delta t|}{A} = \frac{m v_i}{A t_{coll}}. \tag{3.107}$$

This is also called a pressure or a stress. Such decelerations to zero speed are totally inelastic. In totally elastic or partially elastic collisions, the objects bounce back and the momentum transfer and the resulting forces are greater. The force integrated over time, called the *impulse*, equals the change in momentum $m\Delta v$.

Partially Elastic Collisions

So far we have implicitly assumed that the object is colliding with an infinitely massive body. Also we have not described the elasticity of the collision quantitatively.

In a collision, linear momentum and energy are conserved. However, if the collision is not elastic, kinetic energy is not conserved and heat is produced. Let us consider two objects with mass m_1 and m_2 that move with speeds in the x direction v_1 and v_2 before the collision and v_1' and v_2' after the collision. For motion to more positive x, $v > 0$, and to more negative x, $v < 0$. Conservation of linear momentum gives

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'. \tag{3.108}$$

The fraction of the speed returned to a body as a result of a collision is characterized by the coefficient of restitution e (or COR).

$$e = -\frac{v_1' - v_2'}{v_1 - v_2}. \tag{3.109}$$

Table 3.10 Coefficient of restitution (e) for balls

Ball	Surface	e	Speeds
"Super ball"	Hardwood floor	0.89	Slow
Golf	Floor	0.83–0.89	Slow
Tennis	Racket	0.76–0.88	Slow
Racket	Floor	0.74–0.88	Slow
Basketball	Hardwood floor	0.76	Slow
Volleyball	Hardwood floor	0.74	Slow
Soccer	Floor	0.69–0.80	Slow
Squash	Plywood	0.48–0.60	Slow
Baseball	Ash boards on concrete	0.563	58 mph
	Wood	0.588	25 mph
	Wood	0.584	18 mph
	Wood	0.46	89 mph
	Typical wood bat	0.55	Typical pitch
Softball	Hardwood floor	0.31	Slow

Using data from [1, 66, 83, 134, 137]

For a stationary massive object 2, we see that $e = v_1'/v_1$ and so it is 1 for a totally elastic collision, for which $v_1' = -v_1$, and 0 for a totally inelastic collision, for which $v_1' = 0$. Table 3.10 shows typical values for balls. The coefficient of restitution typically decreases with speed.

Inserting (3.109) into (3.108) gives

$$v_1' = \frac{(m_1 - em_2)v_1 + (m_2 + em_2)v_2}{m_1 + m_2} \tag{3.110}$$

and

$$v_2' = \frac{(m_1 - em_1)v_1 + (m_2 + em_1)v_2}{m_1 + m_2}. \tag{3.111}$$

3.10.2 Consequences of Collisions

Falls and Crashes

What are the results of a collision? We will examine two types of consequences: potentially fatal collisions due to brain trauma and the breakage of body parts [106, 131].

Let us examine the forces on a head during a collision. Consider a head of 3 kg mass moving at 1 m/s that hits a steel beam and stops in 0.01 s (=10 ms). From

(3.105), this corresponds to a distance of 5 mm. The deceleration felt by the head is $a_{\text{decel}} = (1\,\text{m/s})/(0.01\,\text{s}) = 100\,\text{m/s}^2 \simeq 10\,g$, which is very large. The force felt by the head is $ma_{\text{decel}} = 3\,\text{kg} \times 100\,\text{m/s}^2 = 300\,\text{N}$ (~70 lb) [20]. We will see the consequences of this soon. Years ago, before seat belts were standard in cars, car dashboards were made of hard metal instead of the softer foam-based materials used now. Imagine the head injuries from the rapid deceleration of heads on these dashboards during car crashes!

Now let us examine the forces in our legs when we jump from a ledge barefoot, a ledge that is only $h = 1\,\text{m}$ high. This may not seem to be a dare-devil activity, but if we land stiff-legged and on one leg, the consequences are startling. The deceleration during this type of landing occurs during the compression of the padding on the bottom of our foot.

The gravitational potential energy before the jump is mgh, which equals the kinetic energy before impact with the ground, $mv_{\text{i}}^2/2$, where v_i is the initial speed in the collision. We find

$$v_{\text{i}} = \sqrt{2\,gh} = \sqrt{2 \times 9.8\,\text{m/s}^2 \times 1\,\text{m}} \simeq \sqrt{20\,\text{m}^2/\text{s}^2} = 4.5\,\text{m/s}. \tag{3.112}$$

If the foot padding is compressed during impact by $\Delta x = 1\,\text{cm} = 10^{-2}\,\text{m}$, then $t_{\text{coll}} \sim 5\,\text{ms}$.

Using (3.104) and (3.105), the deceleration is

$$a_{\text{decel}} = \frac{v_{\text{i}}^2}{2\Delta x} = \frac{1}{2}\,\frac{20\,\text{m}^2/\text{s}^2}{10^{-2}\,\text{m}} = 10^3\,\text{m/s}^2 = 100\,g, \tag{3.113}$$

which is very high. For a $m_{\text{b}} = 70\,\text{kg}$ person, this corresponds to a force of $m_{\text{b}}a_{\text{decel}} = (70\,\text{kg})(10^3\,\text{m/s}^2) = 7 \times 10^4\,\text{N}$ on one foot.

This is a compressive force, and it has its greatest effect on the thinnest bones absorbing the impact. The tibia near the ankle has a radius $r \sim 1\,\text{cm}$ and a cross-sectional area $A \sim 3\,\text{cm}^2 = 3 \times 10^{-4}\,\text{m}^2$. During impact the force per unit area in this region of the tibia is

$$\frac{F}{A} = \frac{7 \times 10^4\,\text{N}}{3 \times 10^{-4}\,\text{m}^2} = 2.3 \times 10^8\,\text{N/m}^2 = 230\,\text{N/mm}^2. \tag{3.114}$$

Upon compression bones typical break when subjected to a force per area above $\sim 1.7 \times 10^8\,\text{N/m}^2 = 170\,\text{N/mm}^2 = 170\,\text{MPa}$. This damage threshold is called the ultimate compressive stress (UCS, Chap. 4). The stresses on the tibia exceed this limit, so jumping from 1 m and landing on one stiff leg will break your leg. (Please do not try this!) This also shows us why we need to learn more about the mechanical properties of parts of the body. Problem 3.131 considers the gentler case of using leg muscles to cushion the collision.

These two examples, along with (3.104) and (3.105), suggest how to decrease the impact of a collision. Because the stress during the collision is $mv_{\text{i}}/At_{\text{coll}}$, the effect

of the impact can be lessened by increasing t_{coll} and/or the area of contact A. The collision time can be increased by (a) using better padding at the site of the collision (such as by wearing better sneakers during running), (b) increasing the duration of the impact by adjusting your body during actual impact (as by letting your knees bend during the impact after a jump—the muscles will then bear part of the impact of the collision), or (c) rolling over (as at the end of a parachute jump). (Rolling over at the end of a parachute jump also serves to increase the impact area A.) The saying "roll with the punches" instructs you to adapt to the situation; it is from boxing where it is shorthand for saying you minimize damage from punches by rolling your body at the point of impact to increase the collision time and impact area.

Obviously, damage can occur during falls and collisions. *However, when are such crashes survivable?* It depends on what part of the body makes contact, what type of surface it hits, and the deceleration time. In whole body impact, your fate depends on what part of the body makes contact: the back (supine position), front (prone), or side (lateral). There are some guidelines gleaned from past (unfortunate) experience. You can barely survive lateral falls at 8 m/s and supine falls at 12 m/s. You can survive foot first falls at 12 m/s on concrete, 15 m/s on soil, and 35 m/s on water. You can survive head first falls at 9 m/s on concrete/ice [20]. (Do not try any of these!) Figure 3.61

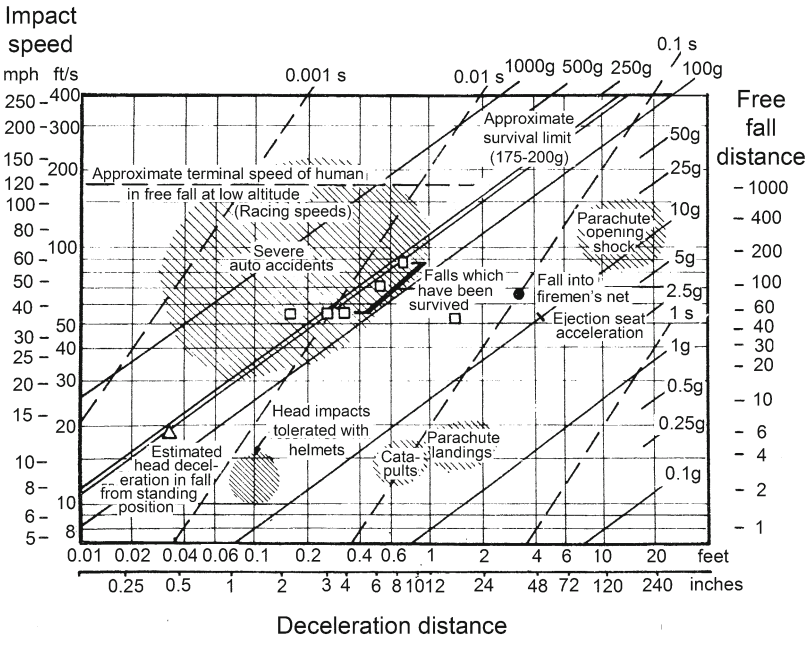


Fig. 3.61 Survivability of collisions. The free fall distance is that for the indicated impact speed, with allowance made for air resistance of the human body near sea level. Also shown are lines for given decelerations and given deceleration times. The band shown for 175–200 g separates the approximate survival and nonsurvival regions (from [131])

shows that human collisions are usually not survivable for decelerations of 175–200 g and greater. Falls under suspicious conditions are discussed in [32].

What is actually occurring in such collisions? Consider an injury to the head. In direct trauma, the head hits an object and the skull can fracture or the brain, which continues moving forward when the skull has been decelerated, hits the interior of the skull, leading to bruising (a *contusion*) or bleeding (a *hemorrhage*). In addition to this primary impact (the *coup*), the brain can bounce off the skull and hit the back of the skull (the *contrecoup*). Some think that this coup/contracoup injury is instead caused by negative gauge pressure that occurs in the part of the brain opposite the impact side (in the contracoup region), which leads to cavitation and this causes the injury. Severe injury can result when the brain is subject to large accelerations even without impact (*indirect trauma*): in inertial injuries due to rotational accelerations and decelerations and in *Shaken Baby Syndrome* nerve cell axons are damaged by stretching (*diffuse axonal injury*) [14]. (*Whiplash* is a neck injury that can be caused in such collisions.)

The gravity of such injuries is qualitatively described by the abbreviated injury scale (AIS). This is a guide to the severity of injury to a body component or organ by its threat to life. It ranges from minor injury (AIS = 1), moderate injury (2), serious injury (3), severe injury (4), critical injury (5), and to unsurvivable injury (6).

There are semiquantitative guidelines for the severity and likely fatality of head injuries that have been determined from the records of past accidents [53, 106]. Using data from experiments on cadavers and animals, the Wayne State University Concussion Tolerance Curve was developed [61, 62, 89], which Gadd converted into an index [54]. The Gadd severity index (GSI) is defined as

$$\mathrm{GSI} = \int \left(\frac{a_{\mathrm{decel}}}{g} \right)^{2.5} \mathrm{d}t, \tag{3.115}$$

expressed in seconds. This is integrated over the entire collision (a frontal blow to the cranium), as in Fig. 3.62a. a_{decel} is expressed in units of g and t is in s. If the deceleration is constant during the course of the collision, the integral is the product of this constant deceleration-based term and the collision time

$$\mathrm{GSI} = \left(\frac{\Delta v}{g t_{\mathrm{coll}}} \right)^{2.5} t_{\mathrm{coll}}. \tag{3.116}$$

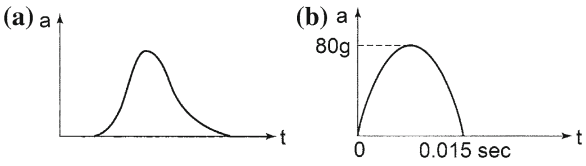


Fig. 3.62 Acceleration versus time in a collision, **a** in general, **b** of the head of a dummy hit by a heavyweight boxer (from [13])

This helps determine the likelihood of skull fracture or concussion for frontal collisions. When the GSI reaches ~1,000 (in units of seconds), there is a 50% chance of fatality. If the GSI is much above 1,000, the collision is usually fatal. A GSI much below 1,000 can still indicate a severe injury. For example, when the GSI ~ 400, there may be a mild concussion that could result in unconsciousness.

Versace recommended a modification of the GSI to remove potential long tails in the deceleration, which are probably unimportant. This truncated index is the head injury criterion (HIC)

$$\mathrm{HIC} = \max\left[\left\{\frac{1}{t_2 - t_1}\int_{t_1}^{t_2}\left(\frac{a_{\mathrm{decel}}}{g}\right)\mathrm{d}t\right\}^{2.5}(t_2 - t_1)\right], \tag{3.117}$$

expressed in seconds [136]. This is the maximum value of this expression between any two times t_1 and t_2 during the collision, separated by not more than 36 ms; in a more recent version (HIC-15) this time separation is 15 ms. One study has shown that the threshold for serious head injury is HIC = 1,000 (in units of seconds), for which the probability of death is about 7% (and is AIS 3). An index of HIC = 1,500 is the threshold for severe/critical injury, with a 26% probability of death (and is AIS 4–5), and this probability is about 100% for HIC = 3,200. Figure 3.63 plots the risk

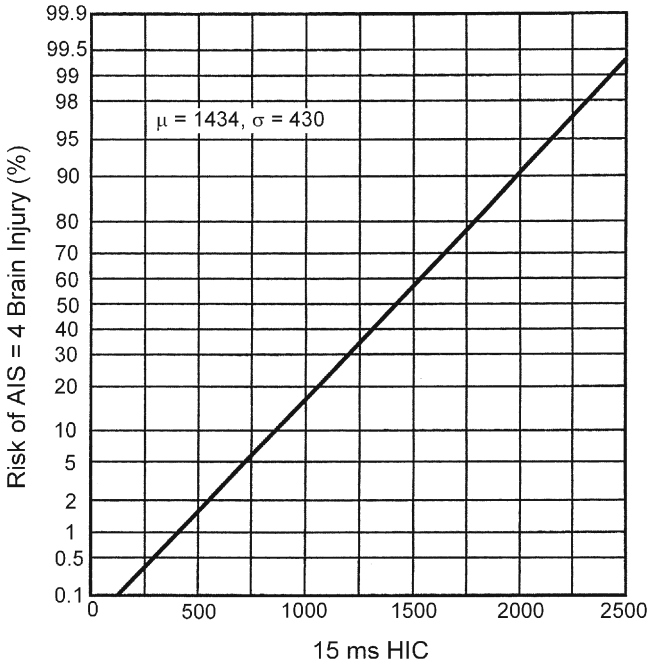


Fig. 3.63 Risk of AIS ≥ 4 brain injury as a function of 15-ms HIC for forehead impacts (from [94])

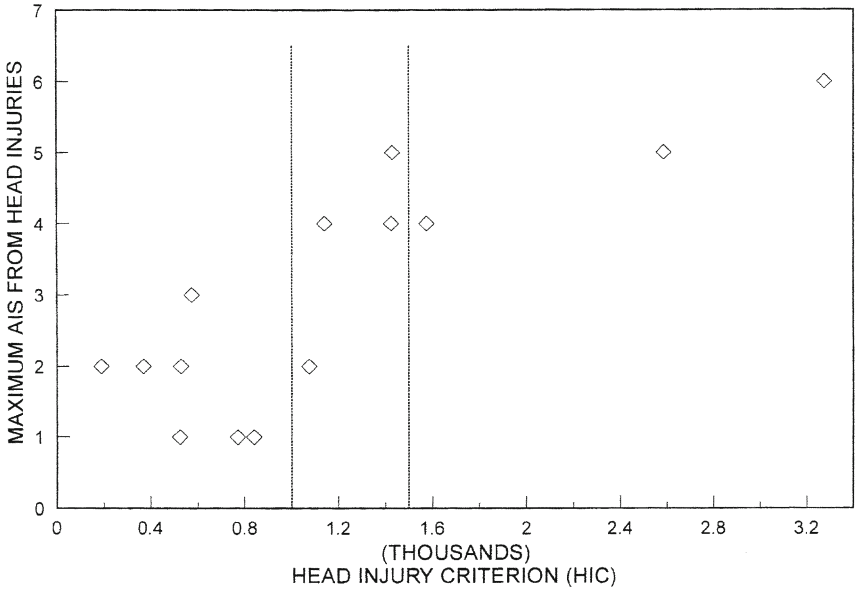


Fig. 3.64 Classification of case-studies of collisions leading to head injury, plotted as the maximum head injury (using the AIS scale) versus the collision condition as quantified by the HIC (from [126], as from [75])

of serious brain injury (AIS $\geq$ 4) for HIC-15, which is the integral over a gaussian probability curve

$$R = \left(\frac{430}{\pi}\right)^{1/2} \int_0^{\text{HIC}-15} \exp\left[-\{(x - 1{,}434)/430\}^2\right] \mathrm{d}x. \tag{3.118}$$

(This integral is also known as the *error function*. See Appendix C.) The risk of serious brain injury is 5% for HIC-15 = 700 and 50% for HIC-15 $\sim$ 1,400–1,500 for this serious injury and for skull fracture. The maximum AIS from head injuries is plotted in Fig. 3.64. The risk of death is plotted in Fig. 3.65. The HIC is a better predictor of the seriousness of a trauma to the brain than is the GSI. Problems 3.134 and 3.135 explore the differences in the GSI and HIC indices for the same accident. The use of such indices to predict the outcomes of accidents is not universally accepted for several reasons. For example, the GSI and HIC assess the effects of linear (rectilinear) accelerations, but not rotational accelerations, which can also be very important. Other criteria have also been proposed [79].

Let us examine the consequences of the elasticity of the head in a car crash. If the head collides with a heavy hard object, the head will rebound elastically. We will also assume the object is much heavier than the head. In the cases we have considered so far we have assumed that $v_\mathrm{f} = 0$. Now we will consider $v_\mathrm{f} = -v_\mathrm{i}$, so

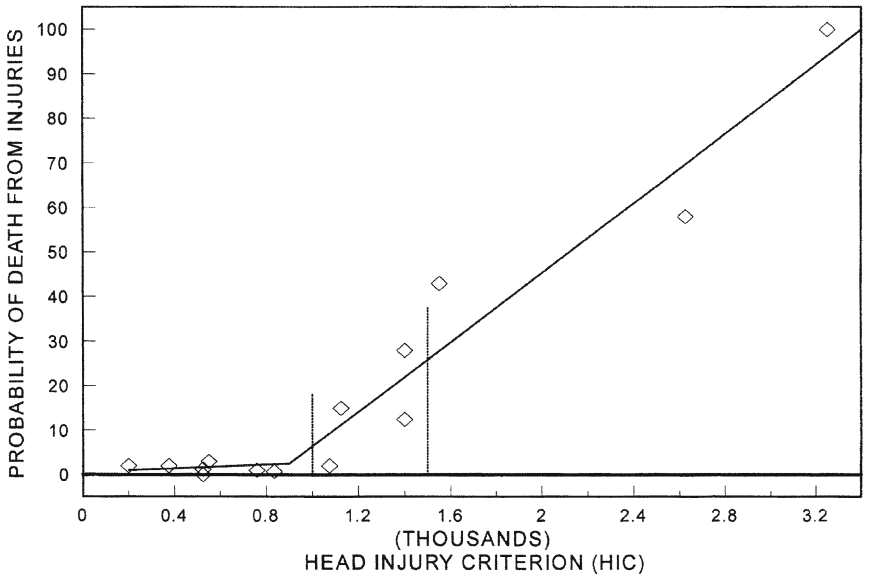


Fig. 3.65 Classification of case studies of collisions leading to death, plotted as the probability of death versus the collision condition as quantified by the HIC. (From [126], as from [75])

$\Delta v = v_\mathrm{i} - v_\mathrm{f} = 2v_\mathrm{i}$ and $a_\mathrm{decel} = \Delta v/\Delta t = 2v_\mathrm{i}/t_\mathrm{coll}$. If this is constant during the collision,

$$\mathrm{GSI} = \left(\frac{2v_\mathrm{i}}{gt_\mathrm{coll}}\right)^{2.5} t_\mathrm{coll} \tag{3.119}$$

for an elastic collision, compared to

$$\mathrm{GSI} = \left(\frac{v_\mathrm{i}}{gt_\mathrm{coll}}\right)^{2.5} t_\mathrm{coll} \tag{3.120}$$

for an inelastic collision. Say $v_\mathrm{i} = 50\,\mathrm{mph} = 73.3\,\mathrm{ft/s} = 22.3\,\mathrm{m/s} = 80.4\,\mathrm{km/h}$. Then t_coll can range from ~2–50 ms. Let us assume a collision time of 10 ms, which gives GSI = 44,000 for a perfectly elastic collision, compared to the much lower 7,800 for a perfectly inelastic collision. This indicates a definitely fatal collision in either case for this example.

Both the locations of the impacts to the head and their magnitudes can be important. In one study, 33% of all impacts endured by collegiate ice hockey players were found to be to the back of the head, 33% to the side, 14% to the forehead, 12% to the facemask, and 8% to the top of the head. The average of the top 1, 2, and 5% of all peak linear decelerations were 105, 90, and 62 g, respectively. While there were many more impacts to the back of the head, impacts to the top of the head were found to be the most severe [63].

A study of college football players showed that the 99th percentile of the highest linear decelerations of the head corresponded to ~95–105 g for most positions, but to ~110–120 g for quarterbacks, which explains why they suffer so many concussions [124].

In another study, impacts to the heads of professional football players during games were reconstructed [115]. Concussions during games were found to correlate strongly with the GSI, HIC, linear decelerations, and changes in the velocity of the head of the person being tackled (in the reconstructed experiments), and less so with head rotational acceleration and not with changes in peak rotational velocity of the head. They were due mostly to impacts to the facemask or side of the helmet, or to the back of the helmet after a fall. Peak average speeds of the helmet of the tackler in helmet-helmet collisions were ~9.3 m/s (20.8 mph), which led to a change of head speed of the tackled player of ~7.2 m/s in collisions that caused concussive effects and ~5.0 m/s in collisions that did not, and led to smaller changes in the heads of the tacklers themselves, of "only" ~4.0 m/s. The deceleration of the head of the tackled player averaged ~98 g (in 15 ms) in collisions leading to concussions, compared to ~60 g for those leading to no injury. The threshold values for concussions were found to be a GSI of ~300 and an HIC of ~250 [115].

Repetitive head collisions, as is common in football, ice hockey, boxing, and several military activities, can lead to a degenerative and incurable disease known as chronic traumatic encephalopathy, or C.T.E., even if none of the collisions is severe enough by itself to cause a concussion. C.T.E. can be in Stage 1, with headaches and loss of attention and concentration, Stage 2, also with depression, explosive behavior, and short-term memory loss, Stage 3, also with cognitive impairment and trouble with executive functions such as planning and organizing, and Stage 4, also with dementia, difficulty finding words, and aggression [12, 56, 92]. Such long-term consequences of repeated head trauma in athletes are becoming increasingly recognized.

In collisions of the head, padding in car dashboards and air bags can clearly help! Padding in bicycle and motorcycle helmets and the helmets of other sports activities can help (and did help in the presented studies)! The helmet padding helps in two ways. It cushions the blow to the head because the kinetic energy of the head is used to deform the padding. This decreases the deceleration of the head. Furthermore, the blow is spread over the inside area of the helmet, which is much larger than the typical collision regions of contact. The desired materials properties and geometries of the helmet shell and interior padding are discussed in Chap. 4. Clearly, the padding must absorb the kinetic energy of the head without returning it, as would happen if you used a spring-like material for the padding.

3.10.3 Contact Sports

For the most part, collisions in football involve a relatively large surface of one person colliding with a relative large surface of another. The goal is to block the opponent

or to tackle the opponent with the football, which means to knock him off balance so he falls to the ground. In boxing, karate and similar contact endeavors, a person delivers blows with a hand and/or foot, using motions that are qualitatively similar to those used for throwing and swinging a bat, described earlier this chapter. As with swinging a bat, the goal is for the body to make contact, but in these activities the contact is with a person. The collision process itself and the fate of the material being hit are both important. The areas involved in the collision are usually smaller than those in football.

Boxing

In boxing, the goal is to land punches on someone else's body, while wearing a glove, which softens the collision by spreading the impact area and duration of the collisions. Damage is done disturbing the balance of the opponent, as it is in wrestling, and by accelerating the head of the opponent, as we just saw (Fig. 3.62), and not necessarily in breaking their bones. The boxing stance, with one foot is in front and the other is in back, provides stability for the boxer delivering blows and the one receiving them.

The long-term decline of cognitive ability in boxers commonly results from repeated blows to the head and the resulting the "punch drunk" symptoms of *dementia pugilistica*, which is like C.T.E. In heavyweight boxing, GSI ∼400 for a solid punch, which suggests a mild concussion consistent with a knockout punch. This is based on integrating the information in force plots (Fig. 3.62b). We can arrive at similar results from first principles.

We first ask: why do boxers wear boxing gloves? These gloves protect the one who throws the punch as well as the one who receives it. Boxing gloves decrease the deceleration a_{decel} of the puncher's fist by increasing t_{coll}, due to the thickness of the padding in the glove, and A, due to the large lateral dimensions of the boxing glove. (This large load area lessens the large forces on specific places on specific bones in the hand, thereby reducing the risk of fracture.) The deceleration is even smaller for yet another reason. The collision of the fist with head becomes much more inelastic (i.e., less elastic) with the glove. Without the glove, a_{decel} would be much larger and the boxer's hand would more easily break (Fig. 3.66) and the damage to the head of the opponent would be greater. The forces involved are tremendously large even with the boxing gloves. Let us examine what happens when a fist hits a head.

A good boxer uses his or her legs and whole body to deliver a punch, but for our model let us consider the motion of the fist and whole arm only and see what happens when it hits the opponent's head; we are saying that their combined mass is the *effective mass* of the puncher in the collision. The other boxer's head is attached to his body (at least before the punch), but let us consider the effect of the punch on the head and neck only, which would be the effective mass of the punched region. The fist is initially moving at $v_{\text{fist,i}}$ and if we ignore the motion of the shoulder the average speed of the arm is $\sim v_{\text{fist,i}}/2$ (which we will use here). The mass of the arm (with fist) is $m_{\text{arm}} = 0.050\, m_{\text{b}}$ (ignoring the mass of the 5 oz gloves and where



Fig. 3.66 X-ray of a boxer's fracture, which results when hitting with a closed fist (which bone is broken?) (from [82], as from [75])

again m_b is the body mass) (Table 1.7). The head (and neck) of the other boxer is $m_{head} = 0.081\, m_b$.

The elasticity of the collision affects the analysis. In elastic collisions, the colliding bodies have different speeds after the collision, and the total linear momentum and kinetic energy are the same before and after the collision. In totally inelastic collisions, the colliding bodies have the same velocity after the collision, and total linear momentum is conserved. Total energy, but not merely kinetic energy, is still conserved. Most collisions are partially elastic and partially inelastic. (We will analyze the collision here without using the coefficient of restitution.)

Say the boxers are wearing gloves. Before the punch collision, the fist is moving at a speed $v_{fist,i}$ and the linear momentum of the arm is $\sim m_{arm}(v_{fist,i}/2)$. The collision is inelastic, meaning that at the end of the punch the fist and head will both be moving at the same final speed v_f (Fig. 3.67a). The average speed of the arm is $v_f/2$. The total momentum after the punch is $m_{arm}(v_f/2) + m_{head}\, v_f$, so

$$\frac{m_{arm}\, v_{fist,i}}{2} = \frac{m_{arm}\, v_f}{2} + m_{head}\, v_f \tag{3.121}$$

and

$$v_f = \frac{v_{fist,i}}{1 + 2\, m_{head}/m_{arm}} = 0.236\, v_{fist,i}, \tag{3.122}$$

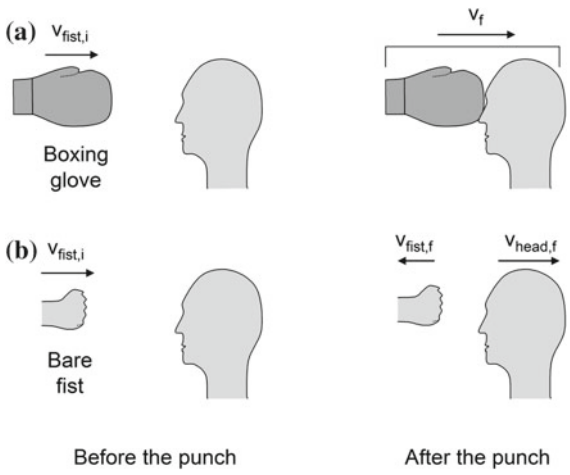


Fig. 3.67 Models of boxing, with **a** boxing gloves, leading to an inelastic collision with the head, and **b** bare fist, leading to an elastic collision with the head

using the arm and head masses. This means that during contact the fist decelerates from $v_{\text{fist,i}}$ to $0.236\,v_{\text{fist,i}}$, while the head accelerates from 0 to $0.236\,v_{\text{fist,i}}$. If the glove compresses by $\Delta x = 1\,\text{cm}$ during the punch, the head accelerates at a rate $a_{\text{coll}} = v_{\text{f}}^2/2\Delta x = (0.236\,v_{\text{fist,i}})^2/(2 \times 1\,\text{cm})$. The collision time $t_{\text{coll}} = v_{\text{f}}/a_{\text{coll}} = 2 \times 1\,\text{cm}/0.236\,v_{\text{fist,i}}$.

Our model for throwing a ball is also a good model for throwing a punch. Instead of the biceps brachii contracting causing θ to decrease during throwing a ball, the triceps brachii contract causing θ to increase. Ignoring the effect of the rest of the body, with similar assumptions we have $v_{\text{fist,i}} \sim 30\,\text{mph}$ (13.4 m/s) for 3 in diameter biceps. Assuming the boxers are wearing boxing gloves, the head of the opponent accelerates to $v_{\text{f}} = 3.2\,\text{m/s}$ at rate $a_{\text{coll}} = 500\,\text{m/s}^2 = 51\text{g}$ in $t_{\text{coll}} = 0.0063\,\text{s}$. If we assume that (3.116) can be used for accelerations as well as decelerations, this leads to a severity index GSI = 117. This indicates damage, but not a fatal blow. With less padding in the glove, say leading to a deformation of $\Delta x = 0.5\,\text{cm}$, the GSI increases to 332. With Δx again 1 cm and with $v_{\text{fist,i}} = 45\,\text{mph}$ (20.1 m/s, our faster estimate), we find $v_{\text{f}} = 4.7\,\text{m/s}$ and the GSI increases to 593, suggesting substantial damage. Boxers' heads clearly do get damaged even when they wear gloves. Because part of the rest of the body is involved in the punch, v_{f} will be larger than estimated here and the GSI will be larger in each case.

Now what happens if the boxers do not wear boxing gloves? Without gloves the collision of the fist and head is assumed elastic, with final speeds $v_{\text{fist,f}}$ and $v_{\text{head,f}}$ (Fig. 3.67b), and the collision distance is much shorter, say 1 mm. Both of these differences contribute to the more dire consequences of bare knuckle boxing, both for the puncher (broken hand) and the punched (more damage to the head). Using the same approximations for arm movement, along with conservation of momentum and kinetic energy, gives

$$\frac{m_{\text{arm}}v_{\text{fist,i}}}{2} = \frac{m_{\text{arm}}v_{\text{fist,f}}}{2} + m_{\text{head}}v_{\text{head,f}}, \tag{3.123}$$

$$\frac{m_{\text{arm}}}{2}\left(\frac{v_{\text{fist,i}}}{2}\right)^2 = \frac{m_{\text{arm}}}{2}\left(\frac{v_{\text{fist,f}}}{2}\right)^2 + \frac{m_{\text{head}}}{2}v_{\text{head,f}}^2, \tag{3.124}$$

and

$$v_{\text{head,f}} = \frac{v_{\text{fist,i}}}{1 + m_{\text{head}}/m_{\text{arm}}} = 0.382v_{\text{fist,i}}, \tag{3.125}$$

which is 1.6× that with gloves. With $v_{\text{fist,i}} = 30\,\text{mph}$ and $\Delta x = 1\,\text{mm}$, the head accelerates to $v_{\text{head,f}} = 5.1\,\text{m/s}$ at a rate $a_{\text{coll}} = 13{,}100\,\text{m/s}^2 = 1{,}340\,g$ in $t_{\text{coll}} = 0.00039\,\text{s}$ and GSI = 25,600. Even though this is a bit of an overestimate since the collision is not perfectly elastic, it is clear that bare knuckle boxing can lead to death (as well as broken knuckles).

Karate

In karate, your bare hand or foot strikes small regions of the bones of an opponent or, in demonstrations, a set of one or several bricks, concrete blocks or wooden boards [16, 44, 144]. Breaking the struck object is a goal, without breaking your own hand or foot. For karate demonstrations, wooden boards should be made of softer woods, such as white pine without knots, rather than harder woods, such as oak, or particle board, and the grains should be parallel to the direction of the striking hand or foot. (Why?)

As we will see in Chap. 4, when a force is applied to the middle of the top of a board supported at either end the top surface is compressed (is pushed together) and the bottom surface is in tension (is pulled apart), by equal amounts (Fig. 4.31). The bones in the hand need to have a higher force/area (stress) threshold for fracture especially for compression (at the ultimate compressive stress (UCS)), than the target has for fracture due to bending. Table 4.1 shows that concrete and pine break easier in tension (at the ultimate tensile stress (UTS)), than in compression, so fracture begins on the bottom surface of the board and the crack propagates upward while the hand is still striking it. Because these materials fracture at levels much lower than bones fracture, the boards break and not the bones in the hand when karate demonstrations are performed properly. Moreover, for example in the hammerfist strike (a name that describes the motion well) the bone closest to contact, the fifth metacarpal, is cushioned by the abductor digiti minimi muscles, and some of the blow is absorbed by this tissue and by the wrist and the rest of the arm, and these factors also lessen the likelihood of bone fracture.

At the point of contact, the punch reaches a maximum speed of 10–14 m/s for downward hammerfist and knife hand strikes, which is attained ∼75% of the way through the strike. During hammerfist strikes on concrete, the fist decelerates to a standstill at decelerations up to $4{,}000\,\text{m/s}^2$ and the collision lasts for ∼5 ms, with peak forces as high as 2800 N. The collision is largely inelastic and the effective

mass of the fist is closer to that of the hand than to that of the combined mass of the hand and forearm.

See the discussion about breaking beams and bones in Chap. 4 to learn more about breaking boards.

3.10.4 Hitting Balls

We will now examine collisions with objects such as baseballs [2, 30, 33, 127, 128, 137] and tennis balls. The physics of hitting such objects starts with the human swinging motion of an object: bat, golf club or tennis racket—to hit them (or the kicking motion of the leg to kick a football or soccer ball), which we discussed earlier in this chapter. The second part is the collision itself, much as in punching in boxing. However, the emphasis in boxing is the effect of the collision on the people delivering and receiving the punch. (As in many other activities in life, in boxing it is definitely better "to give than to receive.") Catching a ball is a type of half collision in which the ball stops as a result of the collision and so it is an inelastic collision.

As a result of the collision, a hit ball will have a given velocity and spin. The duration of the impact itself is brief, ∼0.5 ms for a golf ball and up to ∼5 ms for a tennis ball (which is longer because of the distortion of the strings (the "give" of the racket)) [36]. The distortion of the ball during the collision can be a sizeable fraction of its diameter. Later in this chapter we will consider the flight of the ball. We will also look at bouncing balls, which are collisions of balls with the ground (which is similar to hitting balls with rackets), collisions of balls with other balls, and rolling balls.

To examine what happens when a bat hits a baseball, we use (3.110) with 1 to denote the baseball and 2 to denote the bat to arrive at

$$v'_{\text{ball}} = \frac{(m_{\text{ball}} - em_{\text{bat}})v_{\text{ball}} + (m_{\text{bat}} + em_{\text{bat}})v_{\text{bat}}}{m_{\text{ball}} + m_{\text{bat}}}. \tag{3.126}$$

(In this and subsequent equations in this discussion, we will substitute all masses by their weights (because $W = mg$ and the g cancels out). This will be useful in using the weights of the ball and bat in the very common units of oz.)

To understand the plight of the batted ball we need to know the motion of the body, arms, and wrist during the swing (Fig. 3.68), the speed of the pitched ball, where the ball hits the bat (relative to its center of mass), and the weight of the bat. (Major league bats typically weigh approximately 32 oz; much heavier, 44–50 oz, bats were once routinely used by sluggers.) We know that $W_{\text{ball}} = 5.125$ oz and that the coefficient of restitution for the collision of a baseball with a wooden bat is 0.55.

We should also want to know why right-handed batters stand in the left side of the plate in the batter's box (as seen by the catcher), so the left sides of their faces face the pitcher and why they hold bats with their right hands above their left hands (instead of below, as in the rarely-used cross-handed grip) and therefore they push rather than

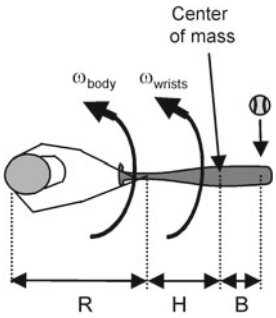


Fig. 3.68 Model of batting a ball (based on [137])

pull during a swing [16]; this is reversed for left-handed batters. The higher hand controls the stability of the bat better, so for a righty it is better for the right hand to be on top, and stability is likely better when pushing (when batting from the left side of the plate) than pulling (from the other side). This added control apparently overrides the two major disadvantages faced by righty batters by standing on the left side of the plate: (1) they are farther from first base and (2) they have more difficulty hitting righty pitchers (who greatly outnumber lefties ("southpaws")), because these pitchers throw curveballs that curve away from the batters and it is harder for these batters to track the throwing arms of these pitchers.

To hit a home run in a major league park, the initial speed of the batted ball (sometimes called the "exit velocity") needs to be $\sim$100 mph, so it could clear a 15 ft tall fence that is $\sim$330 ft from home plate. This includes the effect of drag. (Note that it is best to hit the ball at an angle of $\sim 35°$ to maximize the distance traveled for the initial speed because of drag, which is smaller than 45° optimum angle for range with no drag (see [1])). To hit a ball this hard requires a bat speed of $\simeq$50 mph at its center of mass or $\simeq$58 mph a bit farther out on the bat, at the "sweet spot," which is the most effective location for this collision (see [137]). It is thought the balls hitting this spot will move the fastest because the bat handle will not vibrate or move during impact.

There are two models for determining this sweet spot, here applied to a model bat with uniform cross section. In one approach, this spot is the *center of percussion*. Figure 3.69a shows that when the ball is hit there the linear velocity of the bat handle due to the collision force cancels the rotational velocity of the handle caused by torque about the bat center of mass, so the bat-ball collision does not cause the bat handle to move and there is less reaction force on hands due to the collision. Problems 3.167 and 3.168 show this center of percussion is 2/3 up from the handle of this idealized bat. The physics behind the second approach will become clearer in Chap. 10 where we will discuss the vibrational modes of tubes and rods. Figures 3.69b and 10.5 show that in the first excited mode (the third harmonic) vibration in a tube or bar that is closed or fixed on one end and open or free to move on the other, there is no motion (a "node") 2/3 of the way from the closed end to the open end. The closed end is like the bat handle held by the hands that should not oscillate laterally as a result of the ball-bat collision and this (fixed) node is the sweet spot. Because there is no

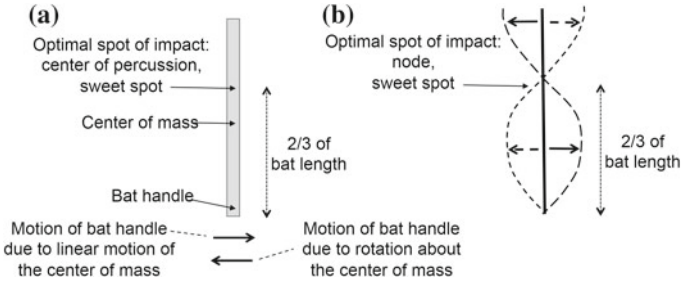


Fig. 3.69 Models of the sweet spot where the ball hits the bat as the **a** center of percussion (so the linear velocity of the bat handle due to the collision force cancels the rotational velocity of the handle caused by torque about the bat center of mass) or **b** node of the first excited mode in a bar that is fixed at one end (the bat handle) and free to move at the other (so the shown lateral oscillatory motion does not occur) (Based on [16, 36])

motion at the point of collision, the bat does not vibrate and energy is not lost into the unwanted vibration of the bat [16]. Both approaches predict a sweet spot at the same place. The concept of the sweet spot can be applied to other ways of hitting balls, such as with tennis rackets.

What is the optimal bat weight? In one model [10, 80, 107, 137] it is assumed that the optimal bat weight is one that requires the least energy input to achieve a given batted-ball speed. Minimizing the total initial kinetic energy of the ball and bat, and using that result in (3.126) gives

$$\frac{W_{\text{bat,ideal}}}{W_{\text{ball}}} = \frac{v'_{\text{ball}} - v_{\text{ball}}}{v'_{\text{ball}} + e v_{\text{ball}}}. \tag{3.127}$$

Using the typical pitch speed of $v_{\text{ball}} = -80\,\text{mph}$ and the batted ball speed to hit a home run $v'_{\text{ball}} = 110\,\text{mph}$, the "optimized" bat weight is $W_{\text{bat}} = m_{\text{bat}} g = 15\,\text{oz}$, which is smaller than the actual weight of bats that are used by adults. (Is the assumption that the initial kinetic energy should be minimized really reasonable?)

Another way to determine the ideal bat weight combines a model with measured data. The fastest speed you can swing a bat has been observed to decrease with bat weight, typically as

$$v_{\text{bat}} = A - B W_{\text{bat}}. \tag{3.128}$$

Substituting (3.128) into (3.126) (and using weights instead of masses), and then differentiating the equation with respect to W_{bat} and setting the result equal to zero, gives the bat weight leading to the fastest batted ball

$$\frac{W_{\text{bat,ideal}}}{W_{\text{ball}}} = \sqrt{1 + (A - v_{\text{ball}})/(B W_{\text{ball}})} - 1. \tag{3.129}$$

In particular, for one slugger the bat speed relation (3.128) has been measured to be [10, 137]

$$v_{\text{bat}}(\text{in mph}) = 63 - 0.39 W_{\text{bat}}(\text{in oz}), \tag{3.130}$$

so here $A = 63\,\text{mph}$, the maximum bat speed for a massless bat, and $B = 0.39$. For $v_{\text{ball}} = -80\,\text{mph}$, the ideal bat weight is 38.5 oz, which is quite reasonable, and the batted ball speed is 95.1 mph. In fact, the batted ball speed varies very little for a broad range of bat weights.

We have seen that using multisegment models with more than one joint is important in analyzing many types of problems in statics and motion. Figure 3.68 shows that it is also very important in hitting balls. The body rotates at an angular speed ω_{body} due to the combined rotation about the hip, shoulders, and ankles and the wrists rotate at an angular speed ω_{wrists}. With the extension of the arms being R, the speed of the hands is $R\omega_{\text{body}}$. With the center of mass of the bat a distance H from the hands, the speed of the bat center of mass due to body rotation is $(R + H)\,\omega_{\text{body}}$. The speed of the center of mass of the bat due to wrist motion is $H\omega_{\text{wrists}}$. Therefore, before hitting the ball, the bat center of mass moves at the speed v_{bat}

$$v_{\text{bat}} = (R + H)\,\omega_{\text{body}} + H\omega_{\text{wrists}}. \tag{3.131}$$

If the ball hits the bat a distance H from its center of mass, then H is replaced by $H + B$, so

$$v_{\text{bat, not CM}} = (R + H + B)\,\omega_{\text{body}} + (H + B)\,\omega_{\text{wrists}}, \tag{3.132}$$

and this is related to the speed of the bat center of mass by

$$v_{\text{bat, not CM}} = v_{\text{bat}} + B(\omega_{\text{body}} + \omega_{\text{wrists}}). \tag{3.133}$$

This shows that the motion of the region where the bat hits the ball can be viewed as the rotation of the bat center of mass plus a rotation of this region about the bat center of mass at angular speed $\omega_{\text{bat}} = \omega_{\text{body}} + \omega_{\text{wrists}}$.

How does this change our analysis of the collision of the ball and bat? Conservation of linear momentum (3.108) still holds for the ball and the bat center of mass. However, you need to use the bat speed at the actual point of impact before and after the collision in the coefficient of restitution equation (3.109). Furthermore, there is a torque on the bat during the collision when the ball does not hit the center of mass of the bat. This is examined further in Problem 3.165.

The goal is not always to hit the ball as hard as possible. In bunting in baseball, the bat is suddenly pulled back just before the pitch arrives, so with good bat control, the ball is a very slow grounder in fair territory that does not reach any of the fielders quickly, unless they are positioned for the bunt. (The batter does this to advance a runner on base into scoring position or to get a single.) In (3.126) v_{bat} is now negative, and v'_{ball} ball becomes very small, but remains positive.

Can the hit ball spin after it is hit? Yes. The spin results from friction between the ball and bat, racket or club during the collision [36, 60], and depends on how it is hit.

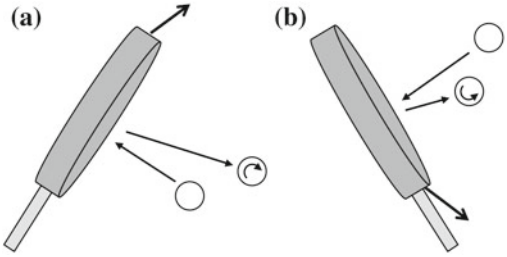


Fig. 3.70 Applying **a** topspin and **b** backspin (slice shot) in tennis and ping pong (Based on [60])

When the center of a swinging bat is slightly below the baseball it imparts backspin and when it it slightly above it it imparts topspin. (We will define these terms later this chapter in the discussion of the flight of a spinning ball). Topspin and backspin can be imparted in tennis and ping pong respectively by an upward and downward motion of the racket, as seen in Fig. 3.70. In fact, the speed on the surface of a ping pong ball due to spinning can exceed the linear speed of its center of mass [16]. We will revisit how a hit ball acquires spin during impact later in this chapter when we analyze how balls bounce (which is a type of collision).

Spin changes the trajectory of the ball, due to the Magnus force. A ball hit with topspin experiences a downward force, so it travels a shorter distance, and one with backspin experiences an upward force, called lift, so it travels a longer distance. Spin also affects how the ball bounces, both directly due to the spin during the collision with the ground and indirectly through its altered trajectory and the altered angle at which it hits the ground. Balls with topspin seem to jump up fast after the bounce, compared to balls with no spin, because they hit at a steeper angle, and this counters the shallower bounce expected from the spin itself. Spin in other directions causes the ball to follow a curved trajectory.

3.10.5 Running

Your foot suffers a collision every time it touches the ground during walking, running, and landing after a jump. During a very slow step—a gentle collision—the forces on the foot slowly build up from 0 to $m_b g$. During running the forces on the foot reach a peak of $\sim$2.0–3.6 $m_b g$ (marathon running: $\sim$2.7 $m_b g$; sprinting: $\sim$3.6 $m_b g$). Does this make sense?

Equation (3.113) show the deceleration during a collision is $a_{\text{decel}} = v_i^2/2\Delta x$. Here, v_i is the initial downward speed of the foot and Δx is the vertical braking distance of the foot. The downward speed is $\sim$1.5 m/s for a running speed of 5 m/s (5.4 min/mile). The braking distance is due to the compression of the fatty pad of the heel of the foot, which is $\sim$4 mm, plus that of the heel of the running shoe, $\sim$8 mm, for a total of $\sim$12 mm. The deceleration is 187.5 m/s^2 and the maximum force on the

Table 3.11 Energy return from elastic sports equipment

Equipment	k (1,000 N/m)	Δx_{max} (cm)	KE_{max} (J)	H_{max} (cm)
Trampoline	5	80	1,600	230
Tumbling floor	50	10	250	36
Gymnastic floor	120	5	150	2
Running track	240	1	12	2
Gymnasium floor	400	0.5	5	1

Δx_{max} and KE_{max} are the maximum deformation and stored energy ($KE_{max} = k\Delta x_{max}^2/2$) and H_{max} is the maximum height of a $m_b = 70\,kg$ person with this energy, $H_{max} = KE_{max}/m_b g$
Using data from [134]

decelerating mass m is $19.1\,mg$. The approximate energy returned to the body as a consequence of this collision was addressed above and is given in Table 3.11.

If the entire body were being decelerated during the collision of the heel (the human body as a rigid body), the force on the heel would be due to the entire mass of the body, $19.1\,m_b g$. That is excessive, so it appears that during the collision the entire body mass is not being decelerated at once by the heel. In the opposite extreme we can assume that only the foot is being decelerated. Because the mass of the foot is $0.0145\,m_b$ (Table 1.7), the force on the heel would be $0.28\,m_b g$ plus that from the slower rest of the body, $0.9855\,m_b g$, for a total peak force of $1.26\,m_b g$. This seems low. Following the lessons learned from "Goldilocks and the Three Bears" (if first too big and then too small, try somewhere in between), we should examine several intermediate cases for this effective mass. Let us say that during the collision the foot plus lower leg mass $= 0.061\,m_b$ (or alternatively the foot plus lower leg plus thigh mass $= 0.161\,m_b$) is being decelerated. Then the peak forces would be $1.165\,m_b g + 0.939\,m_b g = 2.104\,m_b g$ (or $3.075\,m_b g + 0.839\,m_b g = 3.914\,m_b g$ for the entire leg). Comparing this to the cited measurements suggests that during running the initial impact of the collision is borne by your foot and lower leg, with some significant coupling to the thigh. Experience shows us that this is reasonable. Without some coupling to the thigh, your knees (patellar tendons, etc.) would not have the opportunity to hurt when you run.

We have been assuming that the runner lands with the heel striking the ground. The measured force in Fig. 3.33 shows a spike at ~1/30 s for a runner wearing cushioned shoes landing this way. The spike is even greater when running barefoot. However, this impact peak disappears when the runner lands on the forefoot (the front of the foot), whether running with shoes or barefoot [85, 87]. In this mode of running, during landing the foot is more plantarflexed (extended) and this may decrease the effective mass of the body in the collision of the foot with the ground due to the increased compliance of the ankle during the landing. Some think this mode of running may lessen impact-related injuries of the feet and legs during running and may be the preferred way to run.

Running shoes have been improved for many years to address the supposed risk factors for running injuries, by controlling foot pronation and minimizing impact

forces. Some have concluded that even with these improvements, there has been little change in running injuries since the 1970s, and to minimize injuries running shoes should be chosen to provide maximum comfort during running, while not changing the runner's natural movement path [111].

Sliding in Baseball

The goal of sliding into second or third base in baseball is to touch the base and then maintain contact with it before being tagged by the fielder. By sliding, you are at ground level and therefore harder to tag. You also have a greater chance of maintaining contact with the base after "colliding" with it, without bouncing off of it. Even better, by sliding you decelerate and collide with the bag at a slower speed, so there is less chance of breaking your foot (or hands in the ill-advised head-first slide). However, because you are decelerating during the slide, it takes a longer time to reach the base, $\Delta t_{\text{slide,extra}}$, and it is more likely you will be tagged out than if you had continued at the same initial speed v_{i} and then suddenly stopped at the base without overrunning it (and damaging your ligaments, etc.). You can (1) control "when" you start your slide—which really means the distance before the base where you start to slide, Δx_{slide}, (2) try to control your (final) speed at the base, $v_{\text{f,slide}}$ (the speed of the collision with the base), and (3) try to control the effective kinetic coefficient of friction in your deceleration μ_{slide}. (You slow down faster with more body area and weight on the ground and with your spikes dug in more. You decelerate slower on mud than on dry dirt.) Assuming a normal force equal to your body weight, the frictional force has a constant magnitude $\mu_{\text{slide}} m_{\text{b}} g$ and so $a_{\text{coll}} = -a_{\text{decel}} = -\mu_{\text{slide}} g$.

Using (3.102), we see that $\Delta t_{\text{slide}} = (v_{\text{i}} - v_{\text{f,slide}})/\mu_{\text{slide}} g$, where Δt_{slide} is the duration of the slide. (Clearly, $\Delta t_{\text{slide,extra}} = \Delta t_{\text{slide}} - \Delta x_{\text{slide}}/v_{\text{i}}$.) Also, using (3.103) we see $\Delta x_{\text{slide}} = v_{\text{i}} \Delta t_{\text{slide}} - 1/2 \mu_{\text{slide}} g \Delta t_{\text{slide}}{}^2$. This is examined more in Problems 3.144 and 3.145.

3.10.6 Jumping

Several forms of jumping involve full collisions with a surface. The spring constant (stiffness) of this surface greatly affects how much energy can be returned to the body. This is seen in Table 3.11, which shows different surfaces, and for each their force constant and maximum deformation, the maximum stored energy, and the maximum height a 70 kg person could attain with this maximum stored energy. This is obviously significant for jumping on trampolines and gym floors.

3.10.7 Chewing Food

Chewing food can be thought of as a collision in the body, of the upper and lower jaws with food in between them. Breaking down the food into smaller pieces so it

can fit in the mouth, *incision*, is accomplished mostly by the incisors, along with tearing action by the canines as needed [86]. This occurs on both sides of the jaw at the same time, with little or no lateral movement. Incision is followed by grinding the food into even smaller pieces, *mastication* or *comminution*. This is accomplished mostly by the molars and the bicupsids (premolars). It usually occurs on only one side of the jaw at a given time, and involves several motions of the lower jaw relative to the upper jaw, including lateral motion, with net forces of 200–400 N over areas of $\sim$1 cm^2. The statics of the jaw are discussed in Chap. 2. The effect of chewing on food is described in the discussion of the mechanical properties of food in Chap. 4, and how processing food affects chewing food and extracting nutrients from it is discussed there and in Chap. 6.

3.11 Sustained Acceleration

We have seen that collisions involving the head and other parts of the body can lead to damage from the direct impact and from accelerations induced by the collision. Sustained linear and rotational (or rotary) accelerations of the entire body are also generally important, and they are particularly significant for astronaut space flight and for fighter pilot excursions [50, 51].

We will define linear accelerations using the coordinate system in Fig. 1.1, with $+g_z$ corresponding to the normal gravity acceleration g we feel on Earth, and consider the effect of these accelerations [49, 50]. For such positive vertical accelerations, it is difficult to raise oneself at 2.5 g_z; it is impossible to raise oneself and difficult to raise one's arms and legs, and vision dims after 3–4 s from 3–4 g_z; and vision is blacked out after 5 s, hearing and consciousness is later lost, and convulsions are possible from 4.5–6 g_z. As we will see in Chap. 7, one reason for these extreme responses is that blood pumped by heart cannot reach the brain. At $+1\ g_z$, the (systolic) arterial pressure is 120 mmHg at the heart, 96 mmHg at the head, and 170 mmHg at the foot. At $+5g_z$, this pressure is still 120 mmHg at the heart, but 0 mmHg at the base of the brain and 370 mmHg at the foot. This means your brain does not get blood (which means you become unconscious) and you need an extra 250 mmHg pressure to pump the blood in the veins in your legs and feet back to the heart. Even though your body compensates for this a bit (by increasing blood pressure, increasing the heart rate, decreasing the cardiac volumetric output, and so on), so you may be able to withstand $+5\ g_z$, the situation is still dire. Vision is affected at lower accelerations than for general brain function because blood flow to the retina has the barrier of the intraocular pressure, which is about 20 mmHg higher than the intracerebral pressure. (Vision may be affected in ways similar to *retinal dysfunction ischemia*.)

Negative vertical accelerations also lead to physiological consequences: there is severe facial congestion, throbbing headaches, and after 5 s blurring of vision

from $-2\,g_z$ to $-3\,g_z$; and most subjects cannot even tolerate 5 s of $-5\,g_z$ because of bloodied eyes, nosebleeds, and hemorrhage.

We can tolerate forward $+g_x$ and backward $-g_x$ acceleration a bit better. A forward acceleration of $2g_x$ is tolerable for at least 24 h and $4g_x$ for at least 60 min. There is increased weight, abdominal pressure, and some difficulty in focusing vision from 2–3 g_x; progressive chest tightness and pain, difficulty in breathing, and blurring of vision from 3–6 g_x; each symptom gets worse from 6–9 g_x; arms cannot be lifted at 8 g_x; the head cannot be lifted at 9 g_x; breathing is difficult and peripheral vision is lost from 9–12 g_x; and there is extreme difficulty in breathing and speaking and a recurrent complete loss of vision at 15 g_x. The effects of backward accelerations are similar, but there is none of the increased pressure on the chest that accompanies forward accelerations, so breathing is easier.

For lateral accelerations, $\pm g_y$, there is discomfort at 3 g_y after 10 s, and external hemorrhage and severe headaches afterward for 5 g_y for 14.5 s.

Rotational acceleration about the body x-axis is called *rolling*, about the y-axis *pitching*, and about the z-axis *yawing* or *spinning*, using the coordinate system in Fig. 1.1. Sometimes the term *tumbling* is used instead of pitching and rolling. Tumbling (about the pitch axis) centered about the heart is safe for rotations rates of at least 120 rpm for about 3 s and for successively longer times at slower rates, such as 70 rpm for 400 s [50, 138].

3.12 Flight of the Ball

We have seen how people can throw and hit balls. *What is the trajectory of these balls and similar projectiles?* The trajectory of a ball depends on its dimensions (mass, shape, surface topography), its initial speed, direction and rotation speed (imparted by a hand or racket or golf club), the effects of air (as we will see, due to drag, lift, the Magnus force due to spinning in air, air conditions, and wind), storage conditions [96], and the difference between its initial height (at which it is thrown or hit) and final height (at which it lands, is caught, goes over a fence or goal post, and so on).

If the ball is not caught, the first stage of flight in air is followed by bouncing and then rolling and maybe skidding before the ball comes to rest. Only the flight phase is important in football passing, basketball shooting (when the backboard is not used), and in baseball pitching and throwing and catching fly balls. The flight and bouncing phases are important in baseball for hit ground balls and fly balls that are not caught by fielders and in basketball for shots off the backboard and bounce passes. In golf, the bouncing and rolling phases are as important as the flight phase. In bowling, the initial bouncing phase and subsequent rolling phase are important. We will look at the flight of the ball, first in vacuum and then in air, and then investigate bouncing and rolling. This is discussed very well in [16, 35–37, 60, 135]. In particular, the books

by Daish [36] and de Mestre [37] solve the physics at a high level. The former book is a bit more comprehensive, while the latter tackles some more difficult problems.

If the velocity vector component of a ball is v_x in the forward direction and v_z in the vertical direction, the equations of motion of a nonspinning ball of mass m_b in gravity in a vacuum are:

$$m_\mathrm{b}\frac{\mathrm{d}v_x}{\mathrm{d}t} = 0 \tag{3.134}$$

$$m_\mathrm{b}\frac{\mathrm{d}v_z}{\mathrm{d}t} = -g. \tag{3.135}$$

The solution when the ball starts and ends at the same height is familiar to you from introductory physics courses, and is reviewed in Problems 3.91–3.94. It is extended in Problem 3.96 to when the initial and final heights are different.

3.12.1 The Effect of Air Drag on Moving Balls and Humans

In Chap. 7 we will discuss how fluids resist motion and how the motion of objects in fluids is affected by the frictional force called drag. For balls moving in air with speed v, drag can be understood by going into the reference frame of the ball and looking at the air streamlines, as in Fig. 3.71. The streamlines flowing around a nonspinning spherical object become more densely packed in the boundary layer and the air speeds up and its pressure decreases (as a consequence of Bernoulli's Principle 7.14) [36].

For very, very slow balls the streamlines recombine after they leave the ball in this *laminar* flow regime where drag varies as v (7.58), and it depends on the ball diameter d and air mass density ρ (average molecular weight $\times$ number density)

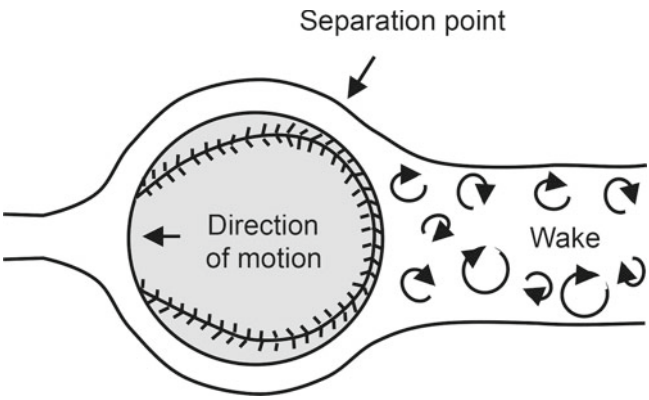


Fig. 3.71 In the rest frame of the ball (i.e., moving along in a frame with the ball), the flow of air around a baseball. The ball is moving to the *left*, so the air is moving to the *right* in the drawing. The air flows along the ball and detaches at the separation point to form a wake of chaotic, swirling flow full of eddies (based on [137])

and viscosity η. As we will see, the relevant parameter is the Reynolds number $Re = \rho v d/\eta$ (7.11), which is $\simeq 640\, vd$ in air, where d is the diameter of the sphere in cm and v is the speed in m/s [36]. (In Chap. 7, the flow speed is called u.) In this regime, $Re < 2000$, as it is for blood flowing in our arteries (see Chap. 8). At the higher ball speeds common in sports (for which $Re \sim 100{,}000$), the streamlines do not recombine after the ball; eddies form and a turbulent wake forms, and this is called the *turbulent* regime ($Re > 2000$). The eddies (Fig. 3.71) have extra energy that must come from a loss of kinetic energy of the ball and this causes the drag force on the ball in the direction opposite to the velocity vector of the ball. As v increases the flow separates from the ball surface earlier, increasing the number of eddies and consequently the drag force increases, usually as v^2 in this hydrodrodynamic flow regime. As we will see in (7.59), $F_{\text{drag,hydrodyamnic}} = \frac{1}{2} C_{\text{D}} A \rho v^2$ where A is the frontal surface area and C_{D} is the drag coefficient.

Because the direction of the drag force is opposite to the velocity vector, which has magnitude $v = \sqrt{v_x^2 + v_z^2}$, it is negative and the fraction of the drag force in the x direction the force is v_x/v and that in the z direction it is v_z/v. The equations of motion of a nonspinning ball in vacuum, (3.134) and (3.135), now become:

$$m_{\text{b}} \frac{\mathrm{d}v_x}{\mathrm{d}t} = -\frac{1}{2} C_{\text{D}} A \rho v v_x \tag{3.136}$$

$$m_{\text{b}} \frac{\mathrm{d}v_z}{\mathrm{d}t} = -g - \frac{1}{2} C_{\text{D}} A \rho v v_z. \tag{3.137}$$

These equations can be solved by numerical integration (Problem 3.101) to find the range, x_{final}, and time of flight (travel time), t_{final}, for an initial speed v_0 and the takeoff or launch angle α, and a given initial and final height. For a nonspinning ball, the motion stays in the same plane (Problem 3.91). (Depending on the direction of the spin, it can be out of plane, as we will see (Problem 3.110).) You can obtain approximate solutions relating the beginning and end of the ball trajectory by using a ballistic table such as Table 3.12 [36, 37] (Problems 3.102–3.106). (This was the easiest way to determine the range of baseballs and other projectiles, such as missiles and bullets, before computers became widely available.) This table provides this approximate solution for the same initial and final heights by relating the specific values of three problem parameters, $b_1 = v_0^2 \sin(2\alpha)/c$, $b_2 = x_{\text{final}}/c$, and $b_3 = t_{\text{final}} v_0 \cos(\alpha)/c$, where $c = m_{\text{b}}/(C_{\text{D}} d^2)$ is the ballistic coefficient. You can work forward from initial conditions to get the final conditions: the range and time of flight. You can also work backward to determine the initial conditions needed to maximize or achieve a given range or travel time. Sometimes a tradeoff between these trajectory goals is necessary. In football you sometimes want to maximize how far the football can be punted (kicked) even if the travel time (hang time) is relatively short, while sometimes times you want to increase the hang time even at the expense of some decrease in range (see Problem 3.95). You will usually need to interpolate between the values given in the table.

For the same initial speed and angle, drag decreases the distance the ball travels, its maximum height and the distance it travels where this maximum occurs; it also makes

Table 3.12 Ballistic table for the flight of a nonspinning ball for a ball beginning and ending at the same height, by relating b_1, b_2, and b_3, as defined in the text

b_1	b_2	b_3
0	0	0
2	0.18	0.20
4	0.34	0.40
6	0.48	0.59
8	0.61	0.78
10	0.73	0.96
12	0.84	1.13
14	0.95	1.30
16	1.05	1.47
18	1.14	1.63
20	1.22	1.79
22	1.30	1.95
24	1.38	2.11
26	1.45	2.26
28	1.52	2.41
30	1.58	2.56

They are given in metric units
Using data from [36, 37]

the trajectory asymmetric (in vacuum it is a symmetric parabola), greatly steepening the angle of descent, as depicted in Fig. 3.72. The maximum travel distance occurs with a takeoff angle of 45° in air and ∼35–40° in air.

In vacuum the range varies as v_0^2, for initial speed v_0 (Problem 3.92), so those who can throw balls faster and hit them harder, can make them travel much farther. With drag this advantage is much less, with each increment of added speed helping

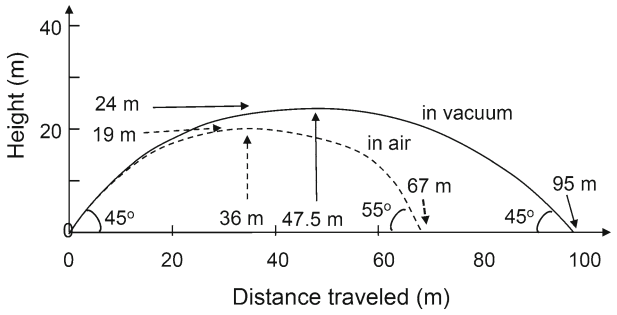


Fig. 3.72 Comparison of the flight of a cricket ball in vacuum and air with initial speed 3 m/s, illustrating the effect of air drag (based on [36])

less, and consequently drag has been called the great equalizer [36]. If a fast initial speed of v_0 is increased by δv ($v_0 \gg \delta v$), the range for takeoff angle 45° increases proportionally to $v_0 \delta v$ in vacuum and only as $\sim \delta v$ with drag [36]. This can be seen by numerical integration of the equations of motion and by measurements. For example, for an initial speed of v_0 (in ft/s), the range of a golf ball before its first bounce (its"carry") in yards has been seen to be $\simeq (3/2)v_0 - 105$. Including bouncing and rolling (its "carry plus run") it is $\simeq (5/4)v_0 - 27$, which, of course, is sensitive to the nature and conditions of the golf course [36].

For Re from ∼1,000 to ∼200,000, C_D is $\simeq 0.45$ for smooth spherical balls; this is the regime usually relevant in sports. For $Re >\sim 200{,}000$, C_D decreases abruptly by about a factor of four because turbulence develops in the boundary layer about the ball enabling more energy to enter the flowing medium from regions away from the ball. This helps the flow overcome the increase in pressure it encounters as the flow passes by, and so separation occurs later in the flow and drag decreases. Re for soccer balls kicked at 30 m/s ($d = 22.1$ cm, $Re \sim 425{,}000$) exceeds this limit, and consequently they travel ∼60 m at the optimal launch angle, instead of the ∼40 m they would travel with $C_D = 0.45$ [36]. Ball roughness decreases the critical Reynolds number. Smooth golf balls moving at this same speed would have Re below this critical value ($d = 4.27$ cm, Re $= 82{,}000$), but their dimples reduce the critical Reynolds number and they travel much farther because C_D is much smaller [36]. The dimples also increase the Magnus force due to backspin and provide much lift (as we will see soon).

Balls travel farther when drag forces are smaller. We have seen how surface roughness can decrease C_D. Because of the area factor, drag increases as d^2 for spherical objects, so, other things being equal, there is less drag on small objects. When footballs travel in the direction of their long axis (which is called a "spiral", when they are thrown like this and with spin about this direction), the streamlines are guided better along the contour and separate later, so drag is only about two thirds of that expected for a spherical ball with the same circular cross section. However, tumbling (end-over-end) footballs offer a larger average area and more drag than those that travel with their noses first. For example, footballs kicked in air at a speed of 55 mph at a takeoff angle of 55°, travel 51 yards when moving nose first and only 34 yards when tumbling (which are both much less than the 63 yards they would travel in vacuum) [16]. Drag decreases when the mass density decreases due to decreased molecular number density or decreased average molecular weight. The former occurs when temperature increases (at a given pressure) and when the elevation increases (lower pressure for a given temperature). The latter occurs when the absolute humidity increases (because H_2O has lower molecular mass than N_2 and O_2); the maximum absolute humidty increases with temperature (see Problem 3.103). Changes in g due to changes in latitude (the earth bulges at the equator so g is smallest there) and (local) elevation affect ball flight little. As we will soon see, air conditions also affect how a ball moves in air due to its spin.

Wind can change the effect of drag on the trajectory of a ball (as well as other effects due to air, including lift). The effect of wind can be treated by (1) moving into the rest frame of the wind (which can be assumed to be constant or variable in time),

so the ball velocity vector **v** is replaced by **v**–**w**, where **w** is the wind velocity vector, (2) solving the relevant equations, such as the transformed (3.136) and (3.137), and (3) then transforming back. Considering the effect of drag alone, a soccer ball initially kicked with a speed 28 m/s at an angle 50° has a range of 47.3 m with a time of flight of 3.7 s. Kicked into a 5 m/s headwind it moves 38.4 m in 3.8 s, with a 5 m/s tailwind it moves 61.0 m in 4.2 s, and with a 5 m/s crosswind it moves 48.4 m forward in 3.9 s with a deflection of 5.9 m into the direction of the wind [37]. Wind can change the trajectory greatly.

While it is not possible to solve the equations of motion of balls and humans with drag analytically in two or three dimensions, it is possible to do so in one dimension for some interesting cases. Dropped balls attain a terminal velocity when the forces due to gravity and drag balance. This is also true for people who are dropped from high heights, with or without parachutes. A skier racing down a planar slope also travels in one dimension, subject to gravity and drag forces, as well as friction. We now analyze the motion of the downhill skier, which easily reduces to the motion of dropped objects. We will talk more about drag in all fluids (including air) in Chap. 7, along with the motion of humans in different fluid, water [135].

You can see from Fig. 3.73 that the force of gravity leading to acceleration down a slope at angle α to the horizontal is $m_\mathrm{b} g \sin\alpha$ and the part that produces the normal force that leads to kinetic friction is $m_\mathrm{b} g \cos\alpha$, where m_b is the mass of the skier (including skis and clothing). The equation of motion of a downhill skier on this planar slope is

$$m_\mathrm{b} \frac{\mathrm{d}v}{\mathrm{d}t} = m_\mathrm{b} g \sin\alpha - \mu_\mathrm{k} m_\mathrm{b} g \cos\alpha - \frac{1}{2} C_\mathrm{D} A \rho v^2, \tag{3.138}$$

where s is the distance along the slope and $v = \mathrm{d}s/dt$, A is the effective area the skier presents, and μ_k is the coefficient of kinetic friction. Eventually the skier attains a terminal velocity, which we can obtain by setting $\mathrm{d}v/\mathrm{d}t$ equal to zero, to give

$$v_\mathrm{terminal} = \sqrt{\frac{2 m_\mathrm{b} g(\sin\alpha - \mu_\mathrm{k} \cos\alpha)}{\rho A C_\mathrm{D}}}. \tag{3.139}$$

As expected, the terminal speed is larger for steeper slopes and lower friction, and when the effective area of the skier decreases. A skier with total mass 80 kg (including

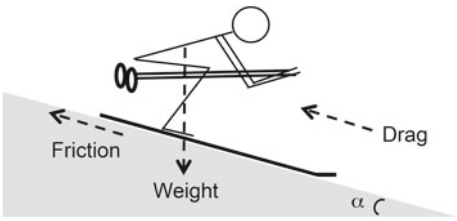


Fig. 3.73 Forces on a downhill skier (based on [135])

clothing and skis) descending a 15° slope attains a terminal speed of 25.1 m/s (with $A = 0.6\,\mathrm{m}^2$, $C_\mathrm{D} = 0.7$, $\mu_\mathrm{k} = 0.05$, $\rho = 1.25\,\mathrm{kg/m}^3$) [135]. Problem 3.66 gives the analytic solution to this one-dimension equation of motion (3.138) as a function of time (and the steps involved in the solution). All of these results are readily applied to dropped balls and other objects by setting α equal to 90°. (When you do this the effect of friction automatically disappears. (Why?))

This analysis of skiing down a one-dimensional downward slope can be extended to skiing down a two-dimensional planar slope. Call the lateral coordinate x and the downhill coordinate on the plane y, which increases down the slope, and the downhill angle to the horizontal α. The skier decides to start at rest at $x = 0$ and $y = 0$ or $(0, 0)$ and end at $x = \beta y_\mathrm{f}$ and $y = y_\mathrm{f}$ or $(\beta y_\mathrm{f}, y_\mathrm{f})$, and so wants to move laterally as she skis downhill. *Which is the fastest route?* We will ignore drag and friction. The answer is not a straight line along the plane, but a cycloid—which is the locus of a point on the circumference of a circle as it rolls on a straight line. The solution to this "brachistochrone" optimization problem is outlined in Problem 3.69 and is depicted in Fig. 3.74 [88]. The shape of this path depends only on β.

Another interesting feature of motion down a slope occurs when there is local curvature in the path [88, 104]. Imagine going downhill in a cart that is in a track and that in some region you enter a region with local curvature R while traveling at a speed v. If you lean into the turn, your center of mass moves closer to the center of curvature by h, which is assumed to be much less than R. Because there are no external torques on the cart, your angular momentum is unchanged, but the radius of curvature decreases to $R - h$, so you leave the turn with a fractional increase in speed of h/R and kinetic energy of $2h/R$. (Where does the increase in kinetic energy come from?) A downhill skier can pump up the speed the same way by entering a trough with radius of curvature R in a crouched position and then standing up in the trough, increasing the center of mass by h (see Problem 3.70).

Does it take less time for a sledder or skier to traverse a valley by going down it and then up it or by going directly across it on a horizontal bridge—if she enters the region with the same horizontal speed and always maintains contact with the surface? With no friction, the valley route is faster because the horizontal speed is always faster in the valley than the final speed when she leaves the valley, which is the same as her initial horizontal speed and the constant speed for the bridge route.

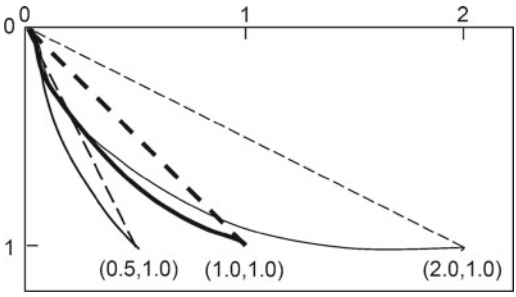


Fig. 3.74 Downhill skiing routes that minimize travel time, with the downhill distance normalized to 1.0. (based on [88])

When there is friction, the horizontal route becomes faster above a critical coefficient of friction [73]. (How would drag affect this?)

3.12.2 The Flight of a Spinning Ball

The rotation of a thrown or batted baseball affects its motion [16, 36, 60, 108]. The ball thrown by a pitcher rotates due to wrist rotation and arm motion, such as the screwdriver action of the radius rolling on the ulna (described in Chap. 1), and by how he grips the ball. The pitcher tries to maximize this rotation speed (ω, in rad/s) when throwing a curve ball and minimize it when throwing a knuckleball (as is seen in Table 3.13 in the problem section). The spin can be topspin (which causes a downward force on the ball), backspin (which causes an upward force called lift), or in another direction (which could cause the ball to curve). The rotation axis vector of the ball is the direction toward you if you see it rotating counterclockwise, as seen in Fig. 3.75, so if a ball is thrown horizontally along the positive y-axis and the z-axis points up, backspin (or bottom spin) and topspin (or forward spin) respectively point along the positive and negative x-axis; the ball will curve if the spin vector has a component along the z-axis. Spin can also be imparted when throwing other types of balls, such as when a football quarterback throws a spiral with rotation about the travel (y) direction and when a basketball player shoots the ball by letting it roll off the fingers so there is backspin (along the x-axis) (though in these specific cases the spin has another interesting, but different effect on the ball trajectory, as we will see). As we learned earlier this chapter and will see in more detail later, spin can also be imparted to balls when you hit them, as with a bat or racket (baseballs, golf balls, tennis balls, and so on) or your feet (soccer balls), due to friction during impact. Spin affects the ball trajectory, which we discuss now, and how a ball bounces, which we will discuss later.

This rotating motion can alter the path of pitched baseballs and confuse batters, help avoid defenders by making a kicked soccer ball follow a curved path (as noted in the movie "Bend it like Beckham") (and see [58]), and make golf balls travel much farther. You will severely underestimate the range of a golf ball if you calculate its trajectory and ignore lift due to its fast backspin.

Why does a spinning ball move this way? A spinning ball feels the *Magnus force* due to this rotation, which changes its path from that of a rotation-free ball. When a nonspinning ball travels in air, the air travels symmetrically on the top and bottom and it leaves a wake in its path. As sketched in Fig. 3.71 in the rest frame of the ball, the air streamlines separate from the ball symmetrically and this wake is symmetrical and there is no force normal to the direction of the ball. For the spinning baseball in the photograph in the rest frame of the ball in Fig. 3.76, the wake is noncentered because the air flows farther along the "top" where the upper surface of the rotating ball is moving in the same direction as the airstream than along the "bottom" where the lower surface of the rotating ball is moving in the direction opposite to that of the airstream. The wake is pushing air "downward" and there is an equal and opposite

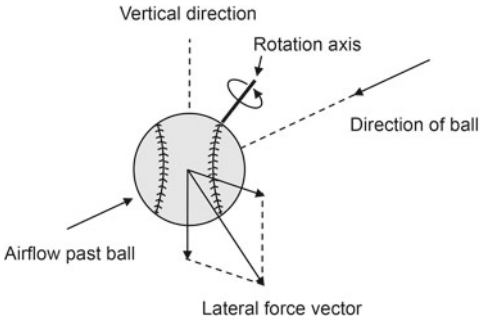


Fig. 3.75 Direction of the Magnus force (based on [137])

reaction that pushes the ball "upward" according to Newton's Third Law. Figure 3.75 shows that the direction of this Magnus force is normal to the rotation axis vector of the ball (whose direction is to you if you see it rotating counterclockwise) and the ball velocity vector, in the direction of their cross-product. If these two vectors point in the x and y directions, respectively, the Magnus force will be in the z direction [16, 36].

The Magnus force explains why thrown balls do not always follow the same trajectories as expected by gravity and drag. Spinning balls can curve, such as with thrown curve balls. Balls thrown with topspin drop faster than expected by gravity alone and those with backspin drop slower. Balls thrown by a right-handed pitcher normally spin counterclockwise, when viewed from above, and curve away from a right-handed hitter. Balls thrown with clockwise spin (curve ball from a left-hander or screwball from a right-hander) curve to a right-handed hitter. Balls rotating at other angles have motion both vertically and horizontally, aside from the drop by gravity. This is discussed more in [1] and [137], and in Problems 3.108–3.113.

Spin imparted in a collision when there is friction between the racket, bat or foot and the ball depends on the location of the collision (off center in a bat, at

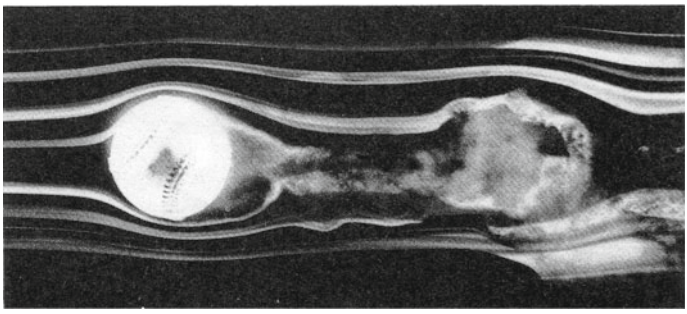


Fig. 3.76 A baseball rotating clockwise in a wind tunnel, with air moving to the right. (Smoke has been added to make the air motion visible.) The streamline detaches later on the top—where the upper surface of the ball moves in the same direction as the streaming air—leading to a downward wake and a net upward force on the ball (from F.N.M. Brown, courtesy of the University of Notre Dame Hessert Laboratory)

angle for a golf ball) and the motion of the racket (tennis and ping pong ball)—and sometimes it is hard to distinguish between the two. In baseball, balls hit to the right side of the park curve to the right and those hit to the left side curve left because of spin. With topspin (see Fig. 3.70), a ping pong ball can clear the net, and then plunge and hit the table, and bounce very high. With backspin, as in a drop shot in tennis, the ball seems to "die" after the bounce. Golf balls driven from a tee that are undercut develop backspin, with the initial lift almost balancing gravity, so the balls travel farther. In golf, the loft of the club (which is the angle the face makes with the vertical), can range from $\sim 10^\circ$ for a driver to $\sim 50^\circ$ for a wedge, leading to spins of $\sim$50, 100 and 130 revolutions per sec for wedge, 5-iron, and 7-iron shots respectively [36]. Approximately 80% of the spin of a golf ball remains after a flight of 5 s. Unfortunately, hitting a golf ball above its center with the bottom of the club produces topspin and the ball plunges faster than expected. Also, spin pointing up and down respectively causes the golf ball to draw or hook to the left and fade or slice to the right. In soccer, sometimes the kicker trying to score a goal wants to impart spin pointing up or down so the ball will curve when a direct shot on goal is not advisable (as when the ball needs to go around defenders) or possible (as in a corner kick, when a line of sight goal is not possible). Billiard balls develop topspin and backspin by hitting them with cues above and below their centers; hitting them on either side causes curving motion.

Deviations from a purely spherical shape can affect the motion of nonspinning balls, such as those due to the general roughness and seams on baseballs (Problem 3.114 addresses throwing knuckleballs and scuffed balls), the rings on the surface of cricket balls (Problem 3.115), and the series of oval holes on one half of Wiffle balls [123], because they redirect air flow. Dimples on golf balls help increase the lift force and likely help the ball spin more when it is hit by the golf club (and also decrease drag).

For nonspherical projectiles, spin affects orientation due to the Magnus force and drag [16, 36, 60]. The long axis of a football or javelin thrown in a tight spiral (spinning fast about its long axis) tracks the velocity vector. This would seem to violate conservation of angular momentum because the direction of spinning angular momentum changes during flight, but can be seen to be due to drag. There is no Magnus force associated with this spinning, because the direction of the spin and the velocity vector are parallel. A wobbly thrown football (which is not a spiral) presents a greater area for drag, which decreases how far it can travel. Spinning is also very important in the motion of the discuss, Frisbee, flying ring and boomerang [37].

Spin also affects bouncing, as we will see soon.

3.12.3 Bouncing, Rolling and Hitting Balls, including those with Spin

We now consider the kinematics of a spherical ball of radius r, mass m_b, and moment of inertia I bouncing on a hard, flat surface, as in Fig. 3.77 [28, 36, 41]. Before it hits

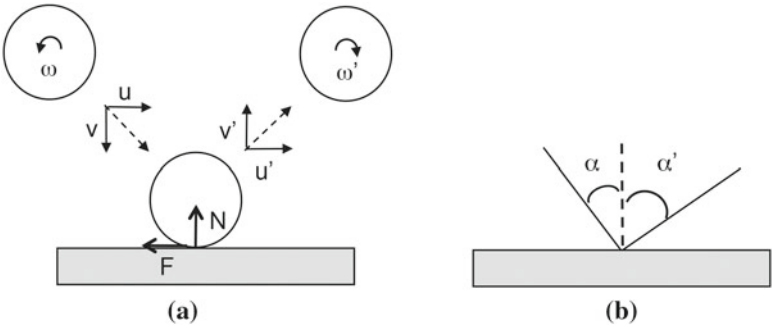


Fig. 3.77 Schematics of a ball bouncing on a hard surface, with all parameters defined as positive as shown (based on [36])

the surface it is spinning at a rate ω, which we define as being positive for backspin, and it has linear velocity components u and v parallel to and normal to the surface, with incident speed $V = \sqrt{u^2 + v^2}$. After the bounce it spins at a rate ω', which we define as positive for topspin, and velocity components u' and v' parallel to and normal to the surface. Because the spinning rate of the incident ball in this example is parallel to the surface, the incident and bouncing ball move in the same plane. The angles of incidence and rebound are given by $\tan\alpha = u/v$ and $\tan\alpha' = u'/v'$.

During the collision there is a normal or reaction force $N(t)$, which causes a change in the linear momentum of the ball normal to the surface. When this force is integrated over the collision time, it gives the impulse N', which is related to the change in normal momentum by

$$N' = m_\mathrm{b}(v + v'). \tag{3.140}$$

Also, during the collision there is the force of friction $F(t) = \mu_\mathrm{k} N(t)$ at the point of contact, where μ_k is the coefficient of kinetic friction. This causes a change in linear momentum parallel to the surface. Integrating this force over the collision duration gives F', where

$$F' = m_\mathrm{b}(u - u'). \tag{3.141}$$

The frictional force also creates a torque about the center of mass of the ball, rF'. When integrated over the course of the collision, there is a change in angular momentum of the ball

$$rF' = I(\omega + \omega'). \tag{3.142}$$

We will assume that the ball is solid and spherical, so $I = (2/5)m_\mathrm{b}r^2$. Because the surface is fixed and massive, (3.108) gives

$$v' = ev, \tag{3.143}$$

where e is the coefficient of restitution.

More information is needed to find u', v' and ω'. If friction is relatively strong, the ball will stop skidding (and sliding) before the collision is over and momentarily roll. This means that the ball does not move at the point of contact because the forward speed of linear translation equals the backward speed due to rotation, or

$$u' = r\omega', \tag{3.144}$$

and so the friction force stops before the collision is over. Equations (3.141) and (3.142) can be solved (Problem 3.122) to give

$$u' = \frac{5u - 2r\omega}{7} \tag{3.145}$$

$$v' = ev \tag{3.146}$$

$$\omega' = \frac{u'}{r} = \frac{5u - 2r\omega}{7r} \tag{3.147}$$

$$e \tan \alpha' = \frac{5}{7} \tan \alpha - \frac{2\alpha\omega}{7V \cos \alpha}. \tag{3.148}$$

If friction is relatively weak and the ball never stops skidding during impact (see Problem 3.124 to find this condition), then $F(t)$ is always proportional to $N(t)$ during impact and this is also true for their integrated values, so

$$F' = \mu_k N'. \tag{3.149}$$

Solving (3.140)–(3.142) (Problem 3.123) gives

$$u' = u - \mu_k v(1 + e) \tag{3.150}$$

$$v' = ev \tag{3.151}$$

$$\omega' = \frac{5\mu_k v}{2r}(1 + e) - \omega \tag{3.152}$$

$$e \tan \alpha' = \tan \alpha - \mu_k(1 + e). \tag{3.153}$$

Figure 3.78 illustrates some of general trends for bouncing balls [36]. As the rate of topspin of the incident ball increases, the ball rebounds at larger angles and with faster speed. As the rate of backspin increases, the rebound angle decreases, and the ball can bounce backwards. The rebound speed decreases for slower rates, but can increase at even faster rates. Backspin deadens the bounce, which is why golfers use backspin in chip shots to slow down rolling on the green, as do tennis players attempting drop shots and basketball players who use the backboard when shooting. So-called bad bounces in baseball usually occur due to irregularities of

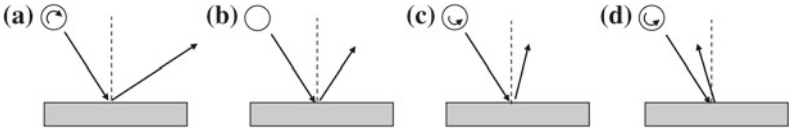


Fig. 3.78 Examples of a ball bouncing on a hard surface, with **a** topspin, **b** no spin, **c** some backspin, and **d** more topspin (based on [36])

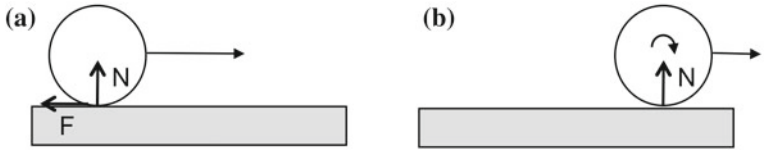


Fig. 3.79 A ball **a** initially skidding **b** begins to roll (based on [36])

the ground; however, balls can also bounce in unexpected ways if they move slowly and spin unexpectedly fast, so the rotation speed of the ball surface due to spinning approaches the linear speed of its center of mass [16]. The bouncing of nonspherical balls, such as footballs, can be unusual [16, 29].

After several bounces, a spherical ball can start to roll. In bowling and bocce, the goal is to roll the ball without bouncing. As in bouncing, the ball may first skid (slide) before rolling (Fig. 3.79), and so there are similarities in the analysis of both problems. A ball rolls forward without skidding when the forward speed of its center of mass v equals ωr, where ω is the rotation speed and r is the ball radius, and the bottom of the ball is no longer sliding on the ground [36]. Before that it skids and friction provides a retarding force of $\mu_k m_b g$ at the point of contact, so its speed decreases from the initial value v_0, as $v(t) = v_0 - \mu_k g t$. Friction also causes a torque $\mu_k m_b g r$, leading to an angular acceleration of $\mu_k m_b g r / I$. We will use the moment of intertia I for a solid sphere, $(2/5)m_b r^2$, so the angular speed increases linearly with time as $\omega(t) = (5/2)\mu_k g t / r$. The rolling condition is $v(t) = \omega(t) r$ after which there is no more kinetic friction, and this occurs at $t = (2/7)v_0/\mu_k g$, which gives the final v of $5v_0/7$ and final ω of $(5/7)v_0/r$. We see that while the final speed is independent of the level of friction, rolling begins earlier with more friction.

One activity where rolling would seem to be important is bowling. However, bowling balls spin and skid most of the way down the lane before they begin to roll and (for good bowlers) hook to the left (for righties), so they approach and hit the pins at an oblique angle and can knock down many pins. The solutions of the equations of motion following translation and rotation of the ball and friction between the ball and the lane [52, 74] show that this desired level of hooking occurs because the ball is not symmetric (its center of mass is not in the middle of the ball and its three moments of inertia are unequal, because it has holes for your fingers) and because the lane is prepared so friction increases down the lane near the pins.

We have seen how spin affects how a ball bounces, both directly due to the spin during the collision with the ground and indirectly through its altered trajectory and

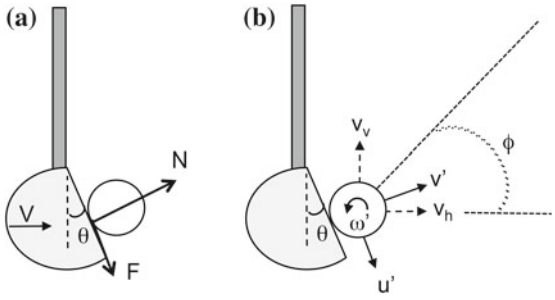


Fig. 3.80 The collision of a golf clubhead and ball just **a** before and **b** after impact (based on [36])

angle at which it hits the ground [16, 36, 60]. Backspin on a basketball makes it bounce off the backboard more gently. Backspin on a tennis ball (in a drop shot), helps deaden its bounce. Balls with topspin seem to jump up fast after the bounce because they hit the surface at a steeper angle, and this counters the shallower bounce that is expected from the spin itself. We now use our results on rolling and bouncing to examine two examples of hitting balls, a golf ball with a club, which leads to motion in three dimensions (and in a plane when there is no sidespin), and a pool ball with a cue, which leads to motion in two dimensions (when the ball does not leave the pool table). Ball rotation is very important in both cases.

Figure 3.80 depicts a golf clubhead hitting a solid spherical golf ball of radius r and mass m_b [36]. The head has a loft (that is, its hitting surface is at an angle) of θ with the normal. During the collision there is a normal force and a force due to friction. The effect of gravity on the ball during the collision is small and will be neglected. The head has mass M and is moving horizontally at a speed V. After the collision, the ball has a speed v' normal to the surface and u' along it, the corresponding total speed $v = \sqrt{v'^2 + u'^2}$ and horizontal and vertical speeds v_h and v_v, and rotates at a rate ω', which is positive for backspin.

To examine motion along the clubhead, we can analyze this problem easily if we first move to a frame moving to the right with speed V so the clubhead is stationary and the ball it hitting it as it would any stationary surface [36]. We will analyze how the ball rolls along the surface during the collision and then move back to the initial frame. In the rest frame of the clubhead, the velocity vector of the ball is $-\mathbf{V}$ as it hits the surface with a speed of $V\cos\theta$ moving to the surface and $V\sin\theta$ moving up along the clubhead. As with a rolling ball, if friction is large enough that the ball stops skidding during impact, it will have a terminal speed upward along the head of $-(5/7)V\sin\theta$ and develop a back spin $\omega' = (5/7)(V/r)\sin\theta$. Moving back into the frame of the ball before the collision, the ball has a speed along the surface of $u' = -(5/7)V\sin\theta + V\sin\theta = (2/7)V\sin\theta$. We now use (3.110) to describe how the moving clubhead surface hits the stationary ball with a normal speed $V\cos\theta$, and see that the speed of the ball normal to the surface is $v' = (1+e)(M/(M+m_\mathrm{b}))V\cos\theta$ after the collision. This leads to

$$v = \sqrt{\left(\frac{(1+e)MV\cos\theta}{M+m_\mathrm{b}}\right)^2 + \left(\frac{2V\sin\theta}{7}\right)^2} \tag{3.154}$$

$$v_v = v'\sin\theta - u'\cos\theta \tag{3.155}$$

$$v_h = v'\cos\theta + u'\sin\theta \tag{3.156}$$

$$\tan\phi = \frac{v'\sin\theta - u'\cos\theta}{v'\cos\theta + u'\sin\theta}. \tag{3.157}$$

These relations can be used to show that as the loft increases from 0 to 50° for a 0.25 kg clubbing head moving at 30 m/s hitting a 0.046 kg golf ball with 20.6 mm radius (with $e = 0.7$), the initial ball speed decreases from 43.2 to 28.5 m/s, the takeoff angle increases from 0 to 37° (always being less than the loft), and the backspin rate increases from 0 to 127 revolutions/s [36]. The range of the ball before bouncing can be determined using these results by including drag and the lift provided by the backspin. For drives from a tee, we would have to relax the assumption that the clubhead moves horizontally when hitting the ball. When the ball is on a tee, contact can be made just after the clubhead passes its minimum, so it can hit the ball with an upswing, which increases the takeoff angle.

Figure 3.81 shows three ways of hitting a pool ball with a cue [31, 36, 135]. When hit in the center, the ball initially does not rotate, so it skids until it starts to roll, as we have seen. When hit above center its rolls with topspin and usually skids at first, but if it is hit above center at a height 70% of its diameter, it rolls with topspin and does not skid. (Why? See Problem 3.169.) When hit below center, it develops backspin and initially skids. This leads to a curious behavior when it hits another pool ball (that is, it is curious if you ignore ball rotation), which can be understood by considering the analogous case of a sliding disk moving with velocity **v** that collides elastically with a stationary disk with the same mass. The first disk becomes stationary and the second one moves at velocity **v** (as is well known, Problem 3.153). When a ball whose center of mass moves with velocity **v**, while it rolls or skids, collides elastically with a stationary ball with the same mass and radius, the center of mass of the first ball is at first stationary but it continues to rotate as before the collision and the center of mass of the second ball initially moves with velocity **v** (Fig. 3.82). The spins of the balls do not change during the collision because there is no torque about the center of mass of either ball. The second ball skids until it begins to roll, as we have

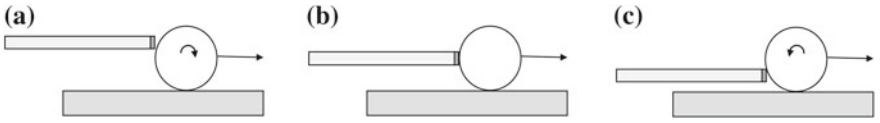


Fig. 3.81 A cue hitting a pool ball **a** above, **b** at, and **c** below its center, shown just after the collision (based on [36])

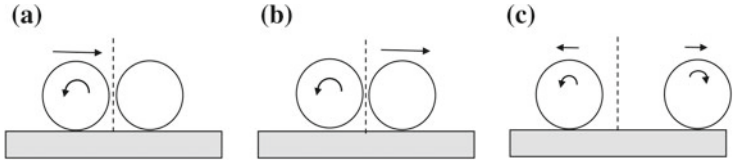


Fig. 3.82 Collision of a cue ball with backspin (*bottom spin*) hitting a stationary, nonpsinning pool ball on a pool table **a** just before, **b** just after, and **c** some time after impact (based on [36, 135])

seen. If the first one had topspin, it will soon move forward due to friction, and if it had backspin, it will soon move backward due to friction (Fig. 3.82), sometimes returning to where it was first hit or even past this point.

Other examples of how bats and rackets impart spin were given earlier in this chapter in our discussion of how spin affects trajectory.

3.13 Summary

Motion of the body can be understood by analyzing the kinematics, dynamics, and energetics of the motion. Understanding body stability and the action of skeletal muscles is also important. Walking, running, jumping, and throwing a ball can be analyzed by using simple models, which can be refined to be more realistic. Similarly, collisions can be modeled to understand body motion and action, as in sports, and how the body is affected when it is involved in a crash.

Problems

Stability, Friction, Human Moments of Inertia, and Muscles

3.1 Consider a person modeled as a one-dimensional being of height H and mass m_b in the two-dimensional world of Fig. 3.8a. The person has a constant mass per unit length and the body rests 25% from the back of her massless feet, which have length $0.152\,H$ (as from Table 1.6) and negligible height. To what angles can the person tilt in the forward and backward directions and still maintain overall balance and stability? (Consider potential rotations about the front and back of the feet, as in Fig. 3.8a.)

3.2 Can the pregnant woman, modeled in the two-dimensional rigid-figure world of Fig. 3.8b, maintain overall balance and stability? Assume the centers of mass of the woman without the womb and the womb itself are both $0.58\,H$ above her feet. The additional mass is $0.25\,m_b$ and has a center of mass $0.1\,H$ in front of the rest of her body. (Assume the other information given about the feet in Problem 3.1.)

3.3 Refer to the gymnast in the arabesque position described in Problem 2.11. In attaining this position her center of mass descends but does not move laterally. How does this affect her overall stability?

3.4 A person is supported by two crutches. Sketch diagrams of the patterns of the person's feet and the bottoms of the two crutches, when the cructches are held either in front of, behind, or to either side of the person. Describe the stability provided by each arrangement.

3.5 When you stand on a moving bus, how do you arrange your feet to try to maintain stability and why?

3.6 In football, lineman often line up near the line of scrimmage before the play starts crouching in a three-point stance (two feet and one hand on the ground) or a four-point stance (two feet and two hands on the ground), and when the play starts they push forward with their legs [60].
(a) Explain why these stances help them push forward well and maintain a low center of mass.
(b) For each stance, sketch a diagram of the ground showing the areas spanned by their feet and hand(s), the area of stability, and the vertical projection of the center of mass onto the ground.

3.7 Explain why the center of mass of a standing child, as determined by the fraction of body height over the soles of the feet, is higher ($\sim$5%) than in adults [16].

3.8 (a) Why is a tripod stable?
(b) Explain why a moving six-legged creature (an insect) can always be stable if it lifts and moves three feet at a time, while keeping its other three feet on the ground [117].
(c) Explain how a four-legged animal (a horse) can move, albeit slowly, always maintaining stability, by keeping three of its feet on the ground at all times. Is this type of motion normal for a four-legged animal? Why?
(d) The two-legged motion of humans cannot take advantage of the stability of a tripod. Still, standing barefoot on one foot or both feet can sometimes approximate a tripod. Explain this and why such tripods are fairly unstable.

3.9 How does the force due to the biceps brachi cause rotary motion in the different throwing phases in Fig. 3.47a due to the force component normal to the lower arm [48]? Show in addition that in the early phases when $\theta > 90°$ there is a component of this force that tends to stabilize the lower arm with the upper arm and that in the later phases when $\theta < 90°$ there is a component that tends to destabilize it, with the force tending to dislocate the lower arm. Use force diagrams to illustrate each of these points.

3.10 The normal force on the head of the femur in the knee is 300 N. Find the frictional force for a normal ($\mu = 0.003$) and arthritic ($\mu = 0.03$) joint.

3.11 What would happen if the coefficient of friction between a hand and the vault were too small during vaulting in gymnastics?

3.12 When we push a heavy object, why do we usually lean into it?

3.13 The accelerating sprinter described in this chapter pushes off with 560 N force. If the net downward force on the foot on the ground is $3\,m_b g$ for this 70 kg sprinter, what is the minimum coefficient of friction needed to prevent slipping? Under what conditions is this possible?

3.14 Use the data from Chap. 1 to show that the moments of inertia of a normally standing person about his or her center of mass are roughly the following, about these axes [66]:
(a) 1 kg-m^2 about a vertical axis.
(b) 11 kg-m^2 about the axis in the transverse plane that is normal to the vertical axis.
(c) 11 kg-m^2 about the axis in the mid-sagittal plane that is normal to the vertical axis.
(d) If a person curls up into a ball the moments of inertia about all three axes will be roughly the same. Estimate them.

3.15 Calculate the moments of inertia of a standard man about the three normal axes though his center of mass—described in Problem 3.14, using each of the models in Problem 1.32.

3.16 (a) For each limb and limb segment in Table 1.9, show that the sum of the squares of the radius of gyration about the center of mass and the distance of the proximal end from the center of mass equals the square of the radius of gyration about the proximal end. (All distances can remain normalized by the segment length.)
(b) Repeat this for the distal end.
(c) Why is this so?

3.17 Use the parallel axis theorem to calculate the moment of inertia of a sprinter's leg about the hip axis, as in (a) Fig. 3.83. The moments of inertia of the upper leg (thigh), lower leg (calf), and foot about their respective centers of mass are 0.1052, 0.504, and 0.0038 kg-m^2, these centers of mass are, respectively, 0.30, 0.45, and 0.53 m from the hip rotation axis, and these segments, respectively, have masses 7.21, 3.01, and 1.05 kg.

3.18 How do the leg joint angles defined in Fig. 3.1 differ from those in our discussion of walking and those defined in the caption to Fig. 3.55?

3.19 Relate the angles defined in Problem 2.38 for the multisegment model of Fig. 2.58 to the joint angles.

3.20 Refer to Fig. 3.5 and describe how activating either muscle 5 or 6 along with muscles 2 and 4 affects hip and knee flexion and extension. Characterize each in terms of an effective redistribution of torque between the hip and knee.

3.21 Refer to Fig. 3.5 and describe the ways to activate leg muscles to extend the knee without applying any torque to the hip. What are the advantages of each?

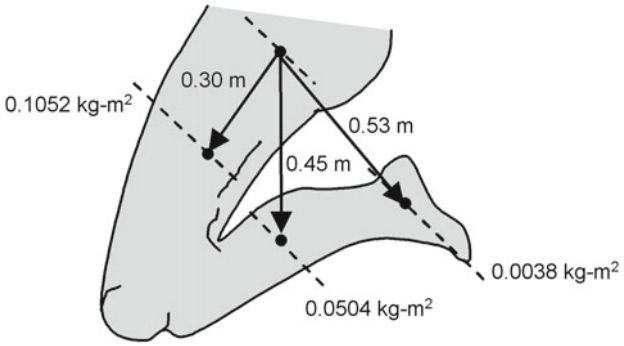


Fig. 3.83 Determination of the moment of inertia of a sprinter's leg about the hip axis (based on [66].) For Problem 3.17

Walking and Running

3.22 Use Fig. 3.10 to determine the fraction of time that both legs, one leg, and neither leg is on the ground during:
(a) Walking
(b) Running.

3.23 Sketch the direction of normal forces on the ground-bound foot during the stages of stance. (Hint: At midstance the normal force is vertical, and before and afterwards that foot is decelerated and accelerated respectively and the body center of mass changes.)

3.24 (a) Use the pendulum model of walking to determine how fast someone walks. Every half period the person takes a step that corresponds to the arc length the foot (at the end of the leg of length L) that traverses 30° during the step (so the step length is $(\pi/6)L$). (This is approximately the maximum swing angle for a fast walk.) (What are you assuming about the swing phase in the gait cycle?)
(b) Calculate the pendulum frequency and this walking speed for a 2 m tall adult and a 1 m tall child. Do the ratios of their frequencies and speeds make physical sense? Explain. (You can use the model that assumes constant leg linear density.)

3.25 Show that it is reasonable that leg length L scales as $m_b^{1/3}$, and therefore the walking speed scales as $m_b^{1/6}$, as in Table 1.13.

3.26 Obtain the moment of inertia about the center of mass of a leg of mass m_{leg} and length L_{leg} with constant linear mass density by:
(a) Changing the integration limits in (3.30)
(b) Using the results of (3.30) and the parallel axis theorem.

3.27 Compare the moments of inertia of the whole leg as determined by the constant linear density model, (3.30) and the more refined model (result after (3.34)), with that determined from the total leg radius of gyration given in Table 1.9. (Use the parallel axis theorem.)

3.28 How can using a walking hiking stick help you when hiking in regard to:
(a) going uphill,
(b) going downhill,
(c) improving your stability (center of mass), and
(d) preventing slipping?

3.29 Why can it be easier to walk and run on wet sand than on dry sand?

3.30 Estimate what fraction of the change in body angular momentum due to leg motion is canceled by swaying your arms out of phase with your legs when you walk. Assume standard parameters for your arms and legs, that they are straight and each has a uniform mass density per unit length:
(a) First assume that $\mathrm{d}\theta/\mathrm{d}t$ is equal in magnitude and opposite in sign for the legs and arms.
(b) Now assume that the forward displacements traversed by the foot and hand $L\mathrm{d}\theta(t)/\mathrm{d}t$ are equal in magnitude and opposite in sign, where L is alternately the leg or arm length.

3.31 Estimate the vertical angular momentum gained by your body with each stride (ignoring the motion of your arms and torso) for a 3.4 m/s running speed.

3.32 Each step in climbing a staircase can be modeled in two parts. First, one foot is placed on the next step and, second, that foot is used to propel the body up. Assume that the center of mass is raised an insignificant amount in the first part, so the whole change in the center of mass occurs in the second part of the step. Also, assume that the durations of both parts of the step are the same. Consider a staircase with 13 steps, each 20 cm high. It takes 3.6 s to go up the staircase "quickly" and 6.0 s to go up "slowly." Calculate the average total (not net) vertical force in the second part of the step for each case, in terms of the body weight W_b. (You can ignore horizontal motion.)

3.33 Determine the total range of motion of thigh, knee, and ankle for each example of running in Table 3.6.

3.34 (a) A 70 kg person runs at a speed of 4.50 m/s. If 100 J of kinetic energy is lost each time a foot touches the ground (Fig. 3.28), what is the speed after stage (a) (if it was 4.50 m/s just before stage (a)). (Ignore vertical motion and potential energy changes throughout this problem.)
(b) How much energy needs to be supplied by the body in the acceleration phase (stage (c)) to account for the loss when the foot earlier hit the ground, if the 93% of the 35 J which is stored in the Achilles tendon and the 80% of the 17 J which is stored in flattening the foot arch in stage (a) are both returned in stage (c) (and nothing else is)?
(c) If the runner takes 3 steps per second and the runner's muscles are 20% efficient in converting energy into the mechanical work of running, how much extra energy is used by the body per hour when it is running? Express your answer in kcal/h, where $1\,\mathrm{kcal} = 1 \times 10^3\,\mathrm{cal}$ and $1\,\mathrm{cal} = 4.184\,\mathrm{J}$.

3.35 Explain why sprinters start a race pushing off from blocks.

3.36 After how many strides does the accelerating sprinter reach 90% of her final speed? What distance does this correspond to (in m)?

3.37 The final speed attained of the accelerating sprinter refers to the upper body and one leg. If the other leg can be considered to be still, what is the speed of the sprinter's center of mass after acceleration?

3.38 We assumed that the pushoff leg of the accelerating sprinter does not contribute any kinetic energy in (3.38)–(3.47). Let us say that the center of mass of that leg is moving at half of the speed of the upper body and other leg:
(a) Is this reasonable? Why?
(b) How would this change (3.38)–(3.47)?
(c) How would it change the numerical value of the calculated final speed?
(d) How would it change the number of strides needed to reach 90% of the final speed?

3.39 We assumed that the pushoff leg of the accelerating sprinter does not contribute any kinetic energy in (3.38)–(3.47). Let us say that this leg rotates while this foot is on the ground, so the top of it moves at the same speed as the rest of the body, but the bottom of it is still. Assume this leg is straight and of uniform linear density. What fraction of the kinetic energy of the leg is lost as it changes from a translating to a rotating leg?

3.40 Do parts (b)–(d) of Problem 3.38 for the case posed by Problem 3.39.

3.41 Plot the data points in Table 6.34 as average speed in miles/hr versus distance in m (with this axis plotted on a log scale), and fill in the curve between these points either by hand or by curve fitting.

3.42 (a) Use the data in Table 6.34 to model the average world record running speeds for men, v_{av} (in mi/hr) for distances D (in m) longer than 400 m as $v_{\text{av}} = K/D^n$ [16].
(b) Estimate what the world record time for running 4,000 m would be expected to be.
(c) Estimate how fast you would expect the speed to be in the middle of that race? (If you are assuming that the speed is constant during the race, why would you think this is a reasonable assumption?)
(d) Compare the 2014 men's world record with those in 1957 ($K = 29.66, n = 0.089$) and 1982 ($K = 29.91$, $n = 0.085$). (Plot all on the same set of axes and discuss.)
(e) Why are shorter distances not used in this type of analysis?

3.43 Show that the maximum (horizontal) running speed of mammals is expected to be roughly independent of mass by using the following approach [95, 130]. Assume that the maximum muscle force is proportional to its cross-sectional area, which is $\propto L^2$ for characteristic dimension L. The body mass is proportional to the volume, and so $m_{\text{b}} \propto L^3$. Proceed as follows:

(a) The work done by leg muscles in each stride is the muscle force $\times$ the muscle contraction distance. Show that this scales at L^3.
(b) The kinetic energy of that limb is $I(\mathrm{d}\theta/\mathrm{d}t)^2/2$, where I is the moment of inertia of the leg and $\mathrm{d}\theta/\mathrm{d}t$ is its angular speed, which is the running speed v divided by the leg length. Show that this energy scales as L^3v^2.
(c) By equating these, show that v is independent of L and therefore m_b.

3.44 Show that the maximum uphill running speed of mammals should decrease with linear dimension L, as $1/L$, and therefore with mass m_b as $1/m_\mathrm{b}^{1/3}$ [95, 130]. Do this by equating the power available from the muscles to the power required to work against gravity. (Hint: The power available from muscles is the work done by leg muscles per stride ($\propto L^3$, from Problem 3.43) divided by the time per stride (which is the leg length divided by the speed, and Problem 3.43 shows the speed is independent of L). The power needed to work against gravity is the body weight times this uphill running speed.)

Jumping

3.45 Neglect air resistance and muscle atrophy and assume the same upward normal reaction force as on Earth:
(a) If someone can increase her center of mass by 0.7 m in a vertical jump on Earth, how high could she jump on the moon?
(b) If an athlete can long jump 25 ft on Earth, how far could she long jump on the moon? (Assume her takeoff angle is 30°.)

3.46 (a) A person tries to touch the highest point possible by fully extending his/her arm during a vertical jump. You are told that before takeoff, the ankle and knee joints are at peak flexion, with angles 85° and 90° and that after takeoff and at peak height, these angles are 149° and 178°. From the context of the information given, how are these angles being defined? Are these definitions the same as those in Fig. 3.1?
(b) Make a sketch of the body at these positions, which should include the ankle, knee, and hip pivot points (in this 2D sketch) and these angles.
(c) Where do these angles fall within the ranges of motion of the two joints?

3.47 (a) During a vertical jump the center of mass of a 70 kg person is 0.65 m from the ground in the crouch phase and 1.05 m at takeoff. During the extension phase the average (total) force exerted by the floor (summed on both feet) is 1,600 N. (Remember that part of this counters the body weight and the remainder—the net vertical force—counters the forces due to muscles.) When necessary, assume that this force is constant during the extension phase. Find how high the center of mass rises during free vertical flight, the speed at takeoff, and the (temporal) duration of the extension phase.
(b) About 43% of the mass of the average person is muscle and essentially all of it is skeletal muscle mass. This average person in (a) does weight training, which increases his body mass by 5 kg. Assume all of it goes into muscle proportionately

throughout the body and that the total vertical force is proportional to the total body muscle mass. How high does his center of mass rise now during a vertical jump?
(c) Repeat (b) (starting with the person in (a)) if the "weight training" instead leads to adding only 10 kg of muscle.
(d) Repeat (b) (starting with the person in (a)) if the "weight training" instead leads to adding only body fat—with no change in muscle mass.
(e) Find how high the center of mass rises in (a) if the person takes a wonder drug that decreases the extension phase by 0.05 s.
(f) On the moon, how high does the person in (a) rise in a vertical jump (from a hard floor)? Assume $g_{\text{Moon}} = g_{\text{Earth}}/6$ and that the muscles function exactly as on Earth.

3.48 Redo Problem 3.47b if the person now does not gain weight, but instead either increases his muscle mass fraction from 43 to 53% or decreases it to 33%.

3.49 (a) If the person does a vertical jump jump from a rubber pad or sand instead of a hard floor the center of mass would rise less. Why?

3.50 The world record pole vault (6.16 m) exceeds that calculated here (5.4 m). Compare this world record to the calculation assuming world class speed, 10.4 m/s, and $h_{\text{min}} = 0.02$ m. Why are they different?

3.51 In a high jump the athlete takes a running start and then hurls himself over a horizontal pole. (In the older, straddle high jump (face down when over the bar), the athlete's center of mass is ~150 mm over the bar, while in the newer Fosbury flop method (face up when over the bar), the center of mass is slightly under the bar. Assume here that the center of mass needs to clear the bar by 100 mm during a successful high jump (H_3 in Fig. 3.41)):
(a) If a 70 kg athlete's initial center of mass is 0.9 m high (H_1), how high can the bar be for a successful jump ($H_1 + H_2 - H_3$) if the athlete runs at 7.0 m/s and all of his initial kinetic energy (corresponding to motion in the horizontal direction) is converted into kinetic energy corresponding to motion in the vertical direction and then into potential energy?
(b) Since the world record high jump is 2.45 m (in 2014), how much of the initial kinetic energy could not have gone into kinetic energy associated with vertical motion?
(c) Measurements show that when 76 kg athletes run 6.7 m/s and high jump over a bar that is 2.0 m high, their horizontal speed over the bar is 4.2 m/s. How much energy is still not accounted for? What happened to it? (Note that the total kinetic energy (in two-dimensions, x and z) is the sum of that for horizontal and vertical motion $= mv_x^2/2 + mv_z^2/2$.)

3.52 In a high jump using the Fosbury flop method (face up when over the bar, as in Fig. 3.41), the center of mass can be slightly below the bar. Find H_3 if the athlete height is 1.96 m, $H_1 = 1.40$ m, $H_2 = 0.97$ m, and the height of the bar is 2.30 m (as for the jumper Dwight Stones, [66]).

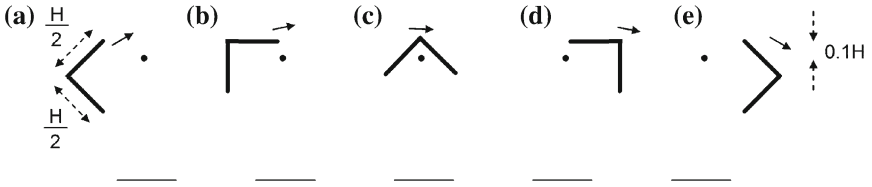


Fig. 3.84 Simple one-segment model of a high jump using the Fosbury flop method for a person with height H, with the body modeled as two segments of equal length, normal to each other. In (**a**)–(**e**), the *top* of the modeled body is always 0.1 H above the bar (which is the *dot*). This is a reasonable assumption for (**c**), because the center of the person's chest is high enough for her to clear the bar. For Problems 3.53 and 3.54

3.53 For the high-jumper modeled with a two-segment model of the Fosbury flop in Fig. 3.84, calculate how much lower the center of mass of the jumper is than the bar for positions (a)–(e). How much lower is it for position (c)?

3.54 Assume the two segments of the Fosbury flop jumper in Fig. 3.84 are at an obtuse angle, rather than the right angle shown in the figure. (This is more realistic.) Also assume that the general orientation of the jumper is otherwise the same in each part of the figure. Find the maximum angle for which the center of mass of the jumper will always remain below the bar.

3.55 (a) The length of the long jump has three parts, the takeoff distance (L_1, the center of mass precedes the foot at takeoff, take as 0.24 m here), the flight distance (L_2, the distance the center of mass travels during flight), and the landing distance (L_3, the distance the heel lands in the sand in front of the center of mass, take as 0.53 m), as in Fig. 3.42. Calculate the lengths of world-class male and female long jumps with speeds at takeoff of 9.8 and 8.6 m/s, respectively, and a takeoff angle of 20.0°.
(b) We have ignored drag here, as well as the difference in the height of the center of mass at takeoff and landing. How are the results in (a) affected if the center of mass at takeoff is 60 cm above that during landing?

3.56 A 70 kg world-class long jumper accelerates to a speed of 10.5 m/s and then jumps the longest distance possible:
(a) If drag due to air resistance is neglected, show that this occurs at a 45° takeoff angle, and find the length of this longest possible long jump. (Assume that the long jump distance is the same as the horizontal distance traveled by the center of mass. (They can be slightly different because of the different takeoff and landing arrangements of the body, as is addressed in Problem 3.55)).
(b) Compare this distance to the world record of 8.95 m (in 2014), and qualitatively account for any differences between this record jump and your calculated value. (Consider that many long jumpers take off at an angle closer to 20°.)

3.57 If drag due to air resistance is neglected, a projectile, initially at ground level, travels farthest with a 45° takeoff angle. Including air resistance the takeoff angle is closer to ∼35°. Given this, does it makes sense that many long jumpers take off at an angle closer to 20°?

3.58 Show that the maximum jumping height of the center of mass of mammals is expected to be roughly independent of mass. Do this by first showing the acceleration a due to muscles varies as $1/L$, for characteristic mammal linear dimension L. Then show that the launch speed, $v = \sqrt{2as}$—where s is the vertical distance traveled during acceleration—is independent of L and so the height of the jump $v^2/2g$ is expected to be independent of mass. Use the same scaling rules for the maximum muscle force and body mass as in Problem 3.43. (How does s scale with L?)

Other Motions

3.59 Explain why circus performers doing tightrope and high-wire acts carry a long horizontal pole [16].

3.60 (a) A figure skater, rotating at an angular frequency of f (in revolutions per s, which can be ∼0.6 revolutions/s) with outstretched arms, rotates even faster after pulling in her arms. Why?
(b) The moment of inertia of this person changes by a factor α by this pulling action. What is the skater's new rotation speed?
(c) Estimate this factor. (You can use Tables 1.6–1.9 and Problems 1.32 and 1.33 for guidance.)
(d) Does this maneuver increase the kinetic energy of the skater? Is so, where does this extra energy come from?

3.61 Explain why your pumping motions during your forward and backward motions on a swing enable you to go higher and higher. (Show how both types of pumping decrease your moment of inertia about the rotation axis and that there are no external torques.)

3.62 Describe the joint action of each joint in the dip phase of a pushup.

3.63 (a) Sketch four ways of doing a pushup with your hands pushing up on a horizontal surface (to improve your arm strength): (i) with your knees (and not feet) and hands on level ground, (ii) with your feet (and not knees) on the ground and your hands on the fourth stair on a nearby upstairs stairway, (iii) with your feet on the ground and your hands on the second stair on the stairway, and (iv) with your hands and feet both on level ground (which is the conventional pushup). Show that in each case the normal force that is exerted on your hands in equilibrium has magnitude $N = m_b g(L/H)$, for body weight $m_b g$ and horizontal distances from your hands and the center of mass to the rear point of contact (knees or feet) of H and L respectively.
(b) Show that you can successfully do a pushup only when you exert a force $N > m_b g(L/H)$.

(c) Show that a larger N is successively needed from cases (i) to (iv), so the pushup becomes successively harder to do and your workout becomes more meaningful. (Make reasonable assumptions about these distances as needed.)

3.64 Explain why the two people in a canoe paddle on opposite sides of the canoe when they want to go straight and why they paddle on the same side when they want to turn (and do so on the side opposite from the turning direction).

3.65 A platform diver bounces on the platform and after takeoff does a triple somersault. Describe the motion of the center of mass during this complicated maneuever.

3.66 (advanced problem) How far does a downhill skier travel as a function of time [135]?
(a) Show that you can rewrite (3.138) as

$$m_{\mathrm{b}} v \frac{\mathrm{d}v}{\mathrm{d}s} + \frac{1}{2} C_{\mathrm{D}} A \rho v^2 = m_{\mathrm{b}} g \sin\alpha - \mu_{\mathrm{k}} m_{\mathrm{b}} g \cos\alpha \tag{3.158}$$

and then as

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{2}v^2\right) + \frac{\rho A C_{\mathrm{D}}}{m_{\mathrm{b}}}\left(\frac{1}{2}v^2\right) = g(\sin\alpha - \mu_{\mathrm{k}}\cos\alpha). \tag{3.159}$$

(b) Show that multiplying both sides by the integrating factor $\exp(\rho A C_{\mathrm{D}} s/m_{\mathrm{b}})$, gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{2}v^2 \exp(\rho A C_{\mathrm{D}} s/m_{\mathrm{b}})\right) = g(\sin\alpha - \mu_{\mathrm{k}}\cos\alpha)\exp(\rho A C_{\mathrm{D}} s/m_{\mathrm{b}}). \tag{3.160}$$

(c) Now show that integrating this equation gives

$$v(s)^2 = \frac{2m_{\mathrm{b}} g(\sin\alpha - \mu_{\mathrm{k}}\cos\alpha)}{\rho A C_{\mathrm{D}}}(1 - \exp(-\rho A C_{\mathrm{D}} s/m_{\mathrm{b}})), \tag{3.161}$$

assuming that $v = 0$ when $s = 0$, meaning the skier starts at rest at $s = 0$, or, using (3.139),

$$v(s) = v_{\mathrm{terminal}}\sqrt{1 - \exp(-\rho A C_{\mathrm{D}} s/m_{\mathrm{b}})} \tag{3.162}$$

or

$$\frac{\mathrm{d}s}{\mathrm{d}t} = v_{\mathrm{terminal}}\sqrt{1 - \exp(-\rho A C_{\mathrm{D}} s/m_{\mathrm{b}})}. \tag{3.163}$$

(d) After putting all the s terms on one side of the equation and all t terms on the other and integrating, show that the distance traveled s (after a time t) and t are related by

$$t = \frac{1}{k v_{\mathrm{terminal}}} \ln\left(\frac{1+z}{1-z}\right), \tag{3.164}$$

where $k = \rho A C_{\mathrm{D}}/m_{\mathrm{b}}$ and $z^2 = 1 - \exp(-ks)$.

3.67 Use the results of Problem 3.66 to show that for the example in the text where the downhill skier attains a terminal velocity of 25.1 m/s, the skier travels 2,500 m in 108 s [135].

3.68 Show that (3.161) in Problem 3.66 reduces to the expected results when there is no friction or drag.

3.69 (advanced problem) This problem determines the path of a downhill skier that takes the minimum time [88]. With the lateral coordinate x and the downhill coordinate on the plane y, which increases down the slope, and the downhill angle to the horizontal α, the skier decides to start at rest at $x = 0$ and $y = 0$ and end at $x = \beta y_f$ and $y = y_f$. The distance along the path is called s, so $ds = \sqrt{1 + y'^2}\, dx$ where $y' = dy/dx$.
(a) Show that the skier has speed $v = \sqrt{2g'y}$, where $g' = g \sin\alpha$. (This depends on y and not x. We will be assuming that the skier can change directions at any y without loss of speed, and will ignore drag and friction.)
(b) Show that skier arrives at the destination at time $t = \int_0^{y_f} \sqrt{(1 + y'^2)/2g'y}\, dx$.
(c) Variational analysis shows that t is a minimum if $F - y'\partial F/\partial y'$ is a constant where $F = \sqrt{(1 + y'^2)/2g'y}$. Show that this means that $y(1 + y'^2) = C$, a constant.
(d) By substituting $y' = \cot\theta$, show that $y = (C/2)(1 - \cos 2\theta)$.
(e) Show that $dx/d\theta = (1/y')dy/d\theta = C(1 - \cos 2\theta)$.
(f) With $C = 2A$, $2\theta = \phi$, and $R = A/2g'$, show that $x = R(\phi - \sin\phi)$ and $y = R(1 - \cos\phi)$. This is cycloid obtained by a point on a circle of radius R rolling on a straight line.

3.70 (a) You are going downhill at speed v in a cart on a track that enters a turn with radius of curvature R or on skis and enter a trough with radius of curvature R, and move so the center of mass (of you and the cart or of you, respectively) moves h closer to the center of radius ($h \ll R$). If your angular momentum does not change during this turn, show that you have a fractional increase in speed of h/R and kinetic energy of $2h/R$ [88, 104].
(b) Explain why there are no external torques during this motion, so the angular momentum does not change.
(c) Where does the increase in kinetic energy come from?
(d) Sketch where the center of curvature is relative to the slope is both cases, before and during the turn.

3.71 (a) When a downhill skier raises her center of mass by 0.5 m at the bottom of a trough with radius of curvature 4 m, find the fractional increase of her kinetic energy.
(b) If she has a speed of 10 m/s, find the normal force she feels during this maneuver, expressed in terms of her body weight.

3.72 A Nordic ski jumper takes off with a horizontal speed of 15 m/s from a tower that is 10 m above the slope that descends at a 45° angle.
(a) How far down the slope does the skier land, measured along the slope from the base of the tower? Ignore the effect of air.

(b) If the effect of air is included, a 70 kg jumper with a frontal area of 0.7 m^2 will land 65.9 m down the slope (using an air density of 1.007 kg/m^3, $g = 9.81$ m/s^2, and drag and lift proportional to the square of the speed and $C_D = 1.78$ and $C_L = 0.44$) [37]. How does your answer compare to this and what does this mean?

3.73 A skier jumps from a ramp with speed of 15 m/s^2 at a 45° angle. If the skier lands at the same height as she takes off (so the snow line is horizontal and the ramp height is negligible), how far is the jump and how long does it take? Analyze this first by ignoring drag and lift then by including only drag (using the information in Problem 3.72 and Table 3.12). How important is drag?

Throwing Balls and Other Objects, and Ball Flight

3.74 Use (7.11) to show that $Re \simeq 640\, vd$ in air.

3.75 Show that $Re \sim 140{,}000$ for a cricket ball (7.2 cm diameter) traveling at 30 m/s.

3.76 Find the Reynolds number for the following balls traveling a 30 m/s:
(a) soccer (diameter of 8.7 inches, 22.1 cm),
(b) tennis (2.5 inches, 6.35 cm),
(c) golf (1.68 inches, 4.27 cm),
(d) squash (1.6 inches, 4.07 cm),
(e) ping pong (table tennis) (1.5 inches, 3.82 cm) [36].
(f) How do these compare with the critical value?

3.77 Explain how energy generated by the rotation about the shoulder can be transferred to rotation of the elbow when throwing a ball, by treating the upper arm and the forearm as a double pendulum.

3.78 (a) Use the one-segment model in this chapter that predicts you can throw a baseball at 45 mph to determine how fast can you throw a football. A National Football League (NFL) football must weigh between 14 and 15 oz (and have a mass between 0.40 and 0.43 kg). Is this consistent with an NFL quarterback throwing a ball at 68 mph? Does this make sense?
(b) Would it make sense if you modified the one-segment ball throwing model so the baseball is released at 100 mph and then used it for throwing a football?

3.79 In the one-segment model for throwing a ball in this chapter, what should the angle of the upper arm be relative to the vertical so that the velocity vector of the released ball is horizontal (with a release at $\theta = \pi/4$)?

3.80 (a) Modify the one-segment model of throwing a baseball so that the whole arm is always straight and is rotated about the shoulder by the deltoid muscles.
(b) How fast is the ball released, assuming 3 in diameter deltoid muscles and the other assumptions in the final model in this chapter?

3.81 (advanced problem) Show that when α_1 and ω_1 are zero the two-segment model torque equation for segment 2 (3.100) reduces to that for a one-segment model (3.68) for segment 2. Why should this be so?

3.82 (advanced problem) Show that when m_2 and I_{2p} are zero the two-segment model torque equation for segment 1 (3.101) reduces to that for a one-segment model (3.68) for segment 1. Why should this be so?

3.83 (advanced problem) Show that when θ_{21}, ω_{21}, and α_{21} are zero the two-segment model torque equation for segment 1 (3.101) reduces to that for a one-segment model (3.68) for the rigid, straight assembly of segment 1 and segment 2. Why should this be so? (You will need to use the parallel axis theorem (3.27), which relates the moment of inertia about the axis through the center of mass to that about another axis.)

3.84 How do the torques we calculated in our pitching model compare to those in Figs. 3.50 and 3.51? Should they be comparable?

3.85 Estimate the relative final angular rotation speeds of the elbow and wrist in throwing a ball by using Fig. 3.49, and then estimate the relative contributions to the speed of the pitched ball from the motion of the lower arm and the wrist. Assume the ball is a distance of half of the length of the hand from the wrist joint.

3.86 Estimate the mass of the internal rotator muscles (in kg) in the shoulders of very large people, who are good baseball pitchers with rapid humerus rotation during pitching. These muscles (and their volumes in cm^3) are: pectoralis major (676), latissimus dorsi (550), anterior deltoid (one third of the 793 total), teres major (231), and subscapularis (319), and the muscle density is 1.06 g/cm^3 [121].

3.87 Can the muscles in the shoulder account for all of the power generated by the shoulder joint during a baseball pitch? Assume that the work done during shoulder rotation during the pitch is 132 J during the acceleration phase that lasts 0.034 s.
(a) Find the average shoulder joint "kinetic energy" power (or kinetic power).
(b) Assume that the maximum power generated by muscles is 250–500 W/kg and the muscle mass is as determined in Problem 3.86. (This mass is also given in the solutions.) How many times larger is the average shoulder kinetic power than the maximum muscle power?
(c) The muscle power given in (b) is for isotonic muscle conditions (see Chap. 5). Prestretching may increase this by a factor of 1.75. How many times larger is the average shoulder kinetic power than this new maximum muscle power?
(d) What can you conclude about the shoulder muscles being responsible for humerus rotation during baseball pitching [121]?

3.88 To throw a football well, you need to grasp it tightly. A regulation National Football League football has an internal gauge pressure of 12.5–13.5 psi, weight of 14–15 oz, circumference at the center of $\simeq$22 inches, and an $\simeq$11 inch tip to tip distance. Use Problem 7.14 to estimate the range of pressures and forces a quarterback's hands need to exert to grasp a football well, assuming his hand makes contact over 2.5 $inch^2$ of the ball.

3.89 (a) A football is found to be inflated to a gauge pressure 2 psi below the allowed minimum. The football may have been inflated correctly by officials, and then deflated (by releasing internal air) by someone trying to help a quarterback with weak or small hands. Estimate the pressure and force that needs to be exerted by the quarterback's hands to grasp this football, assuming his hand makes contact over 2.0 inch^2 of the ball. (Refer to Problem 3.88.)
(b) A reporter says that the pressure in such deflated footballs could be determined by weighing them, to see if they are still within the official limits. Explain why this will not work. Devise a better way to determine the pressure.
(c) With constant external pressure of 14.7 psi the internal gauge pressure of the footballs could decrease because the ambient became colder. How cold would it need to be for the gauge pressure to have decreased by 2 psi, if they were inflated correctly when the ambient temperature had been 70 °F?

3.90 Would using the deflated footballs of Problem 3.89 help running backs grasp onto the ball better, so they could avoid fumbling (losing) the ball?

3.91 A baseball pitcher throws a ball with initial velocity vector component $v_{x,0}$ in the forward direction and $v_{z,0}$ in the vertical direction. (Ignore ball deceleration due to drag):
(a) Show that after a time t, its coordinates are: $x = v_{x,0}t$ and $z = v_{z,0}t - gt^2/2$.
(b) After it travels a distance d in the x direction, show that the ball seems to have fallen a distance $gd^2/2v_{x,0}^2$ from its initial trajectory.
(c) Show that the ball always stays in the $y = 0$ plane.

3.92 If the initial velocity vector component of a ball is $v_{x,0}$ in the forward direction and $v_{z,0}$ in the vertical direction, show that it travels a distance $2v_{x,0}v_{z,0}/g$ (which is called its range), which equals $v_0^2 \sin(2\alpha)/g$, for initial speed v_0 and angle α.

3.93 (a) Show that a ball travels farthest when released at a 45° takeoff angle, and when it is released at speed v, it travels a distance of v^2/g. (This is an overestimate because air drag is omitted. When drag is included, the best angle is really 35°. Also, we are ignoring any difference in the heights of the release point and the ground. This will be examined later in the problems.)
(b) How far does the baseball travel when thrown at 90 mph?
(c) If the ball is wet and weighs 7 oz instead of 5 oz, how far will it travel? Assume the ball leaves the hand with the same linear momentum as the dry ball does in part (b).

3.94 (a) A ball kicked or thrown travels the farthest with takeoff angle 45°. For a given angle $\alpha_1 < 45°$, find the angle $\alpha_2 > 45°$ at which the ball travels the same (lesser) distance. Ignore drag.
(b) What are the travel times at each angle? Are they equal?
(c) With drag included, do you expect the travel distances at these two angles still to be equal? Why?

3.95 (a) When a punter kicks (punts) a football to the opposing team, the goal is usually to kick it the farthest, with one constraint: His "covering" teammate, who

is next to the ball and starts running in the direction of the ball when it is punted, should not arrive after the opposing player (punt returner) can catch the ball, so he can tackle the returner before he can run with (return) the ball. The kicker kicks the ball an initial speed v at takeoff angle α and his teammate runs at speed u (essentially staying under the traveling ball). For a given v and u ($< v$), find the optimal takeoff angle, and the punt distance and travel (hang) time at this angle. (Drag significantly changes these results [16], but ignore it here.)
(b) Find the optimal angle, punt distance (in yards) and hang time (in s) for a kickoff speed of 55 mph, if the covering player can run 40 yards in 4.3 s.

3.96 The farthest distance an object can be thrown is with a takeoff angle of 45°, ignoring drag, only if it is thrown from ground height. Otherwise this is just an approximation, as we will now see:
(a) If an object is thrown with an initial speed v and angle α, from a height $z = h$ show that at a later time: $x(t) = vt\cos\alpha$ and $z(t) = h + vt\sin\alpha - gt^2/2$.
(b) Show that it reaches the ground, $z(t) = 0$, after it has traveled a distance $x_{\text{final}} = (v^2\cos\alpha/g)(\sin\alpha + \sqrt{q^2 + \sin^2\alpha})$, where $q = \sqrt{2gh/v}$. (Note that $q^2 = mgh/(mv^2/2)$, which is the initial potential energy divided by the initial kinetic energy.)
(c) By setting $\mathrm{d}x/\mathrm{d}\alpha = 0$, show that the ball will travel farthest for an initial angle given by $\tan\alpha_{\max} = 1/\sqrt{1+q^2}$ and it travels a distance $x_{\text{final,max}} = (v/g)\sqrt{v^2 + 2gh}$.

3.97 A baseball is thrown from a height of 6.5 ft from the ground with an initial speed of 90 mph. Ignoring drag, find how far it travels if:
(a) it is released at 45°, and you ignore the difference in the release height and the ground.
(b) it is released at 45°, and you take into account the difference in the release height and the ground (see Problem 3.96).
(c) it is released at the optimal angle for distance (which you also must find), and you take into account the difference in the release height and the ground (see Problem 3.96).

3.98 A baseball pitcher throws a ball at 90 mph releasing it 6 ft in front of the pitcher's rubber (the stripe on the pitcher's mound), at a height of 5 ft. The ball arrives at the plate, which is 60 ft 6 in from the pitcher's rubber at a height of 3 ft. (Ignore ball deceleration due to drag):
(a) How long does is take the ball to arrive?
(b) What angle does the initial velocity vector of the ball make with the horizontal?
(c) When the ball arrives at the plate, how much does it appear to have fallen relative to the path it would have taken with its initial velocity vector? (Ignore any effects due to ball rotation.)

3.99 Why are pitchers often tall? Consider where they release the ball and batter reaction times, how fast they can throw (if arm rotation speed were independent of height), drag during flight, and, for those who throw sidearm, the angle the ball can cross the plate.

3.100 A batter hits a baseball so it initially has a speed of 90 mph and travels 45° to the horizontal. Find how far it travels and its travel time assuming it travels in vacuum and has the same initial and final heights. Assume a baseball weighs 5 1/8 ounces and has a circumference of 9 1/8 inches (regulation values).

3.101 Repeat Problem 3.100, now for travel in air (at 70 °F, no humidity, at sea level), by numerically integrating (3.136) and (3.137). Assume C_D is 0.45. Compare your results with those of Problem 3.100.

3.102 Repeat Problem 3.101, by using the ballistic table, Table 3.12. (Calculate b_1 and then determine b_2 by interpolation in this table to obtain the range (in feet).) Compare your results with those in Problem 3.100. (If you are also doing Problem 3.101 compare your results with those of both problems.)

3.103 A batter hits a baseball so it initially has a speed of 90 mph and travels 45° to the horizontal. Let us explore how far it travels under several different air conditions, assuming its initial and final heighs are the same, by using the ballistic table, Table 3.12. Assume a baseball weighs 5 1/8 ounces and has a circumference of 9 1/8 inches (regulation values), and C_D is 0.45. Air drag is proportional to the mass density of the air ρ, which itself depends on the number density of each constituent (from its partial pressure) times its molecular mass.
(a) First find the range in vacuum. (You cannot use the ballistic table for this. You might want to consult other chapter problems for guidance.)
(b) Now, calculate b_1 and then determine b_2 by interpolation in this table to obtain the range (in feet) for the following conditions: (i) 70 °F, no humidity, at sea level (where atmospheric pressure is 760 Torr = 760 mmHg); (ii) 30 °F, no humidity, at sea level; (iii) 100 °F, no humidity, at sea level; (iv) 100 °F, maximum humidity (water vapor pressure is 49 mmHg, the rest is air), at sea level; (v) 70 °F, no humidity, in Denver (where atmospheric pressure is 626 mmHg).
(c) How important are drag and these variations in air conditions? As part of this compare your results to those expected for the ball traveling in vacuum.
(d) Why do people say that balls travel well on hot humid days and at high elevations, and poorly early in the baseball season (i.e. the early spring)?

3.104 Determine the range and time of flight of a soccer ball kicked with initial speed 28 m/s at angle 50° by using the ballistic table, Table 3.12 [37]. The mass of the soccer ball is 0.42 kg and its diameter is 0.22 m. Assume $C_D = 0.2$ (because of the very high Re).

3.105 A soccer ball kicked at 45° initially travels at a speed of 60 m/s. Assume it does not spin.
(a) Show that its Reynolds number exceeds the critical value, at least at the beginning of its flight.
(b) How far would it travel in vacuum?
(c) How far would it travel if throughout the flight the drag coefficient C_D were assumed to be the value 0.45, for smooth balls with a Reynolds number below the

critical value?
(c) How far would it travel if throughout the flight the drag coefficient C_D were assumed to be 0.15 (because the Reynolds number exceeds the critical value)?

3.106 Determine the range and time of flight of a golf ball hit with initial speed 61 m/s at angle 20° by using the ballistic table, Table 3.12 [37]. The value of m_b/d^2 for a golf ball is 27 kg/m^2 [37]. Assume $C_D = 0.2$ (because of the very high Re.)

3.107 Consider a football kicked 100 feet per second at an angle of 45° [57]. At sea level, a spiral kick travels 78 yards with no wind and 65 yards with a headwind of 20 mph. In Denver these corresponding distances are 82 and 70 yards, and the distance is 92 yards with a 20 mph tailwind. A tumbling kick at sea level has a range of 40 yards with no wind, 20 yards with this headwind, and 62 yards with this tailwind. In Denver the tailwind decreases the range from 45 yards to 25 yards. Explain why the order of these ranges makes sense.

3.108 A baseball is rotated by a half revolution by the pitcher during the last 0.04 s before release, due to a constant rotational acceleration due to wrist action and the rotation of the radius on the ulna. Show that the released ball spins at a rate of 1,500 revolutions/min.

3.109 The pitcher tries to maximize the rotation speed of the ball (ω, in rad/s) when throwing a curve ball, as seen in Table 3.13. A spinning ball feels the Magnus force (or lift) due to this rotation, which changes its path from that of a rotation-free ball. This lift force has magnitude $= \pi\rho r^3\omega v/2$ for air mass density ρ, ball radius r, spinning rate ω (in rad/s), and ball speed v. (From [137]):
(a) The Magnus force on a baseball has been determined from measurements to have magnitude (in lb) $= 6.4 \times 10^{-7} f v$, where f is in revolutions/min and the speed v is in ft/s. Is this consistent with the magnitude of the lift force just given?
(b) How many revolutions do a fast fastball, a slow curve ball, and a medium-speed, medium-rotation-rate knuckleball undergo if each is traveling at a constant forward speed and travels 55 ft before it reaches home plate?
(c) Sketch a diagram showing the initial velocity vector of the ball, how the ball rotates, and the direction of the Magnus forces when a right-handed pitcher rotates

Table 3.13 Speed and spinning of pitches by professional-level pitchers

Pitch	Speed (mph)	Typical spin rate (rpm)
Fastball	85–95	1,600
Slider	75–85	1,700
Curve ball	70–80	1,900
Change-up	60–70	1,500
Knuckleball	60–70	25–50

The term rpm stands for revolutions per minute
Using data from [137]

the ball in the direction of the screwdriver motion and releases it so that the rotation axis is vertical.
(d) If the Magnus force is constant, how much has the ball moved laterally—relative to the expected trajectory with no spinning—when it reaches the plate. (Ignore changes to speed and direction due to drag and gravity, so assume the ball is moving forward and horizontally at 80 mph. Also assume that changes in the trajectory due to the Magnus force are so small that you can assume this trajectory in your calculation.) (To learn more about drag forces see Chap. 7.)

3.110 As in Problem 3.91, a baseball pitcher throws a ball with initial velocity vector component $v_{x,0}$ in the forward direction and $v_{z,0}$ in the vertical direction, but now it has additional constant forces in the vertical z direction $\alpha_z g$ and lateral y direction $\alpha_y g$ due to its rotation and the Magnus force:
(a) Show that after a time t, its coordinates are: $x = v_{x,0}t$, $y = \alpha_y g t^2/2$, and $z = v_{z,0}t - (1 - \alpha_z)gt^2/2$.
(b) After it travels a distance d in the x direction, show that the ball seems to have fallen a distance $(1-\alpha_z)gd^2/2v_{x,0}^2$ and moved laterally a distance $\alpha_y g d^2/2v_{x,0}^2$ from its initial trajectory.
(c) Show that the ball always stays in the tilted plane $z = (v_{z,0}/v_{x,0})t - ((1 - \alpha_z)/\alpha_y)y$.
(d) Explain why the ball seems to move laterally when $\alpha_y \neq 0$, it falls even faster than expected when $\alpha_z < 0$, and it falls slower than expected—"the rising fastball"—when $\alpha_z > 0$.

(Note that this additional force vector has been assumed to be constant in this simple model. More specifically, the Magnus force has been assumed to be that for a ball thrown with a constant velocity in the x direction. Changes in the force direction and magnitude due to changes in ball velocity due to gravity and the Magnus force itself—and the addition of drag, make the ball motion even more complex.)

3.111 (a) Show that a 80 mph baseball curve ball subjected to a horizontal force of 2.6 oz ($\alpha_y = 0.5$) curves laterally by 2.1 ft from its expected path (assuming it travels a distance of 60 ft).
(b) In what direction is the spin (given by the right-hand rule) if the ball veers to the right, as seen at the plate. (This is a curve ball as thrown by a right-handed pitcher and a screwball as thrown by a left-handed pitcher.)
(c) How fast is the ball rotating?

3.112 Can the Magnus force described in Problem 3.111 explain the "rising fastball" —for which the batter seems to think that the pitched fastball is rising? This could be possible if the ball spun fast enough in the correct direction to give the ball a lift force exceeding g. (This corresponds to $\alpha_z > 1$ in Problem 3.110.) Calculate it for a translational speed of 100 mph and a rotation speed of 2,300 rpm (typical of some excellent pitchers), and show that under the best of circumstances, the Magnus force can account for at most about $\sim 2g/3$ (or $\alpha_z \sim 2/3$ in Problem 3.110), and so a "rising fastball" is really a consequence of a hitter's perception and expectation [9, 137].

3.113 A right-handed pitcher has similar wrist action when throwing overhand and side arm. When throwing overhand (arm motion in a vertical plane) his ball drops very fast. When throwing side arm (arm motion in a horizontal plane), it moves to the left. When throwing at three-quarter overhand (arm motion in a plane that bisects these two planes), it moves a bit down and to the left. Why?

3.114 In throwing a knuckleball, the pitcher is not concerned with the effects of the Magnus force, but with the sideways force on a ball that is not a perfect sphere because of the stitches on the ball (Fig. 3.85). The pitcher purposely tries to minimize the baseball rotation rate when throwing a knuckleball (Table 3.13), because this force will average to zero for a ball with a moderate rotation rate—and one that is too small for a significant Magnus force. With a slow rotation rate the lateral motion will seem irregular. (This irregular motion also occurs for a ball that has been scuffed—accidentally or on purpose.) The magnitude of this lift force is $C_{\mathrm{L}} A \rho v^2/2$, where C_{L} is the lift coefficient and A is the cross-sectional area, which is similar to (7.63) [137]:
(a) For a knuckle ball, this force is $2.16 \times 10^{-5} v^2$ (in lb) with v in ft/s. Show that this means that $C_L = 0.42$.
(b) For a scuffed ball, this force is $1 \times 10^{-5} v^2$ (with the same units). Show that this means that $C_L = 0.194$.
(c) Estimate how much a knuckleball can move laterally if it does not rotate at all? (If it does not rotate at all, its motion is predictable, and this is not desirable.)

3.115 Redraw Fig. 3.85 for a nonrotating cricket ball (a smooth ball with seams that are stitched at its "equator"). Assume the cricket seam is at a $\sim 45°$ angle with the direction of the flow from the left. Discuss how this seam can affect motion. (Hint: In the rest frame of the ball, the exiting streamlines will not flow along the stitches.)

3.116 Why does a football thrown spinning about its long axis (a spiral) experience no Magnus force?

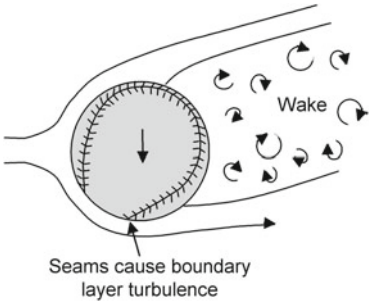


Fig. 3.85 The flow past a spinless ball can be asymmetric. In this example, the *top* is smooth and the stitches (or scuff mark) are on the *bottom*. The seams cause boundary layer turbulence, which delays airstream separation on the *bottom* surface. The wake moves upward and there is a downward force on the ball (based on [137])

3.117 Most basketball players shoot free throws from the free throw line in an overhand motion, but some professional players, notably Wilt Chamberlain and Rick Barry, have shot them underhanded. In both cases the basketball must go through an 18 in diameter horizontal hoop, with the hoop center 15 in. in front of the backboard. The hoop is 10 ft high on a backboard that is 15 ft from the free throw line. A basketball weighs 21 oz, has a circumference of 30 in, and therefore a diameter of 9.7 in.

One study has shown that for a release height of 7 ft, an error in release angle has the minimum affect on accuracy for release angles between 49° and 55°, and this corresponds to entry angles between 38° and 45°, both relative to the horizontal [66, 77]. More recent analysis suggest that the optimal release angle is between 51° and 56° with release speeds between 20.5 and 24.0 ft/s. Spin due to bending elbows and knees and snapping the wrist at the moment of release adds lifts due to the Magnus force [66].

Assume the ball goes through the middle of the hoop at a 41° angle to the horizontal (which is the entry angle). Assume that the 6 ft tall basketball player releases the ball with arms extended vertically for the overhand motion and with outstretched straight arms at shoulder level for the underhand motion. Sketch the motion and determine the initial speed and angle of release for both cases. (Hint: Use the anthropometric data in Chap. 1.)

3.118 A shot putter releases a 7.25 kg shot at a height 2.2 m from the ground. If the shot is released at the angle that maximizes the travel distance (see Problem 3.96), it travels 23.06 m. Find this angle and the initial shot speed [90]. (Ignore drag.)

3.119 (a) Find the initial kinetic energy and the maximum increase in the potential energy of the shot in Problem 3.118.
(b) If the putt is 20% efficient, how much energy is used in putting the shot (in J and kcal).
(c) If the putt is pushed from its resting point on the shoulder as the arm is extended, it is accelerated along a distance corresponding to the length of the whole arm, or about 0.7 m. During this acceleration phase (at this extension and release angle), what is the increase in potential energy and how does it compare to the energies in part (a)?
(d) What is the force exerted to accelerate the shot, if this force exerted during this extension phase is constant?
(e) If the muscles used to accelerate the shot can exert 20 N of force for each cm^2 of effective cross-sectional area, what is the net cross-sectional area of these muscles? If this were a muscle with a circular cross-section, what would its diameter be?

3.120 A hammer thrower releases a 7.25 kg hammer at a height 3.5 m from the ground. If the hammer is released at the angle needed so it can travel a maximum distance (see Problem 3.96), it travels 102 m. Find this angle and the initial hammer speed [90]. (Ignore drag.)

3.121 (a) Find the initial kinetic energy and the maximum increase in the potential energy of the hammer in Problem 3.120.

(b) If the throw is 20% efficient, how much energy is used in throwing the hammer (in J and kcal).

3.122 Derive (3.145)–(3.148) for the flight of a bouncing ball with enough friction so it stops skidding during the impact, using the supplied relations.

3.123 Derive (3.150)–(3.153) for the flight of a bouncing ball with too little friction to stop skidding during the impact, using the supplied relations.

3.124 Show that skidding stops during the course of a bounce when $\mu_k v(1+e) \geq \frac{2}{7}(u + r\omega)$, by equating (3.145) and (3.150).

3.125 A solid ball with diameter 70 mm moving at 5 m/s hits a hard surface ($\mu_k = 0.6$, $e = 0.7$) at angle of 30° [36].
(a) If the ball is not spinning, show that it has a rebound speed of 3.51 m/s and angle 30°.
(b) If it is spinning, show that as long as its spinning rate is less than 371 rad/s it will stop skidding before the end of the collision (see Problem 3.124).
(c) Show that as the top spin of the incident ball increases from 100 to 200 to 300 rad/s, its rebound speed increases from 4.11 to 4.84 to 5.66 m/s and rebound angle increases from 42 to 51 to 58°.
(d) Find the rebound speed and angle for a ball incident with a backspin of 100, 200 and 300 rad/s.

3.126 Show that a solid spherical ball travels a distance $(12/49)v_0^2/\mu_k g$ while it is skidding.

3.127 Using the notation of the chapter, find the final speed of a hollow spherical ball when it starts to roll in terms of its initial speed. (It has a moment of inertia about an axis through its center of mass of $(2/3)m_b r^2$.)

3.128 The forces exerted by muscles are proportional to their physiological cross-sectional area PCA. For each of the following cases, the result is roughly proportional to $(\text{PCA})^x$ (ignoring the force needed to counter the effects of gravity on the body, etc.). Find x in each case. (It should be either 0.5 or 1):
(a) The weight lifted by a weightlifter.
(b) The speed of a ball thrown by a pitcher.
(c) The farthest a ball can be thrown.
(d) How fast a bat can be swung. (Is this similar to case (b))?
(e) The speed of a ball after being hit by a bat. (Assume the pitcher and hitter have muscles proportional to PCA.)
(f) How far a ball can be hit.

Collisions of Humans

3.129 A professor runs to a waiting elevator and slips on a small drop of water. He lands on his knees (between the patellas and patella tendons) and his open hands. Explain why this is preferable to landing only on his knees.

3.130 (a) Show that the straight lines denoting specific deceleration times and decelerations in Fig. 3.61 are expected from simple kinematics, for constant deceleration rates. Note that this figure is a log-log plot.
(b) The slopes of the lines for constant deceleration times in this figure seem to be twice those for constant decelerations. Does this makes sense?

3.131 Let us say that as we land at the end of a jump we use the muscles in our knees to help cushion the collision. (Earlier in this chapter we considered a case where we ignored this potential cushioning.) The center of mass of a person of mass m_b falls a distance h before the person's feet make contact with the ground, and then the person's center of mass is lowered by a distance s and the reaction force from the ground F is used to cushion the collision:
(a) By using conservation of energy, show that the required force is $F = m_b g(1 + h/s)$.
(b) If we land on both feet and want to keep stresses to no more than 10% of the UCS, show that the maximum force felt by the feet during the collision should be no more than 10^4 N. (Use the analysis in the text for landing with stiff legs.)
(c) If we slow the fall by bending our knees by 0.5 m, what is the maximum height from which a 70 kg person can safely land? (Do not try this!!!)

3.132 In what ways are the GSI and the impulse during a collision similar and yet different?

3.133 A skater falls on ice [13]. Assume that his head hits the ice with a speed corresponding to free fall from a height of 6 ft.
(a) Assuming a constant force during impact, show the GSI for this collision is GSI $= (t_f/\Delta t)^{2.5}\Delta t$, where t_f is the time of the free fall and Δt is the collision time.
(b) Calculate the value of the GSI for collision times $\Delta t = 1$, 2, 5, and 10 ms.
(c) Let $H = 6$ ft be the height of the free fall and Δs be the total compressional distance of the collision, which is the flattening experienced by the skin and/or protective padding. Show that the GSI can be expressed as GSI $= (H/\Delta s)^{1.5}\sqrt{2H/g}$.
(d) Calculate the GSI for $\Delta s = 0.1$, 0.2, 0.5, and 1.0 cm.
(e) Identify the range of values of Δt in case (b) and Δs in case (d), for which serious injury can be expected.

3.134 Explore how the GSI and HIC-15 indices differ for constant deceleration, when:
(a) The magnitude of the deceleration is αg and it lasts for a time τ.
(b) $\alpha = 70$ and $\tau = 15$ ms.
(c) $\alpha = 40$ and $\tau = 60$ ms.
(d) What can you say about the weighting of parameters for each index and how each depends on the duration of the collision?

3.135 Let us explore how the GSI and HIC indices change when the rate of deceleration is not constant during the collision. Consider an initial speed v and a total collision time τ (<15 ms). It will be simpler if we express v as $\beta g\tau$, where β is a

constant. The deceleration $a(t)$ is expressed as $\alpha(t)g$. For each of these cases, first confirm that the speed decreases to 0 in a time τ and plot $\alpha(t)$ on the same graph, and then calculate general expressions for the GSI and HIC for each:
(a) $\alpha(t) = \beta$ is constant during the time τ.
(b) $\alpha(t) = 1.5\beta$ from time 0 to $\tau/2$ and 0.5β from time $\tau/2$ to τ.
(c) $\alpha(t) = 2(1 - \tau/t)\beta$, so the deceleration decreases linearly from time 0 to τ.
(d) How do the GSI and HIC differ? Which is more sensitive to changes in deceleration?

3.136 In the text the GSI was calculated for an elastic and inelastic collisions of a head with $v_i = 50$ mph and a collision time of 10 ms. For what collision times would the elastic collision be expected to be fatal and the inelastic collision expected to be survivable?

3.137 Modify (3.116) for a partially elastic collision with a coefficient of restitution e, again with a very massive object.

3.138 (a) Calculate the GSI for the partially elastic collision of a head moving with $v_i = 25$ mph and a collision time of 20 ms with a very massive object, for a general coefficient of restitution e, and then specifically for $e = 0$ (totally inelastic collision), 0.5, and 1.0 (totally elastic collision).
(b) Over what ranges of e will the collision definitely be fatal, likely be fatal, likely to not be fatal but likely result in a significant injury, and likely to lead to a relatively minor injury only?

3.139 (a) A pitcher throws a baseball at the head of an 80 kg batter (who has an average-sized head and who is not wearing a batting helmet). It hits his head at a speed of 90 mph at normal incidence and the collision is elastic. Consider the collision of the ball with the head only (ignore the rest of the body) and calculate the GSI assuming that during the collision the baseball decelerates at a constant rate, the baseball deforms its linear dimension (diameter) by 6%, and the head does not deform at all.
(b) What is the fate of the batter?

3.140 Some professional soccer players have the same type of loss of cognitive ability as do boxers, likely due to the repeated heading of soccer balls. What is the GSI when a 82 kg (180 lb) soccer player hits or redirects the ball with his/her head (a header)? Assume that the 430 g ball is moving at 50 mph and hits the head at a normal angle, so it bounces back on the original path. Say that the collision with the head is elastic and the ball is squeezed by 3 cm in the collision. Treat the collision as with the head (and not the rest of the body).

3.141 Figure 4.74 shows the deceleration and the square wave function approximation to this measured deceleration for a head impacting a helmet with initial speed 5.63 m/s:
(a) Does this square wave approximation stop the head in the time shown?
(b) Calculate the GSI and HIC-15 for the square wave function approximation.

3.142 In professional football, a linebacker and fullback, both weighing 245 lb, are racing toward each other running at 30 ft/s. (Actually, the linebacker is running to the fullback and the fullback is trying to run away from him.) They collide and then both decelerate at a constant rate and become stationary in 0.2 s:
(a) What is the deceleration of each? (Also express your answer in g.)
(b) What is the force on each during the collision?

3.143 (a) A big defensive lineman in professional football weighing 310 lb runs at a speed of 24 ft/s into a small quarterback weighing 189 lb, who is initially still. Using conservation of linear momentum, what are their speeds after this collision and in what directions are they moving? Assume all frictional forces during the collision can be neglected, so kinetic energy is also conserved during the collision.
(b) Alternatively, assume that the lineman holds on to the quarterback during and after the collision, i.e., "tackles" or "sacks" him. In this case, what fraction of the initial kinetic energy is lost? If the forces with the ground can be neglected, where did this lost kinetic energy go?

3.144 A base runner has an initial speed of 9 m/s and slides to a stop over 4 m. What is the coefficient of friction during the slide?

3.145 (a) When you slide into a base at speed $v_{\text{slide,final}}$ your foot stops after crushing into the base a distance d. Find this constant deceleration and, assuming the decelerating force is transmitted to the whole body, this collision force F_{coll}.
(b) Assume this force causes a fracture when this force F_{coll} is distributed over a bone area A and this force per unit area F_{coll}/A exceeds the UCS of the bone. Find the threshold sliding speed $v_{\text{slide,fracture}}$ for fracture.
(c) For reasonable values of A and the ultimate compressive stress (UCS) (see Chap. 4), find $v_{\text{slide,fracture}}$.

3.146 (a) Show that the minimum kinetic energy an object of mass M must have to impart a kinetic energy $\text{KE}_{\text{transferred}}$ to a stationary mass of mass m during a collision is $\text{KE}_{\text{initial}} = ((m + M)/M)\text{KE}_{\text{transferred}}$ assuming an inelastic collision and $\text{KE}_{\text{initial}} = ((m + M)^2/4mM)\text{KE}_{\text{transferred}}$ assuming an elastic collision. These results can be used to model the breaking of boards in karate [144]. We assume the hand (as a fist) with mass $M = 0.7$ kg that strikes the board is the effective mass of the body in this collision.
(b) Assume the amount of energy needed to break a given wooden board is 5.3 J (with mass $m = 0.14$ kg) and a given concrete board is 1.6 J (3.2 kg). Show that the model assuming elastic collisions predicts 9.5 J is needed (corresponding to a strike speed of 5.2 m/s) to break the wood board and 2.7 J (2.8 m/s) is needed to break the concrete board, and the model assuming inelastic collisions predicts 6.4 J (4.3 m/s) is needed to break the wood board and 8.9 J (5.0 m/s) is needed to break the concrete board.
(c) Measurements show that the wood board breaks with an initial kinetic energy of 12.3 J (6.1 m/s) and the concrete board with an initial kinetic energy of of 37.1 J (10.6 m/s), so it is easier to break this wood board than this concrete board in practice.

What can you conclude about the model predictions, specifically regarding which model agrees the observation that the wood board breaks easier and the absolute numerical predictions themselves?

3.147 Repeat Problem 3.146 now assuming the combination of the hand and forearm is the effective mass that strikes the boards, so $M = 2\,\text{kg}$ [144].

3.148 Why would it not be advisable to use oak boards in karate demonstrations of breaking boards? (see Table 4.1).

3.149 A movie shows one man shooting another. The shooter is firm and steady when firing the gun and the other man recoils greatly when the bullet hits him. Does this make sense?

3.150 A bullet of mass 18 g traveling at 400 m/s hits someone's head and comes to rest in the middle of it. Treat the head as a uniform sphere of radius 8.5 cm and mass 5 kg. Assume that the head undergoes no trauma from the bullet and does not change its shape, and that it can be treated as separate from the rest of the body, and so it is the effective mass of the target in this collision.
(a) Find the speed of the head after the bullet stops.
(b) Assuming the bullet decelerates at a constant rate, find the collision time.
(c) Find the GSI for the head due to this collision.
(d) Ignoring the obvious physical damage done by the bullet, what is the fate of the person due to this collision?

3.151 Using the conditions of Problem 3.150 for a bullet lodging in a head, find
(a) the kinetic energy of the bullet before the collision,
(b) the thermal energy deposited in the head by the bullet (which is the difference of total kinetic energies before and after this inelastic collision, assuming no other effects),
(c) the fraction of the initial bullet kinetic energy this represents, and
(d) the temperature rise in the head right after the bullet comes to rest (assuming that the head remains in tact, the specific heat of the head is the same as for the body, that the temperature in the head is always constant, and there is no heat flow from the head).

3.152 When a bullet hits a head, the head is expected to recoil in the direction of the bullet, away from the shooter. Let us consider the gruesome situation when the bullet goes through the brain and exits it, and a jet of blood and brains follows the bullet out of the brain [142]. The brain (or head), bullet and jet can be treated as three objects. Assume the jet has 10% of the initial kinetic energy of the bullet and that it has a mass 15× that of the bullet. Show that if these circumstances could be modeled in this manner, the head would be expected to recoil in the direction opposite to the flight of the bullet, toward the shooter.

Hitting and Kicking Balls

3.153 Show that after a (point-like) object collides elastically with an identical object that is initially at rest, the first one is stationary and the second one moves with the same velocity as the first one had before the collision.

3.154 In terms of the coefficient of restitution e, find the fraction of kinetic energy that is recovered in the collision when a ball falls vertically and bounces from a horizontal floor.

3.155 A "super ball," basketball, volleyball, and softball are dropped from a height of 1.83 m (6 ft) onto a hardwood floor. How high do they bounce?

3.156 A baseball is found to bounce to a height of 0.46, 0.51, and 0.55 m from a height of 1.83 m (6 ft), when it has been previously cooled for 1 h in a freezer, left at room temperature, and heated for 15 min at 225 °C, respectively. What are the coefficients of restitution for the balls in the three cases? [66]

3.157 How far can a batted baseball travel? Assume the ball speed and bat speed are both 90 mph, $e = 0.46$, and the bat is much heavier than the ball. (Go into the rest frame of the bat, analyze the collision, and then return back.) (Your answer will be an overestimate because drag is omitted.)

3.158 If the same torque is generated by a pitcher in throwing a ball and a batter in hitting a ball, would the ability to throw a 90 mph fastball mean you could swing a bat with a 60 mph speed of the center of mass of the bat? Assume the moment of inertia for throwing the ball is that of a straight arm about the shoulder and that for hitting a ball is the body with two outstretched arms and a fully extended bat (as in Fig. 3.68).

3.159 Derive (3.127) by using the method described preceding it in the text.

3.160 Use our simple torque model for throwing a ball to show how you would expect the final bat speed to depend on bat weight. Is the dependence used in the text, (3.128), reasonable in terms of its linearity and the magnitude of its coefficients?

3.161 Derive (3.129) by using the method described preceding it in the text.

3.162 Use (3.126)–(3.130) to find the range of bat weights for which the batted ball speed for this example is within 2 mph of the maximum speed (obtained with the optimized bat weight).

3.163 For the example of optimizing the bat weight for a ball with speed −80 mph and a bat speed that decreases with bat weight, find the bat speed before and after the collision with the baseball.

3.164 Use (3.129) to determine the optimal bat weight and batted ball speed for a major leaguer whose bat speed varies as v_{bat} (in mph) $= 48 - 0.34 W_{\text{bat}}$ (with W_{bat} in oz). Over what range of bat weights is the batted ball speed within 5% of this maximum speed?

3.165 Equation (3.126) gives the speed of the baseball after it collides with a bat at its center of mass. In this problem we show that the analysis is a bit different if it hits the bat elsewhere, as in Fig. 3.68. We will use notation similar to that in the chapter, so v'_{ball} is the speed of the ball after the collision and v'_{bat} is the speed of the bat center of mass after the collision. (Before the collision these variables are unprimed):
(a) Conservation of linear momentum, (3.108), still holds. Show this is now

$$m_{\text{ball}} v_{\text{ball}} + m_{\text{bat}} v_{\text{bat}} = m_{\text{ball}} v'_{\text{ball}} + m_{\text{bat}} v'_{\text{bat}}. \tag{3.165}$$

(b) The total angular momentum of the bat and ball is conserved in the collision. The rotational kinetic energy of the ball is $\simeq$5–10% of its translational kinetic energy (show this), so we can neglect the rotation of the ball. Show that conservation of angular momentum gives

$$I_{\text{bat}}(\omega'_{\text{bat}} - \omega_{\text{bat}}) + B m_{\text{ball}}(v'_{\text{ball}} - v_{\text{ball}}) = 0, \tag{3.166}$$

where I_{bat} is the moment of inertia of the bat about its center of mass, and ω_{bat} and ω'_{bat} are the angular velocity of the bat before and after the collision, respectively.
(c) The coefficient of restitution equation, (3.109), now involves the speed of the bat at the point of impact. Show that it now becomes

$$e = -\frac{v'_{\text{ball}} - v_{\text{bat}} - B\omega'_{\text{bat}}}{v_{\text{ball}} - v_{\text{bat}} - B\omega_{\text{bat}}}. \tag{3.167}$$

(d) Solve these three equations to show that the speed of the ball after the collision is

$$v'_{\text{ball}} = \frac{(m_{\text{ball}} - e m_{\text{bat}} + m_{\text{ball}} m_{\text{bat}} B^2/I_{\text{bat}}) v_{\text{ball}} + m_{\text{bat}}(1+e)(v_{\text{bat}} + B\omega_{\text{bat}})}{m_{\text{ball}} + m_{\text{bat}} + m_{\text{ball}} m_{\text{bat}} B^2/I_{\text{bat}}} \tag{3.168}$$

(e) Show that this reduces to (3.126) when the ball hits the bat center of mass ($B = 0$). For more details see [17, 18, 137].

3.166 A ball is pitched at 85 mph and the batter retracts the bat to bunt the ball. How fast does he have to retract it (in mph) so the ball rebounds with a speed of 5 mph? (Assume all motion is in one dimension. Ignore the redirection of the ball to the ground by the bat and any rotation of the pitched ball.)

3.167 A force F is applied to a bat (such as in a collision with a ball), from the left to the point labeled as the sweet spot or center of percussion in Fig. 3.69a [36]. Let us examine the motion at a point x away (the bat handle in the figure), which we will call the pivot. The center of mass of the bat (as in the figure) is y away from the pivot. The bat has a mass m and moment of inertia I about its center of mass. We assume, without losing generality, that the force is constant over a time t.
(a) Show that the effect of the force on the center of mass is to move the bat to the right (in the figure) with a linear speed Ft/m.

(b) Show that the effect of the torque is to cause a rotation about the center of mass with a rotational speed $\omega = F(x - y)t/I$, and that this leads to a rotational speed of the bat handle of $F(x - y)yt/I$ in the direction shown in the figure (to the left).
(c) Explain why there is no motion of the pivot (the bat handle) when the bat is struck so that $x = y + I/my$ for this collision at the center of percussion.

3.168 A bat of mass m and length L has a uniform cross section, so it has a moment of inertia of $mL^2/12$ about its center of mass. Use the results of Problem 3.167 to show that the center of percussion is a distance $2L/3$ from the bat handle, which is considered to be one of the ends of the bat, or 2/3 of the way from the bat handle [36].

3.169 A pool ball of mass m_b, radius r, and moment of inertia $(2/5)m_b r^2$ is hit with a cue that provides a horizontal force at a height x above the table. Show that when the ball is hit at a height $x = (7/10)(2r)$, which is 70% of the ball diameter, the ball rolls (with topspin) without any initial skidding. (Hint: One way of solving this problem is to show that this requirement is formally equivalent to hitting the object at its center of percussion, as in Problem 3.167 and Fig. 3.69a, with the point of contact between the ball and table identified as the pivot [36].)

3.170 A slow-footed basketball player who likes to shoot off the backboard wants to deflate a well-inflated basketball, but his fast-footed point guard teammate (who dribbles (bounces) the ball) objects to this. Why?

3.171 Why do football punters and placekickers want to kick balls that are very well inflated?

3.172 Kicking a football, such as a "place kick," is a form of a collision – of the foot with the football:
(a) If the average force on the ball is 450 lb during the kick and the kick lasts for 8 ms, how fast does the ball move after the kick? Assume the football weighs 0.91 lb.
(b) How far does the ball travel (in yards) if its takeoff angle is 45° and drag is neglected?

3.173 A football player kicks a football at a takeoff angle of 45°; it lands 50 yd away (ignoring air resistance). Assume the person has a mass of 80 kg and height of 1.9 m (use Tables 1.6 and 1.7, and Fig. 1.15), and the mass of the football can be ignored during the kick. Also, assume that at the end of the kick, the kicker's foot is moving at the same speed as the football. Examine the kicking leg by itself, so it is the effective mass of the body in this collision, assuming that it has a constant mass per unit length, and find the average power generated by the leg during the kick in W and in hp if the duration of the kick is 0.2 s. If the body can produce this mechanical motion with 15% efficiency, what average power does it need, in W and in hp, to achieve this motion?

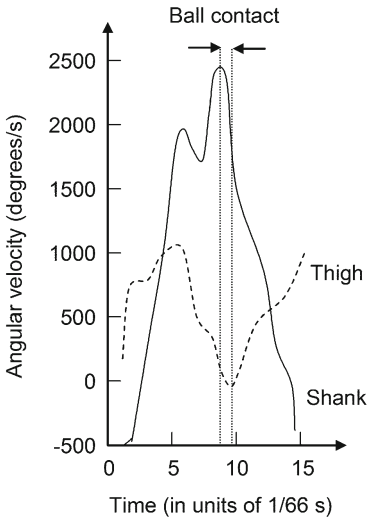


Fig. 3.86 The angular speed of the hip joint (thigh) and knee (shank) during kicking of a soccer ball (The unusual units of 1/66 s are from the 66/s frame rate of the photographs taken by [118].) (based on [97, 118]). For Problem 3.174

3.174 Assume that after a soccer ball is kicked, the ball moves with the same speed as the player's foot. For a soccer player with the dimensions of the average human in Chap. 1, how fast does the soccer ball move with thigh and shank (lower leg) motion as in Fig. 3.86?

3.175 Does Problem 3.174 assume that the collision is inelastic, partially elastic, or elastic? Is this a good assumption? If not, how would you correct the approach in that problem?

3.176 Explain why a baseball hit to the right side of the park develops spin that makes it curve to the right and one hit to the left side develops spin that makes it curve left.

3.177 Using a racket to impart topspin and backspin is very important in ping pong and tennis. Explain why topspin is imparted in the collision depicted in Fig. 3.70a and backspin in Fig. 3.70b, and describe the flight of the ball [60].

3.178 The club head of the driver in a golf drive moves in a circular arc with diameter equal to the shaft length 1 m with a speed 45 m/s. Find the centrifugal pull on the hands if the gravitational pull of the driver head is 2 N. (You will be able to understand why you need to grip the club strongly during the drive by comparing this result with the gravitational pull [36].)

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